

Physics 7501 (Autumn 2026)
Quantum Mechanics 1 (graduate core course)

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Abstract. In this elementary course of quantum mechanics, we will learn the mathematical basis of quantum mechanics (i.e. linear algebra), the postulates of quantum theory, and how to use quantum mechanics to understand microscopic quantum phenomena.

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0 Course Information

Instructor

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Contact Information

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Office Hours

Friday 10:00–11:00 am in PRB 2015

Lecture Logistics

- **Location:** Scott Lab N050
- **Time:** Wed. Fri. 11:10 am – 12:30 pm
- **Schedule:** See Google Spreadsheet

Teaching Assistant & Guided Group Work

- **Graduate Teaching Assistant (GTA):** Mr. Hyungjun Jeon (jeon.258@osu.edu)
- **GTA Office Hours:** Thursday 3:00–4:00 pm in PRB 2015
- **Guided Group Work (GGW) Sessions:** Thursdays 9:30 am – 11:00 am in PRB 0118. For questions regarding GGW sessions, please contact Dr. Beatriz E Burrola Gabilondo (burrolagabilondo.1@osu.edu).

Prerequisites

The course will assume physics and mathematics knowledge at the advanced undergraduate level (e.g., basic quantum mechanics, linear algebra, and differential equations), which we will review as needed.

Lecture Notes

Lecture notes will be uploaded to Carmen as PDF files generated from LaTeX sources at least 4 days before each lecture.

Main Textbooks (Recommended but Not Required)

- *Modern Quantum Mechanics* by J.J. Sakurai and Jim Napolitano (3rd Edition, Cambridge University Press, 2020)
- *Quantum Mechanics* (Vol. 3, Lectures on Theoretical Physics) by David Tong (Cambridge University Press, 2025). Problem sets available at <https://www.damtp.cam.ac.uk/user/tong/books/quantum>
- *Principles of Quantum Mechanics* by Ramamurti Shankar (2nd Edition, Plenum Press, 2011)
- *Quantum Computation and Quantum Information* by Michael Nielsen and Isaac Chuang (Cambridge University Press, 2011)

Other References

- *Lectures on Quantum Mechanics and on Topics in Quantum Mechanics*, David Tong, University of Cambridge
- *Quantum Mechanics: Non-Relativistic Theory* by Lev Landau and L. Lifshitz (2nd Edition, Pergamon Press)
- *Lectures on Quantum Mechanics* by Steven Weinberg (Cambridge University Press, 2013)
- *Quantum theory: a foundational approach* by Charis Anastopoulos (Cambridge University Press, 2024)
- *Quantum Mechanics* by Arjun Berera and Luigi Del Debbio (Cambridge University Press, 2021)

Reading Assignments

For each lecture, you will be expected to complete reading assignments before class. These will be announced on Carmen. On days when reading assignments are discussed in class, you must answer questions about the reading using Carmen "Quizzes" by 10:00 am the day of class. Your answers will not be graded for correctness; full credit is given for submitting on time. Late submissions will not be counted.

Homework

Homework will be assigned weekly on Carmen (except during holidays or exams). Submissions are due on Carmen by 11:59 pm on Fridays (or occasionally Wednesdays).

HW Quizzes

To verify individual effort in solving homework problems, a 10-minute quiz will take place during the last 10 minutes (12:20–12:30 pm) of lecture on homework due dates. Handwritten answers will be scanned and uploaded to Carmen/Gradescope.

Grading Scheme

All homework, quizzes, and exams will be graded by the GTA. Refer to Table 1 for the grading allocation.

Table 1: Grading basis for the course.

Item	Time	Assignments	Percentage
Reading Quizzes	10:00 am on lecture day	23	6%
Homework	Friday at 11:59 pm	11	18%
HW Quizzes	Last 10 mins of Friday lecture	11	18%
Mid Term Exam	Oct. 9 (Fri.), 11:30 am–1:30 pm	1	29%
Final Exam	Dec. 17 (Thu.), 12:00–2:00 pm	1	29%

Exams

- **Mid Term Exam:** In-person exam scheduled for 11:30 am–1:30 pm on Friday, Oct. 9 in PRB 1080 (Smith Seminar Room).
- **Final Exam:** In-person exam scheduled for 12:00–2:00 pm on Thursday, Dec. 17 in PRB 1080 (Smith Seminar Room).

Course Topics

- Linear algebra: mathematics of quantum mechanics
- Postulates of quantum mechanics

- Quantum entanglement
- Schrödinger equation
- Heisenberg's uncertainty principle
- One-dimensional potentials
- Introduction to scattering theory
- The harmonic oscillator and coherent states
- Squeezed states and quantum optics
- Adiabatic evolution and Berry's phase
- Quantum particles in electromagnetic fields

Shared Values

Please note The Ohio State University Shared Values.

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Academic integrity is essential to maintaining an environment that fosters excellence in teaching, research, and other educational activities. The Ohio State University and the Committee on Academic Misconduct (COAM, <https://oaa.osu.edu/resources/policies-and-procedures/committee-academic-misconduct>) expect that all students understand the University's Code of Student Conduct (<https://trustees.osu.edu/bylaws-and-rules/code>). Any activity that compromises academic integrity or subverts the educational process will be reported to COAM.

Artificial Intelligence and Academic Integrity

Generative AI tools are permitted as learning aids to support your understanding (*e.g.*, asking for alternative explanations or math reviews). However, every calculation, derivation, and explanation submitted must reflect your own understanding. You may not copy or lightly modify AI-generated solutions. AI tools are strictly prohibited during timed assessments, exams, and quizzes. When AI is permitted and used for homework, include a brief acknowledgment detailing the tool and how you verified its output.

Religious Accommodations

Ohio State accommodates students' religious beliefs and practices. Requests for accommodations must be communicated in writing to the instructor within the first 14 days of the semester. See the official policy at OAA Religious Holidays Policy.

Disability Statement & Illness

Students requesting disability accommodations should register with Student Life Disability Services (<https://slds.osu.edu/>) and notify the instructor as early as possible. If you are ill and need to miss class, notify the instructor immediately.

Intellectual Diversity

Ohio State is committed to open inquiry and intellectual diversity. Classroom discussions aim to foster critical thinking on course material without promoting any single point of view.

Grievances and Problem Solving

To resolve grievances concerning grades or academic practices, contact the instructor first, followed by the department chairperson, college dean, and provost (Faculty Rule 3335-8-23).

Creating an Environment Free from Harassment and Discrimination

The Ohio State University does not tolerate harassment, discrimination, or sexual misconduct. Reports can be submitted to the Civil Rights Compliance Office (CRCO) at <http://civilrights.osu.edu/>, by phone at 614-247-5838, or via email at civilrights@osu.edu.

1 Postulates and Mathematics of Quantum Mechanics

1.1 Key experiments that established quantum mechanics (8/26)

- Refer to Weinberg Chap 1. For further reading, try Anastopoulos Chap 2, Shankar 3.

Quantum mechanics was first motivated by a series of experiments that cannot be explained by classical mechanics and statistical physics. These exceptions to classical physics happen in the microscopic world, when the scale of physical quantities, such as energy, angular momentum and electric charge becomes comparable to certain characteristic scales.

The quantization of energy was observed in both light and atomic spectra, which motivated Planck, Einstein and others to propose a discretized (“quantized”) energy levels for photons and atoms. Bohr was able to explain the Rydberg formula[1] of hydrogen atom spectral lines under the assumption of a quantized angular momentum for electrons, which was later observed in the Stern-Gerlach experiment[2]. After de Broglie first proposed the wave nature of microscopic particles such as electrons, Davisson and Germer[3] were able to directly observe it in the diffraction pattern of electrons scattered by a nickel crystal. These experiments call for a new theory framework that defeats the intuition of classical physics.

1.1.1 Energy quantization of light in terms of photons

Young’s double-slit experiment clearly demonstrated the wave nature of light, whose energy should change continuously in classical electromagnetism. However, this point of view is seriously challenged by black-body radiation.

1.1.1.1 Black-body radiation and Planck’s formula

Consider light in a cubic box (a simplified model of a black body) of volume L^3 . A standing wave in the box satisfies the following condition for its wavevector $\vec{k} = (k_x, k_y, k_z)$:

$$\lambda_i = \frac{2\pi}{k_i} = \frac{2L}{n_i}, \quad n_i = 1, 2, 3, \dots, \quad i = x, y, z \quad (1.1)$$

Counting the number of standing wave modes in the box in terms of frequency $\nu = ck/2\pi$

$$\int dk_x dk_y dk_z = \frac{1}{8} 4\pi \int k^2 dk = \left(\frac{\pi}{L}\right)^3 \sum_{\{n_i\}} = \left(\frac{\pi}{L}\right)^3 N_{\text{modes}} \quad (1.2)$$

we have the density of states for light-wave modes per unit volume:

$$\frac{1}{L^3} \frac{dN}{d\nu} = \left(\frac{1}{\pi}\right)^3 \frac{\pi}{2} k^2 \frac{dk}{d\nu} = \frac{8\pi}{c^3} \nu^2 \quad (1.3)$$

Assuming the equipartition theorem in classical physics, i.e. each mode carries an averaged energy of $k_B T$, the black-body radiation energy per unit volume is

$$\frac{dE}{L^3 d\nu} = \frac{k_B T}{L^3} \frac{dN}{d\nu} = \frac{8\pi k_B T}{c^3} \nu^2 \quad (1.4)$$

This is known as the Rayleigh-Jeans formula and it correctly captures the long-wavelength (small ν) behavior of universal black-body radiation, first established by Kirchhoff around 1860. However, it diverges in the short wavelength limit, known as the ultraviolet catastrophe. Experimentally the Wien’s Distribution Law was observed at short wavelengths:

$$\frac{dE}{L^3 d\nu} \sim \nu^3 e^{-\alpha(T)\nu} \quad (1.5)$$

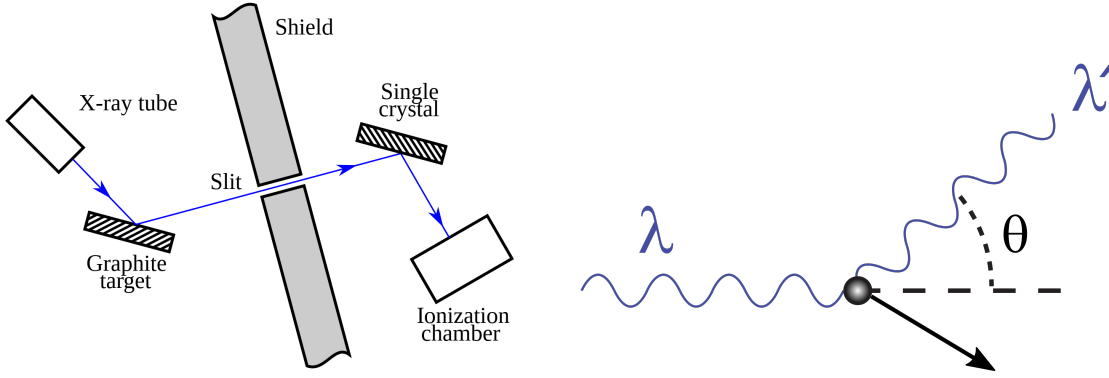


Figure 1.1: (Left) Schematic setup of Compton's experiment. (Right) The light-electron scattering process in the Compton experiment.

In order to reconcile long-wavelength Rayleigh-Jeans formula to short-wavelength Wien's law, Planck came up with his famous formula:

$$\frac{dE}{L^3 d\nu} = \frac{8\pi\nu^2}{c^3} \frac{h\nu}{e^{\alpha\nu} - 1}, \quad \alpha = \frac{h}{k_B T} \quad (1.6)$$

where the Planck constant is

$$h = 6.63 \times 10^{-34} \text{ J} \cdot \text{s} \quad (1.7)$$

Planck's formula (1.6) indicates that the statistical-averaged energy of light at frequency ν is given by

$$U(\nu) = \frac{h\nu}{e^{h\nu/K_B T} - 1} = \frac{\sum_{n \in \mathbb{Z}_+} nh\nu e^{-nh\nu/K_B T}}{\sum_{n \in \mathbb{Z}_+} e^{-nh\nu/K_B T}} \quad (1.8)$$

under Planck's quantization hypothesis that the microscopic states of light have energy $nh\nu$ where $n \in \mathbb{Z}_+$ is a positive integer.

1.1.1.2 Einstein's "light quantum" and photoelectric effect

Motivated by Planck's quantization hypothesis, based on the assumption that light gives out energy in multiples of energy quantum $h\nu$, Einstein predicted the photoelectric effect

$$K_{\max} = eV = h\nu - \Phi \quad (1.9)$$

where $K_{\max}$ is the maximal kinetic energy of emitted photoelectrons when light of frequency ν shines on a metal, and Φ is the work function of the metal. V is minimal voltage that is required to stop photoelectron emission. This experiment was carried out by Millikan[4], who obtained Planck's constant

$$h = e \frac{dV}{d\nu} \quad (1.10)$$

1.1.1.3 Compton effect demonstrated quantization of light in terms of photons

Discovered by Arthur Compton in 1923[5], when the energy of the X ray photon (~ 17 keV) was significantly larger than the binding energy of the atomic electron, the electrons could be treated as being free after scattering. In this case, a *Compton shift* between the wavelengths of

injected light (λ) and the scattered light (λ') was observed to satisfy the following relation with the scattering angle θ (see Fig. 1.1):

$$\lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \theta) \quad (1.11)$$

where h is the Planck's constant.

The Compton shift can be explained by energy and momentum conservation, by assuming the photon energy of $E = h\nu = hc/\lambda$.

1.1.2 Quantization of atomic energy levels

1.1.2.1 Rydberg formula for the spectral lines in a hydrogen atom

After the discovery of dark lines in the spectrum of the sun at the beginning of 18th century, it was later realized that hot atomic gases absorb and emit light only at certain definite frequencies. Building on works of Balmer and others, Rydberg discovered the following rule[1] for the frequency of emitted light from hydrogen gases:

$$\nu = \frac{c}{\lambda} = cR_H \left(\frac{1}{n^2} - \frac{1}{m^2} \right), \quad m, n \in \mathbb{Z}_+. \quad (1.12)$$

This clearly demonstrated the discreteness of atomic energy levels $E_n \propto \frac{1}{n^2}$, further motivating Bohr's theory of the hydrogen atom.

1.1.2.2 Franck-Hertz experiment

In the Franck-Hertz experiment (1914) illustrated in Fig. 1.2, in a vacuum tube fitted with three electrodes (an electron-emitting, hot cathode; a metal mesh grid; and an anode), when a grid voltage is applied between the cathode and the anode, the electric current through the tube filled with thin mercury vapours, due to electrons that pass through the grid and reach the anode, is measured in the experiment. As shown in the middle figure, the current increases rapidly at low voltages, and then sharply drops almost back to zero at 4.9 V, with similar effects observed as 9.8 V and other multiples of 4.9 V. This can be understood by the quantized energy levels of mercury atoms. Inside the single mercury droplet in the vacuum tube, as shown in Fig. 1.2.

1.1.3 Quantization of angular momentum

1.1.3.1 Bohr's explanation of hydrogen atom spectral lines

Rydberg's rule[1] indicates that the photon energy emitted by the hydrogen atom is given by

$$h\nu = E_m - E_n, \quad E_n = -\frac{hcR_H}{n^2}, \quad n \in \mathbb{Z}_+. \quad (1.13)$$

How to explain the $1/n^2$ quantization of hydrogen atom energy levels? According to classical mechanics, for an electron in the Coulomb force field of hydrogen nucleus, its speed v and radius r satisfies

$$\frac{mv^2}{r} = \frac{ke^2}{r^2} \implies E = \frac{1}{2}mv^2 - \frac{ke^2}{r} = -\frac{ke^2}{2r} \quad (1.14)$$

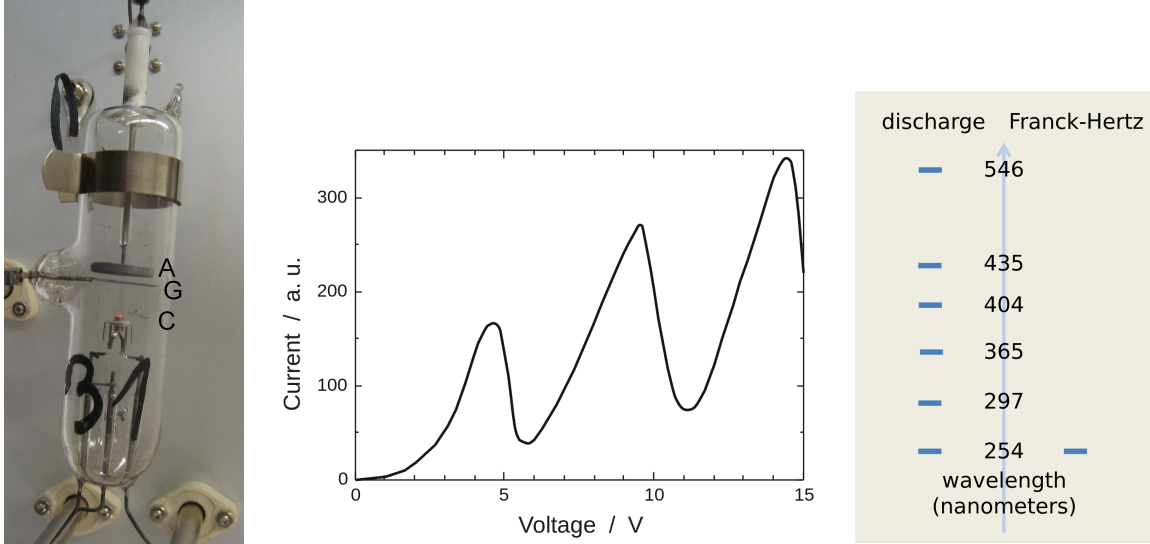


Figure 1.2: (Left) Photograph of a vacuum tube used for the Franck-Hertz experiment. (Middle) Anode current (arbitrary units) versus the grid voltage relative to the cathode. (Right) Wavelengths of light emitted by a mercury vapor discharge and by a Franck-Hertz tube in operation at 10 V. (Picture from Wikipedia)

where k is the Coulomb constant. To fit the n^2 dependence of the orbit radius r , Niels Bohr came up with the following quantization condition for angular momentum:

$$mvr = n\hbar \quad (1.15)$$

where $\hbar$ turns out to be $h/2\pi$. This quantization condition leads to the famous energy levels of electrons in a hydrogen atom:

$$E_n = -\frac{mk^2e^4}{2\hbar^2} \frac{1}{n^2} = -\frac{13.6 \text{ eV}}{n^2} \quad (1.16)$$

1.1.3.2 Stern-Gerlach experiment demonstrated the quantization of angular momentum

In 1922 Stern and Gerlach performed an experiment[2] measuring the deflection of silver atoms by an inhomogeneous magnetic field, as illustrated in Fig. 1.3. Due to the interaction energy $U = -\vec{\mu} \cdot \vec{B}$ of a magnetic moment $\vec{\mu}$ under a magnetic field $\vec{B} = B_z \hat{z}$, a non-uniform magnetic field exerts a force $\vec{F} = -\vec{\nabla}U = \mu_z \vec{\nabla}B_z$ on the moment-carrying silver atoms, therefore deflecting the silver atom beams in the inhomogeneous magnetic field. As shown in Fig. 1.3, in contrast a continuous range of values of the magnetic moment as expected in classical mechanics, the experimental observations pointed to quantization of magnetic moment $\mu_z = g\mu_B S_z/\hbar$, where we defined μ_B as the Bohr magneton:

$$\mu_B = \frac{\hbar e}{2m_e} \quad (1.17)$$

$g \approx 2$ as the electron g -factor and

$$S_z = \pm \frac{\hbar}{2} \quad (1.18)$$

is the quantized spin angular momentum of a silver atom observed in Stern-Gerlach experiment.

This experiment demonstrated that in addition to energy level quantization, other physical observable such as spin angular momentum are also quantized in microscopic scales such as

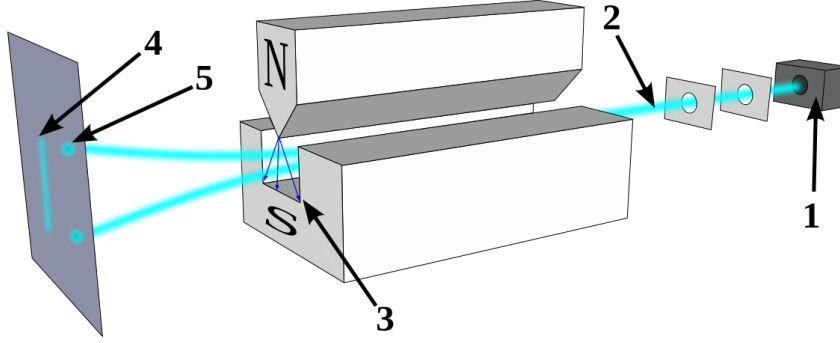


Figure 1.3: Schematic setup of Stern-Gerlach experiment: Silver atoms travelling through an inhomogeneous magnetic field, being deflected up or down depending on their spin. (1) furnace, (2) beam of silver atoms, (3) inhomogeneous magnetic field, (4) the classically expected result, (5) the observed result.

in silver atoms. Moreover, we can consider sequential Stern-Gerlach experiments shown in Fig. 1.4, where silver atoms are deflected by multiple inhomogeneous magnetic fields in a sequence. Remarkably, unlike the classical case of light passing through polarizers, which filter out everything after two polarizers with perpendicular polarization axes, two quantum mechanical “beam splitters” with perpendicular magnetic field directions still allows two splitted beams to pass through, as illustrated in the middle and bottom parts of Fig. 1.4.

1.1.4 Davisson-Germer experiment demonstrated the wave nature of electrons

In 1924 de Broglie formally proposed the wave nature of electrons and other particles, in terms of their de Broglie wavelengths:

$$\lambda = \frac{h}{p} \quad (1.19)$$

where p is the momentum of the particle. If this proposal were true, it means that microscopic particles such as electrons should exhibit interference phenomena in the same way as acoustic and light waves.

In 1927 Davisson and Germer[3] performed an experiment to observe the diffraction pattern of electrons scattered by the surface of a crystal of nickel metal, as shown in Fig. 1.5. The highest intensity was observed at an angle $\theta = 50^\circ$ with a voltage of 54 V, giving the electrons a kinetic energy of 54 eV. To be compatible with the Bragg’s law

$$n\lambda = 2d \sin(90^\circ - \theta/2), \quad n \in \mathbb{Z}_+. \quad (1.20)$$

with the lattice spacing of nickel $d = 0.091$ nm, the $n = 1$ Bragg peak in (1.20) exactly corresponds to the de Broglie wavelength (1.19) of electrons. We will examine the experimental

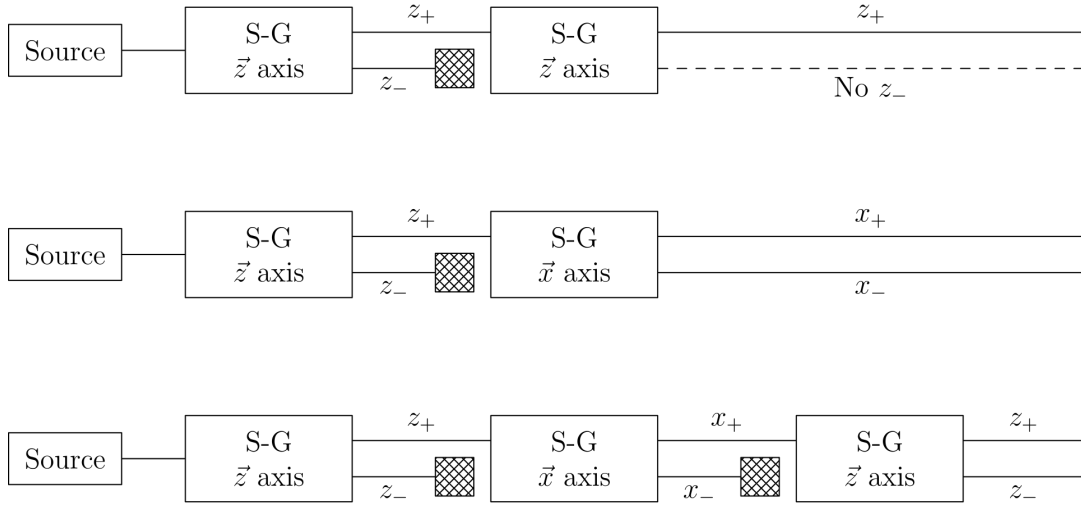


Figure 1.4: Sequential Stern-Gerlach experiments. (Picture from S&N)

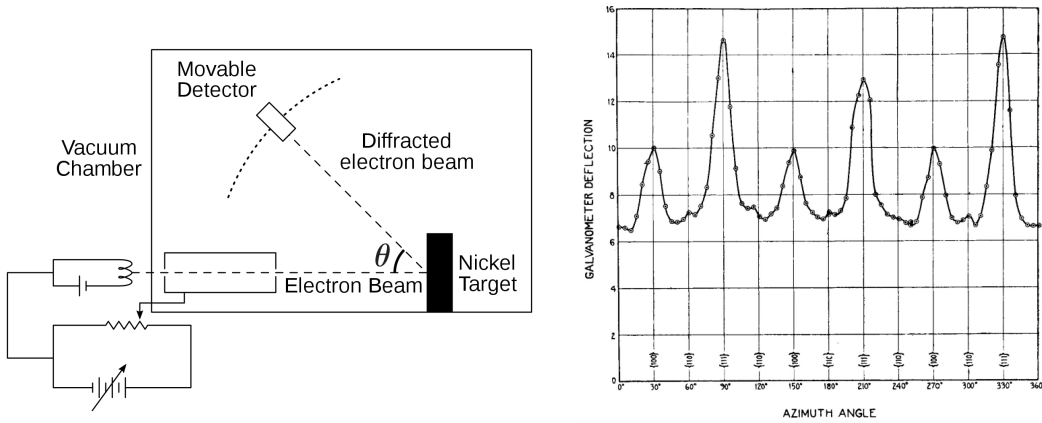


Figure 1.5: (Left) Schematic setup of Davisson-Germer experiment. (Right) Intensity of electron scattering vs. azimuth angle at 54 volts, co-latitude $\theta = 50^\circ$.

1.1 Key experiments that established quantum mechanics (8/26)

results in Fig. 1.5 more carefully in Homework 1.

1.2 Hilbert space and state vectors (8/28)

- Refer to N&C 2.1, Tong 3.1, Shankar 1.1-1.4

How to explain the above experimental observations? 4 postulates, derived (or more precisely, guessed) after a long process of trial and error, form the foundation for a theory framework that explains all observed experimental phenomena, known as quantum mechanics. Below we will introduce these postulates one by one, along the way of introducing linear algebra as the mathematics of quantum mechanics.

Postulate 1.1. *Associated to any isolated physical system is a complex vector space with inner product (i.e. a Hilbert space) known as the state space of the system. The system is completely described by its state vector, which is a unit vector in the system's state space.*

The key mathematical framework for quantum mechanics is linear algebra. The basic objects of linear algebra are vector spaces:

A *vector space* over a *scalar field* $\mathbb{F}$ is a set $\mathbb{V}$ of objects (called *vectors*) equipped two binary operations: a *vector addition* (“+”)

$$|v\rangle + |w\rangle \in \mathbb{V}, \quad \forall |v\rangle, |w\rangle \in \mathbb{V}. \quad (1.21)$$

and a *scalar multiplication*:

$$a|v\rangle \in \mathbb{V}, \quad \forall |v\rangle \in \mathbb{V}, a \in \mathbb{F}. \quad (1.22)$$

Mathematically, the set $\mathbb{V}$ forms an abelian group under vector addition operation and the scalar multiplication defines a ring homomorphism from the field $\mathbb{F}$ into the endomorphism ring of this abelian group. We will break down these terminology below.

It is worthwhile to define a group since we will study symmetry in quantum mechanics later. A group is a set $V = \{u, v, \dots\}$ equipped with a *closed* binary operation $+$: $V \times V \rightarrow V$

$$u + v = w \in V, \quad \forall u, v \in V. \quad (1.23)$$

where 3 group axioms are satisfied:

(1) *Identity element*:

$$\exists 0 \in V, \text{ s.t. } \forall u \in V, u + 0 = 0 + u = u. \quad (1.24)$$

(2) *Inverse*:

$$\forall u \in V, \exists (-u) \in V, \text{ s.t. } u + (-u) = 0. \quad (1.25)$$

(3) *Associativity*:

$$(u + v) + w = u + (v + w), \quad \forall u, v, w \in V. \quad (1.26)$$

An Abelian group is a group whose addition operation is commutative:

$$u + v = v + u, \quad \forall u, v \in V. \quad (1.27)$$

The scalar multiplication is another map $\mathbb{F} \times \mathbb{V} \rightarrow \mathbb{V}$ that is *associative*

$$a(b|v\rangle) = (ab)|v\rangle, \quad \forall a, b \in \mathbb{F}, |v\rangle \in \mathbb{V} \quad (1.28)$$

and *distributive in both the vectors and the scalars*:

$$a(|u\rangle + |v\rangle) = a|u\rangle + a|v\rangle, \quad (1.29)$$

$$(a + b)|v\rangle = a|v\rangle + b|v\rangle, \quad \forall a, b \in \mathbb{F}, |u\rangle, |v\rangle \in \mathbb{F}. \quad (1.30)$$

The above conditions satisfied by vector addition and scalar multiplication are summarized well in Chapter 1.1 of Shankar.

Of particular interests to quantum mechanics is a *Hilbert space*, which is a (n -dimensional) vector space $\mathbb{C}^n$ over the field $\mathbb{C}$ of complex numbers, equipped with a map $\mathbb{V} \times \mathbb{V} \rightarrow \mathbb{C}$ known as an *inner product*:

$$\langle u|v\rangle \in \mathbb{C}, \quad \forall |u\rangle, |v\rangle \in \mathbb{V}. \quad (1.31)$$

In the Dirac notation, $\langle u|$ is known as a *bra*, and $|v\rangle$ is known as a *ket*. The inner product is *skew-symmetric*

$$\langle u|v\rangle = \langle v|u\rangle^* \quad (1.32)$$

positive semidefinite

$$\langle v|v\rangle \geq 0, \quad \langle v|v\rangle = 0 \iff |v\rangle = 0. \quad (1.33)$$

and *linear in ket*

$$\langle u|(\sum_i \lambda_i |v_i\rangle) = \sum_i \lambda_i \langle u|v_i\rangle \quad (1.34)$$

These properties are summarized in Chapter 1.2 of Shankar and Chapter 2.1.4 of N&C.

Note that an inner product also defines a norm for each vector in vector space $\mathbb{V}$:

$$||v\rangle| \equiv \sqrt{\langle v|v\rangle} \quad (1.35)$$

Two important properties of an inner product are

Theorem 1.2. *Cauchy-Schwarz inequality*:

$$\langle v|v\rangle \langle u|u\rangle \geq |\langle u|v\rangle|^2 \quad (1.36)$$

The equality holds if and only if $|v\rangle$ and $|w\rangle$ are linearly dependent.

Theorem 1.3. *Triangle inequality for the norms of vectors*:

$$||v\rangle + |w\rangle| \leq ||v\rangle| + ||w\rangle| \quad (1.37)$$

Triangle inequality can be derived from the Cauchy-Schwarz inequality, which is a part of Homework 2.

Strictly speaking, in addition to being a vector space equipped with an inner product (known as an inner product space), a Hilbert space must also be *complete* in the sense that every Cauchy sequence $\{v^{(k)}\}$ of vectors in $\mathbb{V}$ has a limit that also lies in $\mathbb{V}$:

$$\lim_{k \rightarrow \infty} v^{(k)} \in \mathbb{V}, \quad \text{if } \forall \epsilon > 0, \quad \exists N \in \mathbb{Z}_+, \text{ s.t. } |v^{(m)} - v^{(n)}| < \epsilon, \quad \forall m, n > N. \quad (1.38)$$

For an inner product space of finite dimensions on any complete field $\mathbb{F}$, such as real and complex numbers $\mathbb{F} = \mathbb{R}, \mathbb{C}$, it is automatically a Hilbert space. Below are a few examples of Hilbert spaces, and inner product spaces that are not Hilbert spaces:

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- n -dimensional Hilbert space on complex numbers: $\mathbb{V} \simeq \mathbb{C}^n$
- $n \times n$ complex matrices equipped with the Hilbert-Schmidt inner product forms a Hilbert space $L_{\mathbb{C}^n}$ (see Homework 1)
- n -dimensional Hilbert space on real numbers: $\mathbb{V} \simeq \mathbb{R}^n$
- n -dimensional vector space $\mathbb{V} \simeq \mathbb{Q}^n$ on rational numbers is not a Hilbert space because it is not complete, since a Cauchy sequence of rational numbers can converge to an irrational number.
- The inner product space of continuous square-integrable complex functions on interval $[a, b]$ with the following L^2 inner product:

$$\langle f_1 | f_2 \rangle \equiv \int_a^b f_1^*(x) f_2(x) dx \quad (1.39)$$

is not a Hilbert space due to its incompleteness: a Cauchy sequence of continuous functions can be discontinuous. Its completion is the Hilbert space $L^2[a, b]$.

- For the same reason of incompleteness, the inner product space of polynomials on $[a, b]$ with the same L^2 inner product is not a Hilbert space, even though it's a subspace of $L^2[a, b]$.

A Hilbert space forms the *state space* described in Postulate 1.1. With the help of the inner product structure, we can always choose a complete set of orthonormal basis $\{|i\rangle\}$ that expands the full state space $\mathbb{V}$, such that

$$|v\rangle = \sum_i v_i |i\rangle, \quad \forall |v\rangle \in \mathbb{V}; \quad \langle i | j \rangle = \delta_{i,j}, \quad v_i = \langle i | v \rangle. \quad (1.40)$$

where $\delta_{i,j}$ is the Kronecker delta function. Given a n -dimensional vector space V , one can always find a set of n linearly independent vectors $\{e_j | 1 \leq j \leq n\}$, known as a basis of vector space V , such that any vector $v \in V$ can be uniquely expressed as a linear superposition of the basis vectors: $v = \sum_j v_j e_j$ with $v_j \in \mathbb{F}$. Given the inner product structure in a Hilbert space, a set of complete orthonormal basis can be constructed following the Gram-Schmidt procedure, discussed in details in Chapter 1.2 of Shankar and Chapter 2.1.4 of N&C. In such a complete orthonormal basis of Hilbert space $\mathbb{V} = \mathbb{C}^n$, a Dirac ket $|v\rangle$ can be represented by a column vector $\vec{v} = (v_1, \dots, v_n)^T$, a Dirac bra represented by a row vector, and the inner product reduces to the familiar form in $\mathbb{C}^n$:

$$\langle u | v \rangle = \sum_i u_i^* v_i = (u_1^*, \dots, u_n^*) \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \quad (1.41)$$

The column vector $(v_1, \dots, v_n)$ is known as the *wavefunction* of state $|v\rangle$ in the basis set $\{|i\rangle\}$.

A special set of complete orthonormal basis that plays a particularly important role in quantum mechanics is the eigenstates $|x\rangle$ of position operator $\hat{x}$, where we consider one dimension for simplicity. For any normalized state $|\psi\rangle$, its continuous-variable wavefunction

$$\psi(x) \equiv \langle x | \psi \rangle, \quad \int dx |\psi(x)|^2 = \langle \psi | \psi \rangle = 1. \quad (1.42)$$

determines the probability of detecting the quantum-mechanical particle at position x :

$$\text{Pr}(x) = |\psi(x)|^2, \quad \int dx \text{Pr}(x) = 1. \quad (1.43)$$

1.3 Linear operators and observables (9/2)

- Refer to N&C 2.1; Tong 3.2, 4.1; S&N 1.3-1.4; Shankar 1.5-1.6

Given two vector spaces $\mathbb{V}$ and $\mathbb{W}$, we can also define a linear operator $\hat{O}$ as any function that maps the vector space $\mathbb{V}$ to $\mathbb{W}$, which is linear in its inputs:

$$\hat{O}(\sum_i a_i |v_i\rangle) = \sum_i a_i \hat{O}(|v_i\rangle) \quad (1.44)$$

In the case of $\mathbb{V} = \mathbb{W}$ we call it a linear operator $\hat{O} \in \mathbb{L}_{\mathbb{V}}$ on vector space $\mathbb{V}$, which is of particular interest to quantum mechanics.

Using a complete set of orthonormal basis $\{|j\rangle | 1 \leq j \leq \dim \mathbb{V}\}$ for Hilbert space $\mathbb{V}$, each linear operator has a matrix representation:

$$\hat{O} = \sum_{a,b} O_{a,b} |a\rangle \langle b|, \quad O_{a,b} \equiv \langle a | \hat{O} | b \rangle \quad (1.45)$$

In particular, for the identity matrix which maps every vector to itself, we have the **completeness relation** for basis $\{|i\rangle\}$, also known as the **resolution of identity**:

$$\sum_i |i\rangle \langle i| = \hat{1}. \quad (1.46)$$

In the case of continuous variables, e.g. coordinates for a particle in free space, the summation will be replaced by integral:

$$\int dx |x\rangle \langle x| = \hat{1} \quad (1.47)$$

where $|x\rangle$ is an eigenstate of position operator $|x\rangle$ with eigenvalue x .

We can also define the product of two linear operators $\hat{M}, \hat{N} \in \mathbb{L}_{\mathbb{V}}$:

$$\hat{M}\hat{N} = \sum_{a,b,c} M_{a,b} N_{b,c} |a\rangle \langle c| \quad (1.48)$$

their commutator

$$[\hat{M}, \hat{N}] = \hat{M}\hat{N} - \hat{N}\hat{M} \quad (1.49)$$

and anti-commutator

$$\{\hat{M}, \hat{N}\} = \hat{M}\hat{N} + \hat{N}\hat{M} \quad (1.50)$$

and the *Hermitian conjugate* (or *Hermitian adjoint*) of an operator

$$\hat{O}^\dagger = \sum_{a,b} O_{b,a}^* |a\rangle \langle b| = (O^T)^* = (O^*)^T \quad (1.51)$$

Specifically in Postulate 1.9 about quantum (projective) measurements, each physical observable corresponds to a Hermitian linear operator, represented as a Hermitian matrix in the orthonormal basis:

$$\hat{M} = \hat{M}^\dagger = \sum_{a,b} M_{a,b} |a\rangle \langle b|, \quad (M^\dagger)_{a,b} = M_{b,a}^* = M_{a,b} \equiv \langle a | \hat{M} | b \rangle. \quad (1.52)$$

The diagonal form

$$\hat{O} = \sum_a O_a |a\rangle\langle a| \quad (1.53)$$

where $\{|a\rangle\}$ is a complete set of orthonormal basis of the Hilbert space, is also known as the *spectral decomposition* of a matrix $\hat{O}$. If there is a complete set of orthonormal basis $\{|a\rangle\}$ such that the linear operator $\hat{O}$ can be rewritten as the diagonal form (1.53), the matrix $\hat{O}$ is called “diagonalizable”.

Not all matrices are diagonalizable (i.e. have a diagonal form), two non-diagonalizable examples are

$$\hat{M}_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \hat{M}_2 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad (1.54)$$

$\hat{M}_1$ is known as a Jordan block with only 1 eigenvector, while $\hat{M}_2$ has two eigenvectors with different eigenvalues, but not orthogonal to each other. Generally, any complex square matrix can be decomposed into tensor sums of Jordan blocks by similarity transformations.

Below are a few useful theorems on the diagonalization of matrices:

Theorem 1.4. *The spectral decomposition theorem:* A linear operator $\hat{O}$ on vector space $\mathbb{V}$ is diagonal w.r.t. some orthonormal basis of $\mathbb{V}$ if and only if operator $\hat{O}$ is normal i.e. $\hat{O}^\dagger \hat{O} = \hat{O} \hat{O}^\dagger$.

Refer to N&C Box 2.2 for proof of this theorem.

Theorem 1.5. A normal matrix is Hermitian if and only if it has real eigenvalues.

Theorem 1.6. The eigenkets corresponding to different eigenvalues of a Hermitian matrix are orthogonal.

Theorem 1.7. A positive linear operator must be Hermitian.

Theorem 1.8. *Simultaneous diagonalization theorem:* Two Hermitian linear operators $\hat{A}$ and $\hat{B}$ can be diagonalized simultaneously if and only if $[\hat{A}, \hat{B}] = 0$.

In other words, there is a complete set of simultaneous eigenket $\{|a, b\rangle\}$ such that

$$\hat{A}|a, b\rangle = a|a, b\rangle, \quad \hat{B}|a, b\rangle = b|a, b\rangle. \quad (1.55)$$

if and only if $[\hat{A}, \hat{B}] = 0$.

An Hermitian operator can be diagonalized into a special orthonormal basis $\{|i\rangle\}$, known as its eigenbasis, in which the linear operator is represented by a diagonal matrix

$$\hat{M} = \sum_i \lambda_i |i\rangle\langle i| \quad (1.56)$$

If we use the set $\{m\}$ to label all different (real) eigenvalues of matrix $\hat{M}$, we can rewrite the above diagonal form (1.56) as

$$\hat{M} = \sum_m m \hat{P}_m, \quad \hat{P}_m = \sum_j \delta_{\lambda_j, m} |j\rangle\langle j| \quad (1.57)$$

where operators $\hat{P}_m$ are known as *projection operators* (or *projectors*) onto the eigenspace of $\hat{M}$ with eigenvalue m . Mathematically, a projector $\hat{P}$ is defined as a linear operator satisfying:

$$\hat{P}^2 = \hat{P} \quad (1.58)$$

You will learn more about projectors in HW2.

Given a state $|\psi\rangle$ and a physical observable $\hat{M}$, the (probability-weighted) *expectation value* $\langle\hat{M}\rangle$ of operator $\hat{M}$ w.r.t. state vector $|\psi\rangle$ is

$$\langle\hat{M}\rangle \equiv \frac{\langle\psi|\hat{M}|\psi\rangle}{\langle\psi|\psi\rangle} \quad (1.59)$$

Naturally, the expectation value of any physical observable is a real number, since any physical observable corresponds to a Hermitian operator.

We note that the expectation value (1.59) of any observable is the same for two linearly dependent states $|\psi\rangle$ and $\alpha|\psi\rangle$ for any complex scalar $\alpha \in \mathbb{C}$. This means $|\psi\rangle$ and $\alpha|\psi\rangle$ describe the same physical state. In other words, in a N -dimensional Hilbert space $\mathbb{V} = \mathbb{C}^N$, for a generic state

$$|\psi\rangle \equiv \sum_{j=1}^N \psi_j |j\rangle \quad (1.60)$$

where $\{|j\rangle | 1 \leq j \leq N\}$ is a complete set of orthonormal basis of $\mathbb{V}$, although $\{\psi_j \in \mathbb{C} | 1 \leq j \leq N\}$ correspond to $2N$ independent real variables, only $2N - 2$ of them are physically measurable, because the global phase factor and normalization leads to 2 redundancies that cannot be physically observed.

In the simplest example of a single qubit, we have $\mathbb{V} = \mathbb{C}^2$ for the $N = 2$ -dimensional Hilbert space. There are 4 linearly independent Hermitian operators in $\mathbb{C}^2$, whose matrix representations are

$$\hat{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (1.61)$$

As you will learn in HW1, they (up to normalization) form a complete set of orthonormal basis for the Hilbert space $L_{\mathbb{C}^2}$ of linear operators. In the matrix representation, a generic state vector can be written as

$$|\psi\rangle = \alpha \begin{pmatrix} \cos \theta e^{i\phi} \\ \sin \theta \end{pmatrix}, \quad \alpha \in \mathbb{C} \quad (1.62)$$

θ and ϕ are the two variables in the state vector that can be physically measured. One can carry out measurements for two observables to extract θ and ϕ :

$$\langle\hat{\sigma}_z\rangle = \cos(2\theta), \quad \langle\hat{\sigma}_x\rangle = \sin(2\theta) \cos \phi \quad (1.63)$$

1.3.1 Appendix: Functions of operators/matrices

A function $f(\hat{M})$ of a square matrix $\hat{M}$ can be defined by the following Taylor expansion

$$f(\hat{M}) = \sum_{n=0}^{\infty} \frac{\hat{M}^n}{n!} \frac{d^n f(x)}{dx^n} \Big|_{x=0} \quad (1.64)$$

if the Taylor expansion converges at $x = 0$. In quantum mechanics we will mostly encounter square matrices that can be diagonalized, such as Hermitian and unitary matrices.

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For a diagonalizable matrix $\hat{M}$ with eigenkets $\{|j\rangle\}$ and eigenvalues $\{\lambda_j\}$, the matrix can be written as

$$\hat{M} = \sum_j \lambda_j |j\rangle\langle j| \quad (1.65)$$

and a function of this diagonalizable matrix can be defined as

$$f(\hat{M}) = \sum_j f(\lambda_j) |j\rangle\langle j| \quad (1.66)$$

1.4 Projective measurements (9/4)

- Refer to N&C 2.2.4-2.2.7, Tong 3.4, 4.2, 16.4.1

Postulate 1.9. A *projective measurements* is described by a physical observable, $\hat{M} = \hat{M}^\dagger$ on the Hilbert space of the system being observed. The observable has a spectral decomposition

$$\hat{M} = \sum_m m \hat{P}_m \quad (1.67)$$

where $\hat{P}_m$ is the projector onto the eigenspace of M with eigenvalue m . The possible outcomes of the measurement correspond to the eigenvalues m of the observable. Upon measuring state $|\psi\rangle$, the probability of getting result m is given by

$$\text{Pr}(m) = \langle \psi | \hat{P}_m | \psi \rangle \quad (1.68)$$

If the outcome m occurred, the state of the quantum system collapsed into a new state

$$\hat{P}_m |\psi\rangle / \sqrt{\text{Pr}(m)} \quad (1.69)$$

immediately after the measurement.

Postulate 1.9 is in fact only a special form of quantum measurements, which will be discussed in detail in later parts of this course. Most generally, a generalized quantum measurement is described as follows:

Postulate 2a: A **generalized quantum measurement** is described by a collection $\{M_m\}$ of measurement operators. These are operators acting on the state space of the system being measured. The index m refers to the measurement outcomes that may occur in the experiment. If the (normalized) state of the quantum system is $|\psi\rangle$ immediately before the measurements, then the probability that result m occurs is given by

$$\text{Pr}(m) = \langle \psi | M_m^\dagger M_m | \psi \rangle \quad (1.70)$$

and the state of the system after the measurement is $\frac{M_m |\psi\rangle}{\sqrt{\langle \psi | M_m^\dagger M_m | \psi \rangle}}$. The measurement operators satisfy the completeness equation

$$\sum_m M_m^\dagger M_m = \hat{1}. \quad (1.71)$$

When we choose $\hat{M}_m$ to be projector $\hat{P}_m$ into the eigenspace of observable $\hat{M}$ with eigenvalue m , the above generalized measurement reduces to the a projective measurement. Now it also becomes clear that the expectation value of observable $\hat{M}$ w.r.t. to (normalized) state vector $|\psi\rangle$ is nothing but the probability-weighted average of measurement outcomes:

$$\langle \hat{M} \rangle \equiv \langle \psi | \hat{M} | \psi \rangle = \sum_m m \times \text{Pr}(m) \quad (1.72)$$

One can also find out the *standard deviation* of measuring observable $\hat{M}$ in state $|\psi\rangle$:

$$\Delta(M) \equiv \sqrt{\langle (\hat{M} - \langle \hat{M} \rangle)^2 \rangle} = \sqrt{\langle \hat{M}^2 \rangle - \langle \hat{M} \rangle^2} \quad (1.73)$$

A significant consequence of the collapse part in Postulate 1.9 is the following: starting from any state vector $|\psi\rangle$, if we measure the same observable $\hat{M}$ multiple times, the measurement

outcome does not change any more after the 1st measurement. This is because the state already collapse into an eigenstate of $\hat{M}$ with some eigenvalue m after the 1st measurement.

Another interesting aspect comes from *incompatible measurements* of physical observables that do not commute with each other. If we perform two measurements $\hat{A}$ and $\hat{B}$ that satisfy $[\hat{A}, \hat{B}] \neq 0$, for a generic quantum state, the measurement outcome depend on the order of the two measurements: i.e. measuring $\hat{A}$ after $\hat{B}$ versus $\hat{B}$ after $\hat{A}$ will yield different results. For example, starting from the eigenstate $|\uparrow\rangle$ of Pauli matrix σ_z , if we measure σ_z and then σ_x , there is a probability of 1/2 for us to end up with +1 outcomes for both measurements. However, if σ_x is measured before the σ_z measurement, there is only a probability of 1/4 of +1 outcomes for both measurements.

1.5 Unitary evolution of a closed quantum system (9/9)

- Refer to S&N 2.1; Tong 3.3; N&C 2.1.8, 2.2.2, 4.2; Shankar 4.3

In addition to Postulates 1.1 and 1.9 introduced earlier, we have two more postulates, one regarding how a quantum system evolves with time:

Postulate 1.10. *The time evolution of the state of a closed quantum system is described by the Schrodinger equation:*

$$i\hbar \frac{d|\psi\rangle}{dt} = \hat{H}|\psi\rangle \quad (1.74)$$

where $\hbar = h/2\pi \approx 1.05 \times 10^{-34}$ J·s is a constant, and $\hat{H}$ is a fixed Hermitian operator known as the Hamiltonian of the closed system.

Postulate 1.10 states that the state vector associated with a closed quantum system evolves following the Schrodinger equation (1.74). This postulate can be states in an equivalent manner:

Postulate 3a *The evolution of a closed quantum system is described by a unitary transformation. That is, the state $|\psi(t_2)\rangle$ of the system at time t_2 is related to the state $|\psi(t_1)\rangle$ at time t_1 by a unitary operator U which only depends on the times t_1 and t_2 :*

$$|\psi(t_2)\rangle = U(t_1, t_2)|\psi(t_1)\rangle, \quad U(t_1, t_2) = e^{-i(t_2-t_1)\hat{H}/\hbar} \quad (1.75)$$

One can further generalize this to a time-dependent Hamiltonian $\hat{H}(t)$ where we have

$$U(t_1, t_2) = \mathcal{T}e^{-i \int_{t_1}^{t_2} \hat{H}(t) dt / \hbar} \quad (1.76)$$

where $\mathcal{T}$ denotes time ordering in the integral, as explained in S&N 2.1.

In linear algebra, a unitary operator $\hat{U}$ is defined as any linear operator satisfying

$$\hat{U}^\dagger \hat{U} = \hat{1}. \quad (1.77)$$

which also implies $\hat{U}\hat{U}^\dagger = \hat{1}$. Due to the spectrum decomposition theorem, a unitary operator can be diagonalized, and its eigenvalues all have modulus 1:

$$\hat{U}|\psi\rangle = e^{i\theta}|\psi\rangle \quad (1.78)$$

Three examples of unitary operators in quantum mechanics are the following: (1) unitary time evolution operator $U(t_1, t_2) = e^{-i(t_2-t_1)\hat{H}/\hbar}$ for a closed quantum system discussed above; (2) a symmetry operator in a quantum system as we will discuss later in this course; (3) a quantum (logic) gate in quantum computing. For instance, in a two-dimensional closed quantum system, also known as a qubit, the common quantum gates are summarized in Table 1.1.

Starting from a complete set of orthonormal basis vectors $\{|i\rangle\}$ of Hilbert space $\mathbb{V}$, after the action of a unitary operator $\hat{U}$, we can obtain another complete set of orthonormal basis vectors $\{|i'\rangle = \hat{U}|i\rangle\}$ of the same Hilbert space. In other words, each unitary operator implements a change of basis in the Hilbert space. Therefore, the unitary time evolution merely performs a rotation for vectors in the Hilbert space, since it only changes the state itself but not the inner product of any two states.

In the case of unitary time evolution $U(t_1, t_2)$, the Hamiltonian $\hat{H}$ of a closed quantum system plays a crucial role. In its spectral decomposition:

$$\hat{H} = \sum_a E_a |a\rangle\langle a| \quad (1.79)$$

Single-qubit gate	Symbol	Unitary matrix
Pauli-X (bit flip)	X	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
Pauli-Y	Y	$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$
Pauli-Z (phase flip)	Z	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
Hadamard	H	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$
Phase	S	$\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$
$\pi/8$	T	$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$

Table 1.1: Common single-qubit gates and associated unitary matrices.

the eigenkets $|a\rangle$ are called the *energy eigenstates*, or *stationary states*, and the eigenvalue E_a is called the energy of state $|a\rangle$. In particular, the eigenstate with the lowest energy is known as the *ground state*, while other states with higher energy are known as *excited states*. According to Schrodinger equation (1.74), a generic state vector

$$|\psi\rangle = \sum_a c_a |a\rangle \quad (1.80)$$

evolves as

$$|\psi(t)\rangle = e^{-it\hat{H}/\hbar} |\psi\rangle = \sum_a c_a e^{-iE_a t/\hbar} |a\rangle \quad (1.81)$$

and the different oscillation frequency $\omega_a = E_a/\hbar$ in different components of state $|\psi(t)\rangle$ can result in oscillations of physical observables. Two such examples, i.e. (1) spin precession; and (2) neutrino oscillation are discussed in detail in S&N 2.1, and will be a part of HW3. In particular, a physical observable you will encounter a lot in the future is the *transition probability*:

$$P(i \rightarrow f) \equiv |\langle f | e^{-i\hat{H}t/\hbar} | i \rangle|^2 \quad (1.82)$$

which measures the probability that an initial state $|i\rangle$ evolves into a final state $|f\rangle$ after time t . This quantity is directly related to the outgoing beam intensity in scattering experiments.

The transition probability is the modulus square of the *propagator*

$$G(i, f : t) \equiv \langle f | e^{-i\hat{H}t/\hbar} | i \rangle \quad (1.83)$$

which plays very important conceptual and computational roles in quantum mechanics. We will learn more about the propagator in HW3.

1.5.1 Schrodinger vs. Heisenberg pictures

Regarding the time dependence of physical observables, such as the expectation value of a Hermitian operator $\hat{O}$ at time t , there are two equivalent ways to approach this problem.

In the Schrodinger picture, where the state vector $|\psi\rangle$ evolves with time while the operator $\hat{O}$ remains a constant:

$$|\psi_S(t)\rangle = \hat{U}(t, 0) |\psi_0\rangle, \quad \hat{O}_S(t) = \hat{O} \quad (1.84)$$

In the Heisenberg picture, on the other hand, the state vector remains a constant while the operator evolves with time:

$$|\psi_H(t)\rangle = |\psi_0\rangle, \quad \hat{O}_H(t) = \hat{U}^\dagger(t, 0) \hat{O} \hat{U}(t, 0) \quad (1.85)$$

It is straightforward to show that both pictures yield the same physical observable

$$\langle O(t) \rangle = \langle \psi_S(t) | \hat{O}_S(t) | \psi_S(t) \rangle = \langle \psi_H(t) | \hat{O}_H(t) | \psi_H(t) \rangle = \langle \psi_0 | \hat{U}^\dagger(t, 0) \hat{O} \hat{U}(t, 0) | \psi_0 \rangle \quad (1.86)$$

In general, it is more convenient to use the Heisenberg picture to compute expectation values of an observable, while the Schrodinger picture is more convenient to extract time dependence of the state vector and its related properties, such as entanglement entropy.

1.5.2 *Quantum Zeno effect

Consider a quantum system under repeated measurements of a physical observable $\hat{O}$ with the following spectral decomposition:

$$\hat{O} = \sum_q q |q\rangle\langle q| \quad (1.87)$$

Without loss of generality, let's assume the initial state of the system is $\hat{O}$ eigenstate $|0\rangle$ with eigenvalue q_0 .

Now consider a generic Hamiltonian

$$\hat{H} = H_{0,0}|0\rangle\langle 0| + \sum_{q \neq 0} (H_{q,0}|q\rangle\langle 0| + h.c.) + \sum_{p,q \neq 0} H_{p,q}|p\rangle\langle q| \quad (1.88)$$

that governs the time evolution of the system. We used *h.c.* as a shorthand notation for “hermitian conjugate”.

Starting from the initial state $|0\rangle$, after a very short time Δt , the state evolves into

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar}|0\rangle = (1 - \frac{i}{\hbar}H_{0,0}\Delta t)|0\rangle - \frac{i}{\hbar}\Delta t \sum_q H_{q,0}|q\rangle + O(\Delta t^2) \quad (1.89)$$

Therefore at time Δt , the probability of measurement outcome $\hat{O} = q_0$ is given by

$$|\langle 0 | \psi(t) \rangle|^2 = 1 - \alpha(\Delta t)^2 + O(\Delta t^3), \quad \alpha > 0 \quad (1.90)$$

where the constant α depends on the Hamiltonian $\hat{H}$.

One can further show that if we repeat this 2-step procedure of small time evolution for $\Delta t = \frac{T}{N}$ followed by a measurement of operator $\hat{O}$, after N repetitions, the probability for the state to come back to itself approaches 1 in the large N limit:

$$\Pr(|0\rangle \rightarrow |0\rangle) \geq \lim_{N \rightarrow \infty} (|\langle 0 | \psi(\frac{T}{N}) \rangle|^2)^N = \lim_{N \rightarrow +\infty} e^{-\alpha T^2/N} = 1 \quad (1.91)$$

This is known as the quantum Zeno effect.

1.6 Composite quantum systems (9/11)

- Refer to N&C 2.1.7, 2.2.8, 4.3; Tong 3.6. For generalized measurements, refer to N&C 2.2.6 and Tong 16.4.2

1.6.1 Tensor product

Previously we have focused on a single closed quantum system. When two closed quantum systems are combined together e.g. through some interactions b/w the two systems, how to describe such as composite quantum system? The last postulate addresses this issue, under the mathematical frameworks of *tensor product*:

Postulate 1.11. *The state space of a composite physical system is the tensor product of the state spaces of the component physical systems. Moreover, if we have systems numbered 1 through n , and the system number i is prepared in the state $|\psi_i\rangle$, then the joint state of the total system is $|\psi_1\rangle \otimes |\psi_2\rangle \otimes \cdots \otimes |\psi_n\rangle$.*

Specifically, consider two quantum systems with Hilbert space V of dimension m and W of dimension n , their composite system is described by the tensor product of the two Hilbert spaces:

$$V \otimes W \equiv \{|v\rangle \otimes |w\rangle \mid |v\rangle \in V, |w\rangle \in W\} \quad (1.92)$$

which is a Hilbert space of dimension mn . Linear operators on $V \otimes W$ are given by $A \otimes B$, where A is a linear operator on V and B is a linear operator on W , satisfying

$$(A \otimes B)|v\rangle \otimes |w\rangle = A|v\rangle \otimes B|w\rangle \quad (1.93)$$

An inner product on $V \otimes W$ is naturally defined as

$$\left(\sum_i a_i |v_i\rangle \otimes |w_i\rangle, \sum_j b_j |v'_j\rangle \otimes |w'_j\rangle \right) = \sum_{i,j} a_i^* b_j \langle v_i | v'_j \rangle \langle w_i | w'_j \rangle \quad (1.94)$$

One can therefore define a complete set of orthonormal basis $\{|v_i\rangle \otimes |w_j\rangle\}$ for $V \otimes W$, and the matrix representation of linear operators in $V \otimes W$ is given by

$$(A \otimes B)_{i,j} = A_{i_1,j_1} B_{i_2,j_2}, \quad A \in L_V, B \in L_W. \quad (1.95)$$

Since $A \otimes B$ is a matrix of dimension mn , where m and n are dimensions of matrices A and B respectively, the row and column index of matrix $A \otimes B$ can be written as

$$i = (i_1 - 1)n + i_2, \quad j = (j_1 - 1)n + j_2. \quad (1.96)$$

More properties of tensor product can be found in N&C section 2.1.7.

A *product state* in a composite system $\mathbb{V}_1 \otimes \mathbb{V}_2$ is defined as

$$|\psi\rangle_{12} = |\psi_1\rangle \otimes |\psi_2\rangle \quad (1.97)$$

It has NO quantum entanglement between subsystems 1 and 2.

With the help of tensor product, we can define Hamiltonian and unitary operators for a composite quantum system. We will study the transverse field Ising model for a 2-qubit system

$$H_{\text{TFIM}} = -JX_1 \otimes X_2 - h(Z_1 \otimes \hat{1}_2 + \hat{1}_2 \otimes Z_2) \quad (1.98)$$

as a demonstration. Due to a conserved quantity $Z_1 \otimes Z_2$ that commutes with the Hamiltonian, it can be diagonalized in the $Z_1 \otimes Z_2 = \pm 1$ subspaces respectively.

Name	Controlled-Z (CPHASE)	CNOT	SWAP
Unitary matrix	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

Table 1.2: Common 2-qubit gates and associated unitary matrices.

Name	Toffoli (Controlled-CNOT)	Fredkin (Controlled-SWAP)
Unitary matrix	$\frac{1+Z_1}{2} \otimes \text{CNOT}_{23} + \frac{1-Z_1}{2} \otimes \hat{1}_{23}$	$\frac{1+Z_1}{2} \otimes \text{SWAP}_{23} + \frac{1-Z_1}{2} \otimes \hat{1}_{23}$

Table 1.3: Common 3-qubit gates and associated unitary matrices.

A good set of examples of unitary operators in a composite system are 2-qubit and 3-qubit quantum gates, shown in Table 1.2 and 1.3 respectively, such as a controlled- U gate in the 2-qubit system.

1.6.2 Generalized measurements

Composite systems play an important role in understanding general measurements in quantum mechanics, where we consider the measured system (Hilbert space $\mathbb{V}_s$) and an ancilla (Hilbert space $\mathbb{V}_a$) together as a composite quantum system, whose Hilbert space is the tensor product $\mathbb{V}_s \otimes \mathbb{V}_a$.

Consider an initial state

$$|\Phi\rangle \equiv |\psi\rangle_s \otimes |\phi\rangle_a \quad (1.99)$$

unitarily evolving under a Hamiltonian $\hat{H}$ of the composite system for time t . Next we perform a projective measurement on the ancilla, associated with observable $\hat{1}_s \otimes \hat{O}_a$, where Hermitian operator $\hat{O}$ has the following spectral decomposition:

$$\hat{O} = \sum_q q |q\rangle\langle q| \quad (1.100)$$

The basis vectors $\{|q\rangle\}$ forms a complete set of orthonormal basis for ancilla Hilbert space $\mathbb{V}_a$.

After this measurement, if the measurement outcome is q , the system collapses into a new state $M_q|\psi\rangle_s \otimes |q\rangle_a$ (unnormalized), where linear operator M_q is defined as

$$M_q \equiv \langle q|_a e^{-i\hat{H}t/\hbar} |\phi\rangle_a \in L_{\mathbb{V}_s} \quad (1.101)$$

satisfying the completeness relation

$$\sum_q M_q^\dagger M_q = \hat{1}_s \quad (1.102)$$

The probability for measurement outcome q is given by

$$\text{Pr}(q) = \text{Tr}(M_q|\psi\rangle_s\langle\psi|_s M_q^\dagger) = \langle\psi|M_q^\dagger M_q|\psi\rangle \quad (1.103)$$

Clearly, the completeness relation (1.102) for operators $\{M_q\}$ guarantees the normalization of

probability distribution:

$$\sum_q \text{Pr}(q) = 1. \quad (1.104)$$

If we disregard the ancilla and examine this measurement from the viewpoint of the system, for measurement outcome q , initial state $|\psi\rangle$ becomes $M_q|\psi\rangle$, with a probability $\text{Pr}(q)$ given in (1.103). This is known as a generalized measurement.

If the Kraus operator M_q is very close to identity matrix $\hat{1}_s$

$$M_q = \sqrt{1 - \lambda_q} \hat{1} + \hat{O}_q, \quad \lambda_q \ll 1 \quad (1.105)$$

such a generalized measurement is known as a weak measurement because the measurement only modifies the initial state weakly in the sense that

$$\left| |\psi\rangle - M_q|\psi\rangle \right| \ll 1 \quad (1.106)$$

This is the case when initial state $|\phi\rangle = |q\rangle$ and the evolution time t is much smaller than any energy scale in Hamiltonian $\hat{H}$.

2 Density matrix formulation of quantum mechanics

2.1 Density matrix and mixed states (9/16)

- Refer to N&C 2.4, 2.5, 11.3; Tong 16.3

2.1.1 The density matrix

If we think of a physical system (with Hilbert space A) and its environment (with Hilbert space B) combined together as a (composite) closed quantum system, any state therein is described by a state vector

$$|\psi\rangle = \sum_{i,j} \psi_{i,j} |a_i\rangle \otimes |b_j\rangle \quad (2.1)$$

where $\{|a_i\rangle | 1 \leq i \leq m = \dim A\}$ is a complete set of orthonormal basis for A , while $\{|b_j\rangle | 1 \leq j \leq n = \dim B\}$ is a complete set of orthonormal basis for B .

To study physical properties of system A , the composite state $|\psi\rangle$ in (2.1) is an overkill because it also contains all information about the environment, in which we are not interested at all. It turns out the following quantity, known as the *density matrix* (or *density operator*) for subsystem A

$$\hat{\rho} = \text{Tr}_B |\psi\rangle\langle\psi| = \sum_{i_1, i_2=1}^m \rho_{i_1, i_2} |a_{i_1}\rangle\langle a_{i_2}|, \quad \rho_{i_1, i_2} = \sum_{j=1}^n \psi_{i_1, j} \psi_{i_2, j}^* \quad (2.2)$$

encodes all information about physical system A in composite state $|\psi\rangle$. As shown above it is obtained by tracing out subsystem B in state $|\psi\rangle$. Any physical observable of the system A corresponds to a Hermitian operator $\hat{O}_A \times \hat{1}_B$ in the composite system $A \times B$, and its expectation value is nothing but

$$\langle \hat{O} \rangle_A \equiv \langle \psi | \hat{O}_A \otimes \hat{1}_B | \psi \rangle = \text{Tr}(\hat{\rho} \hat{O}) \quad (2.3)$$

Instead of directly computing the partial trace, there is another efficient way to obtain the density matrix (2.2), known as *Schmidt decomposition*, which we describe below. This approach is especially informative for understanding entanglement between subsystem A and B .

We can treat the wavefunction $\psi_{i,j}$ in (2.1) as a matrix of dimension $m \times n$, which has the following singular value decomposition (SVD):

$$\psi = U^\dagger D V, \quad U^\dagger U = V^\dagger V = \hat{1}, \quad D_{i,j} = \delta_{i,j} \lambda_i, \quad \lambda_i \geq 0. \quad (2.4)$$

where U, V are two unitary matrices of dimensions $m \times m$ and $n \times n$ respectively. D is a $m \times n$ matrix whose matrix elements all vanish except for positive diagonal elements $\{\lambda_i > 0 | 1 \leq i \leq r, r \leq \min(m, n)\}$, known as singular values of matrix ψ . The positive integer $r \leq \min(m, n)$, is the rank of matrix $\hat{\psi}$.

Since unitary matrices implement the change of basis, the SVD procedure introduced a new set of orthonormal basis (known as the Schmidt basis) for both the system A and environment B :

$$|A_i\rangle = \sum_{i'} U_{i,i'}^* |a_{i'}\rangle, \quad |B_j\rangle = \sum_{j'} V_{j,j'} |b_{j'}\rangle \quad (2.5)$$

such that the state vector (2.1) can be written as

$$|\psi\rangle = \sum_{i=1}^r \lambda_i |A_i\rangle \otimes |B_i\rangle, \quad \lambda_i > 0 \quad (2.6)$$

This is known as the Schmidt decomposition of state vector $|\psi\rangle$ in a composite Hilbert space, which is the tensor product of A and B . The rank r of matrix ψ is known as the Schmidt rank (or Schmidt number), and the positive numbers $\{\lambda_i > 0 | 1 \leq i \leq r\}$ are known as Schmidt coefficients.

Note that normalization of state vector $|\psi\rangle$ implies that $\{p_i \equiv \lambda_i^2\}$ leads to a normalized probability distribution:

$$\langle\psi|\psi\rangle = \sum_{i=1}^r \lambda_i^2 = 1 = \sum_{i=1}^r p_i \quad (2.7)$$

In the Schmidt basis $\{|A_i\rangle\}$, the density matrix (2.2) has the following diagonal form:

$$\hat{\rho} = \sum_{i=1}^r \lambda_i^2 |A_i\rangle\langle A_i| = \sum_{i=1}^r p_i |A_i\rangle\langle A_i| \quad (2.8)$$

which has the physical meaning as an ensemble $\{p_i, |A_i\rangle\}$ of pure states. p_i is the probability for the system A to fall into pure state $|A_i\rangle$.

Mathematically, a density matrix is defined as a linear operator $\hat{\rho}$ satisfying 2 conditions:

(1) **Trace condition**

$$\text{Tr } \hat{\rho} = 1 \quad (2.9)$$

(2) **Positivity condition**

$$\hat{\rho} \geq 0 \quad (2.10)$$

The positivity of $\hat{\rho}$ already implies that it is a Hermitian operator i.e. $\hat{\rho} = \hat{\rho}^\dagger$.

All density matrices form a convex set in the sense that $x\rho_1 + (1-x)\rho_2$, $\forall 0 \leq x \leq 1$ is also a density matrix if $\rho_{1,2}$ are density matrices.

In the diagonal basis (2.8) of density matrix $\hat{\rho}$, if the Schmidt rank $r > 1$ i.e. there are at least two independent pure states in the ensemble, we call $\hat{\rho}$ a *mixed state*. If $r = 1$, instead we call $\hat{\rho} = |A_1\rangle\langle A_1|$ to be a *pure state*.

Theorem 2.1. *A density matrix $\hat{\rho}$ corresponds to a pure state if and only if*

$$\hat{\rho}^2 = \hat{\rho} \quad (2.11)$$

As a simplest example of Schmidt decomposition, we consider a single-qubit physical system A and a single-qubit environment B , where a pure state of the combined closed system has the following general form:

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle \quad (2.12)$$

where we have used $|0\rangle \equiv |\uparrow\rangle$ and $|1\rangle \equiv |\downarrow\rangle$ (known as the computational basis) to label eigenstates of the Pauli- $\hat{Z}$ operator. We also used a shorthand notation for tensor product states:

$$|ab\rangle \equiv |a\rangle_A \otimes |b\rangle_B, \quad a, b = 0, 1. \quad (2.13)$$

In this case we have the matrix $\psi_{a,b}$ as

$$\hat{\psi} = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \quad (2.14)$$

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for the SVD procedure.

Similar to the case of pure states, in a composite system $\mathbb{V}_1 \otimes \mathbb{V}_2$, one can also define a *product state* for generic mixed states:

$$\rho_{\text{product}} = \rho_1 \otimes \rho_2 \quad (2.15)$$

and a *separate state*:

$$\rho_{\text{separable}} = \sum_i p_i \rho_{1,i} \otimes \rho_{2,i}, \quad \sum_i p_i = 1. \quad (2.16)$$

A product state features vanishing mutual information (see next section) between subsystems 1 and 2, while a separable state features vanishing quantum entanglement (e.g. in terms of entanglement negativity, see HW4) between subsystems 1 and 2.

2.1.2 Entanglement entropy

In the Schmidt decomposition described previously, if the density matrix $\hat{\rho}$ is obtained by a partial trace over B from state ρ_{AB} in the composite system $A \otimes B$:

$$\hat{\rho}_A = \text{Tr}_B \hat{\rho}_{AB} \quad (2.17)$$

it is called a *reduced density matrix*. Importantly, the reduced density matrix characterizes the correlation and entanglement between subsystem A and B , as we illustrate below.

Consider the following Hamiltonian in a composite system of two qubits A and B :

$$H = -JX_A \otimes X_B - hZ_A \otimes Z_B, \quad h, J > 0 \quad (2.18)$$

It can be simultaneously diagonalized with the conserved quantities $Z_A \otimes Z_B$ and $X_A \otimes X_B$. Its ground state with the lowest energy $E_0 = -h - J$ is given by

$$|Bell_0\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \quad (2.19)$$

while the 3 excited states are given by

$$|Bell_1\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}, \quad (2.20)$$

$$|Bell_2\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}, \quad (2.21)$$

$$|Bell_3\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}. \quad (2.22)$$

These 4 states are known as *Bell states* or *EPR pairs*. They are all examples of bipartite *maximally entangled states* in the two-qubit system, in the sense that their reduced density matrix is proportional to the identity operator:

$$\hat{\rho}_A = \text{Tr}_B |Bell_j\rangle\langle Bell_j| = \frac{1}{2}\hat{1}_A, \quad j = 0, 1, 2, 3. \quad (2.23)$$

Generally, a bipartite pure state $|\psi_{AB}\rangle$ is maximally entangled if and only if

$$\hat{\rho}_A \equiv \text{Tr}_B |\psi_{AB}\rangle\langle\psi_{AB}| = \frac{1}{\dim A} \hat{1}_A \quad (2.24)$$

In order to quantitatively measure the quantum entanglement between two subsystems, we

need to introduce the concept of von Neumann entropy for a density matrix $\hat{\rho}$:

$$S(\hat{\rho}) \equiv -\text{Tr } \hat{\rho} \log \hat{\rho} \quad (2.25)$$

In the diagonal form (2.8) of density matrix $\hat{\rho}$, the von Neumann entropy defined above reduces to the Shannon entropy

$$S = -\sum_j p_j \log p_j \quad (2.26)$$

for a probability distribution $\{p_j\}$, an important concept in information theory. If we take the equilibrium density matrix $\hat{\rho} \propto e^{-\hat{H}/k_B T}$, the von Neumann entropy is nothing but the thermodynamic entropy of the corresponding quantum system described by Hamiltonian $\hat{H}$.

Now for a pure state $|\psi_{AB}\rangle$ in our bipartite system $A \otimes B$, if we consider the reduced density matrix $\hat{\rho}_A$ for subsystem A , its associated von Neumann entropy is called the *entanglement entropy*, because it quantitatively captures the entanglement (or more precisely, correlation) between subsystems A and B . The entanglement entropy has the following properties:

- (1) Positivity: $S(\hat{\rho}) \geq 0$. In particular, $S(\hat{\rho}) = 0$ if and only if $\hat{\rho}$ is a pure state.
- (2) Upper bound: $S(\hat{\rho}_A) \leq \log(\dim A)$, which is saturated by $\hat{\rho}_A = \hat{1}_A/\dim A$. This is why (2.24) is called a maximally entangled state.
- (3) $S(\hat{\rho}_A) = S(\hat{\rho}_B)$, which comes directly from Schmidt decomposition of pure state $|\psi_{AB}\rangle$. If $S(\hat{\rho}_A) = S(\hat{\rho}_B) > 0$, we say that A and B subsystems are (bipartite) entangled in a pure state $|\psi_{AB}\rangle$.
- (4) Subadditivity: $S(\rho_{AB}) \leq S(\rho_A) + S(\rho_B)$ for a mixed state ρ_{AB} . The equality holds if and only if $\rho_{AB} = \rho_A \otimes \rho_B$, which is called a bipartite *product state*. This property allows for defining the *mutual information* between subsystems A and B

$$I(A : B) \equiv S(\rho_A) + S(\rho_B) - S(\rho_{AB}) \geq 0 \quad (2.27)$$

which is a measure of entanglement (or quantum correlations) between A and B in a mixed state. The mutual information reduces to twice the entanglement entropy for a pure state.

- (5) Araki-Lieb inequality: $S(\rho_{AB}) \geq |S(\rho_A) - S(\rho_B)|$.
- (6) Concavity: $S(\sum_i p_i \rho_i) \geq \sum_i p_i S(\rho_i)$ for any probability distribution $\{p_i\}$ and any density matrices $\{\rho_i\}$.
- (7) Strong subadditivity: refer to N&C 11.4.

2.2 Witnessing quantum entanglement (9/18)

- Refer to Tong 16.1.1-16.1.4, S&Z Chapter 18

2.2.1 The EPR “paradox”

Einstein, Podolsky and Rosen (EPR) investigated the singlet Bell state of 2 qubits

$$|\text{EPR}\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \quad (2.28)$$

which was also known as the EPR state. Assuming that Alice has qubit 1 and Bob has qubit 2, they observed that a measurement of Pauli $\hat{Z}_1$ operator by Alice (say, with outcome $q = \pm 1$) will instantaneously collapse the EPR state into a product state $|q\rangle_1 \otimes |-q\rangle_2$, and immediately determine the measurement outcome of Pauli $\hat{Z}_2$ operator by Bob, no matter how far away Alice and Bob are physically separated.

EPR considered the following 3 claims which all seemed reasonable, from the perspective of classical physics:

Locality: Alice’s choice cannot instantaneously alter the physical reality at Bob’s distant location.

Reality criterion: Perfectly predictable results must reflect preexisting properties.

Quantum completeness: The wavefunction completely describes physical reality.

EPR concluded that these three claims cannot all hold simultaneously. Because they retained locality and the reality criterion, they concluded that the quantum state is incomplete, perhaps omitting hidden variables that specify the missing spin values.

The Einstein–Podolsky–Rosen paradox asks whether quantum mechanics can be replaced by a theory that is both local and realistic. Consider two spatially separated observers, Alice and Bob, who receive qubits 1 and 2. Alice may measure either Q or R , while Bob may measure either S or T . Every measurement has outcomes in $\{+1, -1\}$:

$$Q^2 = R^2 = S^2 = T^2 = 1. \quad (2.29)$$

Measurements associated with different parties are compatible, for example $[Q, S] = 0$.

In a local realistic model, measurement outcomes are assumed to have pre-existing values and Alice’s choice does not affect Bob’s value. Thus, for a hidden-variable assignment,

$$E(QS) = E(Q)E(S), \quad (2.30)$$

where each individual value $E(Q), E(R), E(S), E(T)$ is ± 1 .

2.2.2 The CHSH inequality

Define the CHSH combination

$$\mathcal{B} = QS + RS + RT - QT. \quad (2.31)$$

For a local realistic assignment,

$$E(\mathcal{B}) = [E(Q) + E(R)]E(S) + [E(R) - E(Q)]E(T). \quad (2.32)$$

Since $E(R) = E(Q)$ or $E(R) = -E(Q)$, precisely one bracket vanishes and the other equals ± 2 . Consequently,

$$\boxed{|E(QS + RS + RT - QT)| \leq 2.} \quad (2.33)$$

This is the Clauser–Horne–Shimony–Holt (CHSH) inequality.

2.2.2.1 Quantum-mechanical violation

Let the observables be spin measurements along unit vectors $\mathbf{m}_{1,2}$ and $\mathbf{n}_{1,2}$:

$$Q = \mathbf{m}_1 \cdot \boldsymbol{\sigma}_1, \quad R = \mathbf{n}_1 \cdot \boldsymbol{\sigma}_1, \quad S = \mathbf{m}_2 \cdot \boldsymbol{\sigma}_2, \quad T = \mathbf{n}_2 \cdot \boldsymbol{\sigma}_2. \quad (2.34)$$

The two qubits are prepared in the singlet Bell state

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle). \quad (2.35)$$

For this state,

$$\langle \Psi^- | (\mathbf{a} \cdot \boldsymbol{\sigma}_1)(\mathbf{b} \cdot \boldsymbol{\sigma}_2) | \Psi^- \rangle = -\mathbf{a} \cdot \mathbf{b}. \quad (2.36)$$

Hence

$$\langle \mathcal{B} \rangle = -\mathbf{m}_1 \cdot \mathbf{m}_2 - \mathbf{n}_1 \cdot \mathbf{m}_2 - \mathbf{n}_1 \cdot \mathbf{n}_2 + \mathbf{m}_1 \cdot \mathbf{n}_2 \quad (2.37)$$

$$= -(\mathbf{m}_1 + \mathbf{n}_1) \cdot \mathbf{m}_2 + (\mathbf{m}_1 - \mathbf{n}_1) \cdot \mathbf{n}_2. \quad (2.38)$$

Choose coplanar directions with neighboring axes separated by 45° . Then

$$|\langle \mathcal{B} \rangle| = 2\sqrt{2} > 2. \quad (2.39)$$

The value $2\sqrt{2}$ is the maximal quantum value, known as the Tsirelson bound.

2.2.2.2 Experimental test with photon polarization

In an optical realization, a source produces pairs of entangled photons. One photon is sent to polarimeter A and the other to polarimeter B . Each polarimeter selects a measurement orientation, and a coincidence monitor records joint detection events. With

$$|V\rangle = \text{vertical polarization}, \quad |H\rangle = \text{horizontal polarization}, \quad (2.40)$$

polarization measurements implement the two-outcome observables required for a Bell test.

2.2.3 *The GHZ argument

Consider the three-qubit GHZ state

$$|\text{GHZ}\rangle = \frac{1}{\sqrt{2}} (|000\rangle - |111\rangle). \quad (2.41)$$

The state obeys

$$X_A Y_B Y_C |\text{GHZ}\rangle = +|\text{GHZ}\rangle, \quad (2.42)$$

$$Y_A X_B Y_C |\text{GHZ}\rangle = +|\text{GHZ}\rangle, \quad (2.43)$$

$$Y_A Y_B X_C |\text{GHZ}\rangle = +|\text{GHZ}\rangle, \quad (2.44)$$

and

$$X_A X_B X_C |\text{GHZ}\rangle = -|\text{GHZ}\rangle. \quad (2.45)$$

Suppose a local hidden-variable theory assigns predetermined values $s_j^{(x)}, s_j^{(y)} \in \{+1, -1\}$ to

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the X and Y measurements on qubit $j = A, B, C$. The first three GHZ correlations require

$$s_A^{(x)} s_B^{(y)} s_C^{(y)} = +1, \quad (2.46)$$

$$s_A^{(y)} s_B^{(x)} s_C^{(y)} = +1, \quad (2.47)$$

$$s_A^{(y)} s_B^{(y)} s_C^{(x)} = +1. \quad (2.48)$$

Multiplying these equations gives

$$s_A^{(x)} s_B^{(x)} s_C^{(x)} = +1. \quad (2.49)$$

Quantum mechanics instead predicts a product of -1 . This is a direct contradiction, with no statistical inequality required.

2.2.4 *Bell's Original Inequality

Return to the singlet state

$$|\text{EPR}\rangle = \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle). \quad (2.50)$$

For a unit vector $\mathbf{n}$,

$$\mathbf{n} \cdot \boldsymbol{\sigma} = n_x \sigma_x + n_y \sigma_y + n_z \sigma_z, \quad |\mathbf{n}| = 1, \quad (2.51)$$

and

$$\mathbf{n} \cdot \boldsymbol{\sigma} |\pm \mathbf{n}\rangle = \pm |\pm \mathbf{n}\rangle. \quad (2.52)$$

In any common spin basis,

$$|\text{EPR}\rangle = \frac{1}{\sqrt{2}} (|+\mathbf{n}, -\mathbf{n}\rangle - |-\mathbf{n}, +\mathbf{n}\rangle). \quad (2.53)$$

Consider measurement axes $\mathbf{n}_a$, $\mathbf{n}_b$, and $\mathbf{n}_c$. Let P_{ab} denote the probability that the first qubit has outcome $+1$ along $\mathbf{n}_a$ while the second has outcome $+1$ along $\mathbf{n}_b$. Writing $p(s_a s_b s_c)$ for a predetermined hidden-variable assignment,

$$P_{ab} = p(+, -, +) + p(+, -, -), \quad (2.54)$$

$$P_{bc} = p(+, +, -) + p(-, +, -), \quad (2.55)$$

$$P_{ac} = p(+, -, -) + p(-, +, -). \quad (2.56)$$

Nonnegativity of probabilities implies a three-axis probability inequality

$$\boxed{P_{ab} + P_{bc} \geq P_{ac}.} \quad (2.57)$$

This inequality is indeed violated by quantum probabilities. For coplanar directions, parameterize the spin eigenstates by

$$|+\theta\rangle = \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2) \end{pmatrix}, \quad |-\theta\rangle = \begin{pmatrix} -\sin(\theta/2) \\ \cos(\theta/2) \end{pmatrix}. \quad (2.58)$$

For the singlet,

$$P_{ij} = \frac{1}{2} \sin^2 \left(\frac{\theta_i - \theta_j}{2} \right). \quad (2.59)$$

Choose

$$\theta_a = 0, \quad \theta_b = \theta, \quad \theta_c = 2\theta. \quad (2.60)$$

Then

$$P_{ab} = P_{bc} = \frac{1}{2} \sin^2\left(\frac{\theta}{2}\right), \quad P_{ac} = \frac{1}{2} \sin^2 \theta. \quad (2.61)$$

Bell's inequality would require

$$\sin^2\left(\frac{\theta}{2}\right) \geq \frac{1}{2} \sin^2 \theta. \quad (2.62)$$

For $\theta = 60^\circ$,

$$\frac{1}{4} \not\geq \frac{3}{8}, \quad (2.63)$$

which contradicts the local hidden-variable prediction.

2.2.5 Summary

Below is a summary of these entanglement witnesses:

- Local realism leads to the CHSH bound $|\langle \mathcal{B} \rangle| \leq 2$.
- The singlet state can reach $2\sqrt{2}$, violating the CHSH inequality.
- GHZ correlations give a deterministic contradiction with predetermined local outcomes.
- Bell's original probability inequality is also violated by singlet correlations for suitable measurement directions.

2.3 *Generalized measurements and decoherence (9/23)

- Refer to N&C 2.2.6, 2.4.2, 8.2; Tong 16.4; Anastopoulos 8.2

2.3.1 Generalized measurements on a mixed state

Consider the product state $\rho_s \otimes \rho_a$ as the initial state of a composite system $\mathbb{V}_s \otimes \mathbb{V}_a$, consisting of a physical system s unentangled with the experimental apparatus a . We assume the apparatus density matrix ρ_a has the following spectral decomposition:

$$\rho_a = \sum_j p_j |j\rangle_a \langle j|_a, \quad \sum_j p_j = 1, \quad 0 < p_j \leq 1. \quad (2.64)$$

where $\{|j\rangle_e\}$ is a complete set of orthonormal basis for the apparatus Hilbert space $\mathbb{V}_e$.

The system and apparatus evolve together under a unitary operator $\hat{U}$, before a projective measurement is performed for an observable (i.e. Hermitian operator) on the apparatus Hilbert space

$$\hat{O} = \hat{1}_s \otimes \left(\sum_q q |q\rangle_a \langle q|_a \right) \quad (2.65)$$

It's straightforward to show that for measurement outcome q , the initial state ρ_s of the physical system collapses into the following (unnormalized) state

$$\rho_q \propto \sum_j M_{q,j} \rho_s M_{q,j}^\dagger \quad (2.66)$$

with a probability distribution

$$\text{Pr}(q) = \sum_j \text{Tr}(M_{q,j} \rho_s M_{q,j}^\dagger) \quad (2.67)$$

where the measurement operators $\{M_{q,j}\}$ are defined as

$$M_{q,j} \equiv \langle q | \hat{U} | j \rangle \sqrt{p_j} \quad (2.68)$$

satisfying the completeness relation

$$\sum_{q,j} M_{q,j}^\dagger M_{q,j} = \hat{1}_s \quad (2.69)$$

The above discussion is completely in parallel to section 1.6.2, merely generalizing everything to the context of mixed states. Therefore we can formulate the measurement postulate of quantum mechanics for density matrices:

Postulate 2b: A *generalized quantum measurement* is described by a collection $\{M_m\}$ of measurement operators satisfying the completeness relation:

$$\sum_m M_m^\dagger M_m = \hat{1}. \quad (2.70)$$

If the state of the quantum system is ρ immediately before the measurements, then the probability that measurement result m occurs is given by

$$\text{Pr}(m) = \text{Tr}(\rho M_m^\dagger M_m) \quad (2.71)$$

and the (unnormalized) state of the system after the measurement is $\rho_m \propto M_m \rho M_m^\dagger$.

In a generalized measurement, the set of positive operators $\{E_m \equiv M_m^\dagger M_m\}$ is known as a POVM (positive operator-value measure). A generalized measurement is sometimes also known as a POVM measurement.

2.3.2 Examples of generalized measurements

(1) Quantum Zeno effect

In a single-qubit system, consider a repeated 2-step procedure where the 1st step of unitary time evolution for a duration T/N is followed by the 2nd step of a projective measurement on $\hat{X}$:

$$\hat{X} = \hat{P}_+ - \hat{P}_-, \quad \hat{P}_+ = |+\rangle\langle+|, \quad \hat{P}_- = |-\rangle\langle-|. \quad (2.72)$$

where

$$|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle) \quad (2.73)$$

are the eigenstates of Pauli $\hat{X}$ operator.

In this example, the measurement operators associated with the $\pm$ outcome after each step are $\{\hat{M}_\pm\}$ defined as follows:

$$\hat{M}_\pm = \hat{P}_\pm U(T/N), \quad M_+^\dagger M_+ + M_-^\dagger M_- = \hat{1}. \quad (2.74)$$

(2) Detection of a photon

Let $|0\rangle$ denote no photon and $|1\rangle$ denote one photon. A simple photon detection model uses

$$\hat{M}_0 = |0\rangle\langle 0|, \quad \hat{M}_c = |0\rangle\langle 1| \quad (2.75)$$

where the subscript c denotes a click in the photon reader. The photon is annihilated after the click outcome. It's straightforward to check the completeness relation holds for $\{M_0, M_c\}$.

(3) Distinguishing quantum states

Suppose a qubit is prepared in either

$$|0\rangle \quad \text{or} \quad |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle).$$

The task is to determine which state was received.

(i) Projective measurement in the Z basis:

The measurement operators are projection operators

$$\hat{P}_0 = |0\rangle\langle 0|, \quad \hat{P}_1 = |1\rangle\langle 1|.$$

If the outcome is 1, the received state must have been $|+\rangle$ because $\langle 1|0\rangle = 0$. If the outcome is 0, the two possibilities cannot be distinguished.

(ii) Projective measurement in the X basis:

The measurement operators are again projectors

$$\hat{P}_+ = |+\rangle\langle+|, \quad \hat{P}_- = |-\rangle\langle-|.$$

If the outcome is $+$, the states cannot be distinguished. If the outcome is $-$, the received state must have been $|0\rangle$ because $\langle -|+\rangle = 0$.

(iii) A three-outcome POVM:

Consider the POVM $\{E_m = M_m^\dagger M_m\}$:

$$\begin{aligned}\hat{E}_1 &= \frac{1}{2}|1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}, \\ \hat{E}_2 &= \frac{1}{2}|-\rangle\langle -| = \frac{1}{4} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \\ \hat{E}_3 &= \hat{1} - \hat{E}_1 - \hat{E}_2 = \begin{pmatrix} \frac{3}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} \end{pmatrix} \geq 0.\end{aligned}$$

The outcomes have the following interpretation:

1. Outcome 1: the state must have been $|+\rangle$.
2. Outcome 2: the state must have been $|0\rangle$.
3. Outcome 3: the measurement is inconclusive.

2.3.3 Measurements induce decoherence

In its eigenbasis $\{|j\rangle | 1 \leq j \leq \dim \mathbb{V}\}$, a mixed state (2.8) is an ensemble of pure states $\{|j\rangle\}$

$$\hat{\rho} = \sum_j p_j |j\rangle\langle j|, \quad \sum_j p_j = 1. \quad (2.76)$$

while a generic pure state in the same physical system can be written as a (normalized) superposition of the basis vectors $\{|j\rangle\}$:

$$|\psi\rangle = \sum_j \psi_j |j\rangle, \quad \sum_j |\psi_j|^2 = 1. \quad (2.77)$$

But there is a big difference between the (“incoherent”) ensemble summation in the mixed state (2.76) and the (“coherent”) superposition in the pure state (2.77), in the sense that the pure state contains more information in the relative phases of coefficients ψ_j that can be measured experimentally. To make it clear, we use the normalization condition in (2.77) to define the probability distribution $p_j \equiv |\psi_j|^2$, and the normalized pure state can be rewritten as

$$|\psi\rangle = \sum_j \sqrt{p_j} e^{i\phi_j} |j\rangle, \quad \sum_j p_j = 1. \quad (2.78)$$

While the global phase factor is a redundancy with no physical significance, there are $(\dim \mathbb{V} - 1)$ relative phases that can be detected by physical observables.

To illustrate, consider a single qubit example, where we use $(\hat{X}, \hat{Y}, \hat{Z})$ to label the 3 Pauli matrices. We consider a mixed state $\hat{\rho}$ and a pure state ψ , both having 1/2 probability for either $Z = +1$ or $Z = -1$. In the computational basis $|0\rangle \equiv |\uparrow\rangle$, $|1\rangle \equiv |\downarrow\rangle$ that diagonalizes $\hat{Z}$, the mixed state is written as

$$\hat{\rho} = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (2.79)$$

and the pure state as

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{i\phi}|1\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ e^{i\phi} \end{pmatrix} \quad (2.80)$$

If we now measure another observable $\hat{X}$ for both states, we end up with different expectation values:

$$\text{Tr}(\hat{\rho}\hat{X}) = 0, \quad \langle\psi|\hat{X}|\psi\rangle = \cos\phi. \quad (2.81)$$

exactly rooted in the relative phases and associated quantum interferences in the pure state.

As a result, the reduction from a pure state (2.78) to a mixed state (2.76) is a process of decoherence, i.e. a reduction of quantum interference. This can be achieved by a projective measurement (see postulate 1.9), where the measurement outcome is unknown. Consider the implementation of a measurement for physical observable diagonal in basis $\{|j\rangle\}$

$$\hat{O} \equiv \sum_j O_j |j\rangle\langle j| \quad (2.82)$$

on the pure state (2.78), If we don't know the outcome, after a projective measurement, we should view the resulting state as a summation of projector $|j\rangle\langle j|$ weighted by probability $p_j = |\langle j|\psi\rangle|^2$, and hence we end up with the mixed state (2.76). This means a quantum measurement will generally decohere a quantum state, even if we don't know the measurement outcome. This is in sharp contrast to a classical system.

2.4 *Time evolution of open quantum systems (9/25)

- Refer to Tong 16.5.1-16.5.3, N&C 8.3

For a closed quantum system, its unitary time evolution can be easily reformulated using density matrices:

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle, \quad |\psi(t)\rangle = U(t, 0) |\psi(0)\rangle, \quad (2.83)$$

$$i\hbar \frac{d\rho}{dt} = [H, \rho], \quad \rho(t) = U(t, 0) \rho(0) U^\dagger(t, 0). \quad (2.84)$$

However this is not the most general time evolution for a mixed state $\rho(t)$ in an open quantum system, as we will show below.

2.4.1 Quantum channels and Kraus operators

Let system A interact with an environment B . Assume initially the environment is in a pure state $|0\rangle_B$:

$$\rho_{AB}(t=0) = \rho_A(0) \otimes \rho_B(0), \quad \rho_B(0) = |0\rangle\langle 0|_B. \quad (2.85)$$

After joint unitary evolution,

$$\rho_{AB}(t) = U(t, 0) (\rho_A(0) \otimes |0\rangle\langle 0|_B) U^\dagger(t, 0). \quad (2.86)$$

The reduced state in system A is

$$\rho_A(t) = \text{Tr}_B \rho_{AB}(t) \quad (2.87)$$

$$= \sum_k {}_B\langle k| U(t, 0) |0\rangle_B \rho_A(0) {}_B\langle 0| U^\dagger(t, 0) |k\rangle_B \quad (2.88)$$

$$= \sum_k M_k(t) \rho_A(0) M_k^\dagger(t), \quad (2.89)$$

where the *Kraus operators* $\{M_k\}$ defined by

$$M_k(t) = {}_B\langle k| U(t, 0) |0\rangle_B. \quad (2.90)$$

satisfy the completeness relation

$$\sum_k M_k^\dagger M_k = {}_B\langle 0| U^\dagger U |0\rangle_B = \hat{1}_A. \quad (2.91)$$

This indicates the time evolution of an open quantum system A is described by a quantum channel, or a CPTP map $\mathcal{E}$ defined below.

2.4.1.1 Definition of a quantum channel

A quantum channel is a completely positive, trace-preserving (CPTP) map, which can always be represented by a set of Kraus operators $\{M_m\}$:

$$\mathcal{E}(\rho) = \sum_m M_m \rho M_m^\dagger, \quad \sum_m M_m^\dagger M_m = \hat{1}. \quad (2.92)$$

It is trace preserving:

$$\text{Tr } \mathcal{E}(\rho) = \text{Tr } \rho \quad (2.93)$$

convex linear

$$\mathcal{E}(p\rho_1 + (1-p)\rho_2) = p\mathcal{E}(\rho_1) + (1-p)\mathcal{E}(\rho_2), \quad \forall 0 \leq p \leq 1. \quad (2.94)$$

and completely positive. Complete positivity means that for every auxiliary system B ,

$$(\mathcal{E}_A \otimes \mathcal{I}_B)(\rho_{AB}) \geq 0 \quad (2.95)$$

whenever $\rho_{AB} \geq 0$. A Kraus representation is not unique: two sets of Kraus operators $\{M_i\}$ and $\{N_j\}$ represent the same quantum channel if and only if they are related by a unitary matrix $u_{i,j}$:

$$M_i = \sum_j u_{i,j} N_j, \quad u^\dagger u = \hat{1}. \quad (2.96)$$

2.4.1.2 Example: a bit-flip channel

For two qubits with

$$H = -\hbar\omega \sigma_x^{(A)} \otimes \sigma_x^{(B)}, \quad (2.97)$$

the unitary is

$$U(t) = \cos(\omega t) \mathbb{I} + i \sin(\omega t) \sigma_x^{(A)} \otimes \sigma_x^{(B)}. \quad (2.98)$$

With the environment initially in $|0\rangle_B$,

$$M_0 = \cos(\omega t) \mathbb{I}_A, \quad M_1 = i \sin(\omega t) \sigma_x^{(A)}. \quad (2.99)$$

Therefore

$$\rho_A(t) = [1 - p(t)]\rho_A(0) + p(t)\sigma_x \rho_A(0) \sigma_x, \quad p(t) = \sin^2(\omega t). \quad (2.100)$$

Differentiation with respect to p gives

$$\frac{d\rho_A}{dp} = \sigma_x \rho_A \sigma_x - \rho_A, \quad \text{Tr}\left(\frac{d\rho_A}{dp}\right) = 0. \quad (2.101)$$

2.4.2 The Master Equation

2.4.2.1 Lindblad form

Markovian, time-local dynamics has the following Lindblad form

$$\boxed{\frac{d\rho}{dt} = -\frac{i}{\hbar}[H, \rho] + \sum_j \left(2L_j \rho L_j^\dagger - L_j^\dagger L_j \rho - \rho L_j^\dagger L_j \right)}. \quad (2.102)$$

The commutator with Hamiltonian $\hat{H}$ generates coherent evolution. The linear operators $\{L_j\}$ are known as *quantum jump operators*. The remaining terms describe dissipation and decoherence while preserving the trace.

2.4.2.2 Bloch-vector representation

A qubit state can be written as

$$\rho = \frac{1}{2}(\mathbb{I} + \mathbf{r} \cdot \boldsymbol{\sigma}), \quad \mathbf{r} = (x, y, z), \quad |\mathbf{r}| \leq 1. \quad (2.103)$$

2 Density matrix formulation of quantum mechanics

Quantum channels of a single qubit act as affine transformations of the Bloch ball:

$$\mathcal{E}(\rho) = \frac{\mathbb{I} + \mathbf{r}' \cdot \boldsymbol{\sigma}}{2}, \quad \mathbf{r}' = M\mathbf{r} + \mathbf{v}. \quad (2.104)$$

where M is a 3×3 matrix and $\mathbf{v}$ is a constant vector.

2.4.2.3 Bit-flip channel

For

$$\mathcal{E}_X(\rho) = (1 - p)\rho + p\sigma_x\rho\sigma_x, \quad (2.105)$$

the Bloch vector transforms as

$$(x, y, z) \mapsto (x, (1 - 2p)y, (1 - 2p)z). \quad (2.106)$$

The Bloch ball contracts toward the x axis.

2.4.2.4 Phase-flip channel and phase damping

The phase-flip channel is

$$\mathcal{E}_Z(\rho) = (1 - p)\rho + p\sigma_z\rho\sigma_z. \quad (2.107)$$

For

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \rho_0 = |\psi\rangle\langle\psi| = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{pmatrix}, \quad (2.108)$$

one application gives

$$\mathcal{E}_Z(\rho_0) = \begin{pmatrix} |\alpha|^2 & (1 - 2p)\alpha\beta^* \\ (1 - 2p)\alpha^*\beta & |\beta|^2 \end{pmatrix}. \quad (2.109)$$

After n applications the off-diagonal entries acquire a factor $(1 - 2p)^n$. Setting $2p = \Gamma t/n$ and taking $n \rightarrow \infty$ gives

$$\rho(t) = \begin{pmatrix} |\alpha|^2 & e^{-\Gamma t}\alpha\beta^* \\ e^{-\Gamma t}\alpha^*\beta & |\beta|^2 \end{pmatrix}. \quad (2.110)$$

The populations remain fixed while coherence decays.

2.4.2.5 Depolarizing channel

A common parametrization is

$$\mathcal{E}_{\text{dep}}(\rho) = (1 - p)\rho + \frac{p}{3}(\sigma_x\rho\sigma_x + \sigma_y\rho\sigma_y + \sigma_z\rho\sigma_z). \quad (2.111)$$

Using

$$2\mathbb{I} = \rho + \sigma_x\rho\sigma_x + \sigma_y\rho\sigma_y + \sigma_z\rho\sigma_z, \quad (2.112)$$

the Bloch vector transforms as

$$\mathbf{r} \mapsto \left(1 - \frac{4p}{3}\right)\mathbf{r}. \quad (2.113)$$

One Kraus representation is

$$M_0 = \sqrt{1 - p}\mathbb{I}, \quad M_j = \sqrt{\frac{p}{3}}\sigma_j, \quad j = x, y, z. \quad (2.114)$$

2.4.2.6 Amplitude-damping channel

Let $|0\rangle_B$ denote no photon and $|1\rangle_B$ one photon. The joint evolution satisfies

$$U|0\rangle_A|0\rangle_B = |0\rangle_A|0\rangle_B, \quad (2.115)$$

$$U|1\rangle_A|0\rangle_B = \sqrt{1-p}|1\rangle_A|0\rangle_B + \sqrt{p}|0\rangle_A|1\rangle_B. \quad (2.116)$$

The Kraus operators are

$$M_0 = {}_B\langle 0|U|0\rangle_B = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix}, \quad M_1 = {}_B\langle 1|U|0\rangle_B = \begin{pmatrix} 0 & \sqrt{p} \\ 0 & 0 \end{pmatrix}. \quad (2.117)$$

Thus

$$\mathcal{E}_{\text{AD}}(\rho_0) = \begin{pmatrix} |\alpha|^2 + p|\beta|^2 & \sqrt{1-p}\alpha\beta^* \\ \sqrt{1-p}\alpha^*\beta & (1-p)|\beta|^2 \end{pmatrix}. \quad (2.118)$$

For $p = \Gamma t/n$ and $n \rightarrow \infty$,

$$\rho(t) = \begin{pmatrix} |\alpha|^2 + (1-e^{-\Gamma t})|\beta|^2 & e^{-\Gamma t/2}\alpha\beta^* \\ e^{-\Gamma t/2}\alpha^*\beta & e^{-\Gamma t}|\beta|^2 \end{pmatrix}. \quad (2.119)$$

The excited-state population decays as $e^{-\Gamma t}$, while coherence decays as $e^{-\Gamma t/2}$.

2.4.3 Limits of the description

A channel acting only on A does not describe every possible reduced evolution. If A and B are initially correlated, assigning a single completely positive map to arbitrary initial states of A may be impossible. The standard channel construction assumes a fixed, initially uncorrelated environment state. The Lindblad equation adds assumptions such as Markovianity and time locality.

2.5 Appendix: Summarizing the postulates of quantum mechanics

Below we summarize the 4 postulates of quantum mechanics using two different languages: (1) in terms of state vectors for a closed quantum system, (2) in terms of density matrices for an open quantum system.

2.5.1 State vector formulation of quantum mechanics for a closed system

Postulate 2.2. *A quantum system is described by a Hilbert space $\mathbb{V}$, and each state of the quantum system is described by a normalized vector $|\psi\rangle \in \mathbb{V}$ in the Hilbert space, up to a global phase factor.*

Postulate 2.3. *Any physical observable corresponds to a Hermitian operator $\hat{O} = \hat{O}^\dagger$ in the Hilbert space, which has a spectral decomposition $\hat{O} = \sum_q q |q\rangle\langle q|$ with real eigenvalues $\{q\}$. A projective measurement of this observable on state $|\psi\rangle$ can yield a measurement outcome of q with probability $\text{Pr}(q) = |\langle q|\psi\rangle|^2$, and the initial state $|\psi\rangle$ collapses into state $|q\rangle$ after reading out the measurement outcome q .*

Postulate 2.4. *The time evolution of a closed quantum system is described by the Schrodinger equation with a Hamiltonian $\hat{H}(t)$: $i \frac{d}{dt} |\psi(t)\rangle = \hat{H}(t) |\psi(t)\rangle$. Equivalently $|\psi(t)\rangle = \hat{U}(t, 0) |\psi(0)\rangle$ i.e. the state evolves under a unitary time evolution operator $\hat{U}(t, 0)$.*

Postulate 2.5. *A composite quantum system of two subsystem $\mathbb{V}_1$ and $\mathbb{V}_2$ corresponds to a tensor product of the two Hilbert space $\mathbb{V} = \mathbb{V}_1 \otimes \mathbb{V}_2$. The states and operators follow the tensor product rules similarly.*

2.5.2 Density matrix formulation of quantum mechanics for an open system

Postulate 2.6. *A quantum system is described by a Hilbert space $\mathbb{V}$, and each state of the quantum system corresponds to a density matrix $\hat{\rho} \in L_{\mathbb{V}}$, which is a linear operator satisfying the trace condition $\text{Tr } \hat{\rho} = 1$ and positivity condition $\hat{\rho} \geq 0$.*

Postulate 2.7. *A generalized (POVM) measurement on quantum state ρ is described by a set of measurement operators $\{\hat{M}_q\}$ satisfying the completeness relation $\sum_q \hat{M}_q^\dagger \hat{M}_q = \hat{1}$. The probability for measurement outcome q is given by $\text{Pr}(q) = \text{Tr } M_q \rho M_q^\dagger$, and the initial state ρ collapses into (unnormalized) state $M_q \rho M_q^\dagger$ after reading out the measurement outcome q .*

Postulate 2.8. *If the system and environment are not correlated at the beginning, the time evolution of an open quantum system is described by a quantum channel (i.e. a CPTP map). Under further assumptions of Markovianity and time locality, the time evolution is described by a master equation of the Lindblad Form.*

Postulate 2.9. *A composite quantum system of two subsystem $\mathbb{V}_1$ and $\mathbb{V}_2$ corresponds to a tensor product of the two Hilbert space $\mathbb{V} = \mathbb{V}_1 \otimes \mathbb{V}_2$. The states and operators follow the tensor product rules similarly.*

3 A quantum particle in one dimension

3.1 Position, momentum and Schrodinger's wave equation (9/30)

- Refer to Tong 3.2.5, S&N 1.6, 1.7

3.1.1 Position and momentum: from a discrete lattice to the continuum

3.1.1.1 Position basis on a lattice

Consider a one-dimensional lattice with lattice constant a . Its sites are labeled by integers n , and the position eigenstates form the basis of Hilbert space

$$\mathbb{V} = \text{span}\{|n\rangle : n = 0, \pm 1, \pm 2, \dots\} \simeq \mathbb{C}^\infty \quad (3.1)$$

The position operator is

$$\hat{x} = \sum_{n=-\infty}^{\infty} na|n\rangle\langle n|, \quad (3.2)$$

so that

$$\hat{x}|n\rangle = na|n\rangle. \quad (3.3)$$

3.1.1.2 Translation operator

Define the one-site translation operator by

$$\hat{T}_x = \sum_{n=-\infty}^{\infty} |n+1\rangle\langle n|. \quad (3.4)$$

Its adjoint translates in the opposite direction, and

$$\hat{T}_x^\dagger \hat{T}_x = \mathbb{I}, \quad (3.5)$$

so $\hat{T}_x$ is unitary. The position operator and translation operator satisfy

$$[\hat{x}, \hat{T}_x] = a\hat{T}_x. \quad (3.6)$$

The eigenstates of translation are labeled by a dimensionless wave number k :

$$\hat{T}_x|k\rangle = e^{-ik}|k\rangle, \quad 0 \leq k < 2\pi. \quad (3.7)$$

For a periodic lattice with N sites and $|n\rangle \equiv |n+N\rangle$, the change of basis is implemented by a unitary transformation

$$|k\rangle = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} e^{ikn}|n\rangle, \quad k = \frac{2\pi}{N}n_k \pmod{2\pi}, \quad n_k \in \mathbb{Z}. \quad (3.8)$$

known as the discrete Fourier transform (DFT). The inverse transformation is

$$|n\rangle = \frac{1}{\sqrt{N}} \sum_{n_k=0}^{N-1} e^{-ikn}|k\rangle, \quad \langle n_1|n_2\rangle = \delta_{n_1, n_2}. \quad (3.9)$$

We also have the resolution of identity for these two basis sets

$$\sum_{n=0}^{N-1} |n\rangle\langle n| = \sum_{n_k=0}^{N-1} |k\rangle\langle k| = \hat{1} \quad (3.10)$$

3.1.1.3 Momentum as the generator of translation

Because $\hat{T}_x$ is unitary, write

$$\hat{T}_x = e^{-ia\hat{p}/\hbar}. \quad (3.11)$$

The momentum operator is therefore the generator of spatial translations. On a translation eigenstate,

$$\hat{p}|k\rangle = \frac{\hbar k}{a}|k\rangle. \quad (3.12)$$

Using the commutator of $\hat{x}$ with $\hat{T}_x$, or expanding an infinitesimal translation ($a \rightarrow 0^+$), gives the canonical commutation relation

$$\boxed{[\hat{x}, \hat{p}] = i\hbar.} \quad (3.13)$$

3.1.1.4 Continuum and thermodynamic limits

To connect the DFT on a discrete lattice to position and momentum operators in the continuum, we need to simultaneously take the continuum limit

$$a \longrightarrow 0 \quad (3.14)$$

and the thermodynamic limit

$$Na \longrightarrow \infty. \quad (3.15)$$

Introduce continuous variables

$$x = na, \quad p = \frac{\hbar k}{a}, \quad (3.16)$$

with spacings that approach zero in the continuum and thermodynamic limits:

$$\Delta x = a, \quad \Delta p = \frac{2\pi\hbar}{Na}. \quad (3.17)$$

and associated basis vectors:

$$|x\rangle \equiv \frac{1}{\sqrt{a}}|n = \frac{x}{a}\rangle, \quad |p\rangle \equiv \sqrt{\frac{Na}{2\pi\hbar}}|k = \frac{pa}{\hbar}\rangle \quad (3.18)$$

The discrete resolutions of the identity in (3.10) become

$$\int_{-\infty}^{\infty} dp |p\rangle\langle p| = \mathbb{I}, \quad \int_{-\infty}^{\infty} dx |x\rangle\langle x| = \mathbb{I}. \quad (3.19)$$

The continuum states are normalized according to

$$\langle x|x'\rangle = \delta(x - x') = \lim_{\Delta x \rightarrow 0^+} \frac{\delta_{n,n'}}{\Delta x}, \quad \langle p|p'\rangle = \delta(p - p') = \lim_{\Delta p \rightarrow 0^+} \frac{\delta_{p,p'}}{\Delta p}. \quad (3.20)$$

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Finally, the position-momentum overlap is the plane wave

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}. \quad (3.21)$$

3.1.2 The Schrödinger Wave Equation

3.1.2.1 Fourier transformation between representations

Using the continuum completeness relations,

$$|p\rangle = \int_{-\infty}^{\infty} dx |x\rangle \langle x|p\rangle = \int_{-\infty}^{\infty} dx |x\rangle \frac{e^{ipx/\hbar}}{\sqrt{2\pi\hbar}}, \quad (3.22)$$

$$|x\rangle = \int_{-\infty}^{\infty} dp |p\rangle \langle p|x\rangle = \int_{-\infty}^{\infty} dp |p\rangle \frac{e^{-ipx/\hbar}}{\sqrt{2\pi\hbar}}. \quad (3.23)$$

The orthonormality relations follow from

$$\frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} dx e^{i(p'-p)x/\hbar} = \delta(p-p'), \quad (3.24)$$

and similarly for $\delta(x-x')$.

3.1.2.2 Dirac delta function

The Dirac delta distribution satisfies

$$\delta(x) = 0 \quad \text{for } x \neq 0, \quad (3.25)$$

$$\int_{-\infty}^{\infty} \delta(x) dx = 1, \quad (3.26)$$

$$\delta(x) = \delta(-x), \quad (3.27)$$

$$\int_{-\infty}^{\infty} f(x) \delta(x-x_0) dx = f(x_0). \quad (3.28)$$

A useful Fourier representation is

$$\delta(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx} dk. \quad (3.29)$$

The scaling and derivative identities are

$$\delta(ax) = \frac{1}{|a|} \delta(x), \quad \int f(x) \delta'(x-x_0) dx = -f'(x_0). \quad (3.30)$$

Two common regularized representations of Dirac delta function are

$$\delta(x) = \lim_{\epsilon \rightarrow 0^+} \frac{1}{\pi} \frac{\epsilon}{x^2 + \epsilon^2} \quad (3.31)$$

and

$$\delta(x) = \lim_{\epsilon \rightarrow 0^+} \frac{1}{\sqrt{\pi\epsilon}} e^{-x^2/\epsilon}. \quad (3.32)$$

3.1.2.3 Wave functions

For a state $|\psi\rangle$, define its position-space and momentum-space wave functions by

$$\psi(x) = \langle x|\psi\rangle, \quad \tilde{\psi}(p) = \langle p|\psi\rangle. \quad (3.33)$$

They are Fourier-transform pairs:

$$\psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dp e^{ipx/\hbar} \tilde{\psi}(p), \quad (3.34)$$

$$\tilde{\psi}(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dx e^{-ipx/\hbar} \psi(x). \quad (3.35)$$

3.1.2.4 Operators in the position representation

For a function of position,

$$\langle x|V(\hat{x})|x'\rangle = V(x)\delta(x-x'). \quad (3.36)$$

For momentum,

$$\langle x|\hat{p}|x'\rangle = \int dp \langle x|p\rangle p \langle p|x'\rangle \quad (3.37)$$

$$= -i\hbar \frac{\partial}{\partial x} \delta(x-x'). \quad (3.38)$$

Thus, acting on wave functions,

$$\boxed{\hat{p} \longrightarrow -i\hbar \frac{\partial}{\partial x}.} \quad (3.39)$$

More generally,

$$\langle x|U(\hat{p})|x'\rangle = U\left(-i\hbar \frac{\partial}{\partial x}\right) \delta(x-x'). \quad (3.40)$$

3.1.2.5 Time-dependent and stationary Schrödinger equations

The abstract time-dependent Schrödinger equation is

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle. \quad (3.41)$$

If

$$\hat{H}|n\rangle = E_n|n\rangle, \quad c_n = \langle n|\psi(0)\rangle, \quad (3.42)$$

then

$$|\psi(t)\rangle = \sum_n c_n e^{-iE_n t/\hbar} |n\rangle. \quad (3.43)$$

The time-independent (or stationary) Schrodinger equation is

$$\hat{H}|\psi\rangle = E|\psi\rangle. \quad (3.44)$$

3.1.2.6 Wave equation for a particle in one dimension

For a particle with no electromagnetic field,

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}). \quad (3.45)$$

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Taking the position-space matrix element of the stationary equation gives

$$\langle x|\hat{H}|\psi\rangle = \frac{1}{2m}\langle x|\hat{p}^2|\psi\rangle + \langle x|V(\hat{x})|\psi\rangle \quad (3.46)$$

$$= -\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\psi(x) + V(x)\psi(x). \quad (3.47)$$

Therefore,

$$\boxed{-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x).} \quad (3.48)$$

This is the one-dimensional stationary Schrödinger wave equation.

In momentum space, where $\tilde{\psi}(p) = \langle p|\psi$, the same equation becomes

$$\boxed{\frac{p^2}{2m}\tilde{\psi}(p) + V\left(i\hbar\frac{\partial}{\partial p}\right)\tilde{\psi}(p) = E\tilde{\psi}(p).} \quad (3.49)$$

3.1.3 Summary

- Momentum is the generator of spatial translations.
- The continuum position and momentum bases are delta normalized and connected by a Fourier transform.
- In the position representation, $\hat{p} = -i\hbar\partial/\partial x$.
- In the momentum representation, $\hat{x} = i\hbar\partial/\partial p$.
- Projecting the abstract eigenvalue problem onto a position basis gives the Schrödinger wave equation.

3.2 Solving one-dimensional potentials (10/2)

- Refer to Tong 2.1.1, 2.3.2, 2.4.1; S&N Appendix B on “Elementary Solutions to Schrödinger Wave Equation”

3.2.1 General Properties of stationary Schrödinger equations

The stationary Schrödinger equation is

$$\langle x | \hat{H} | \psi \rangle = E \langle x | \psi \rangle. \quad (3.50)$$

For a particle of mass m in a one-dimensional potential $V(x)$,

$$\boxed{-\frac{\hbar^2}{2m}\psi''(x) + V(x)\psi(x) = E\psi(x).} \quad (3.51)$$

This is a linear, second-order ordinary differential equation.

Linearity:

If $\psi_1(x)$ and $\psi_2(x)$ are solutions with the same energy E , then any linear combination

$$c_1\psi_1(x) + c_2\psi_2(x), \quad c_1, c_2 \in \mathbb{C}, \quad (3.52)$$

is also a solution with energy E .

Complex conjugation and time reversal symmetry

If $V(x)$ is real and $\psi(x)$ is a solution with energy E , then $\psi^*(x)$ is also a solution with the same energy. For

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}), \quad (3.53)$$

this follows from

$$\hat{H}^* = \hat{H}. \quad (3.54)$$

This is the stationary manifestation of time-reversal symmetry for a spinless particle in a real scalar potential.

Real eigenfunctions

Energy eigenfunctions can be chosen to be real. Starting from a complex solution $\psi(x)$, define

$$\psi_1(x) = \psi(x) + \psi^*(x), \quad \psi_2(x) = i(\psi(x) - \psi^*(x)). \quad (3.55)$$

Both ψ_1 and ψ_2 are real solutions with the same energy. A nonzero member of this pair can therefore be used as a real eigenfunction.

Smoothness and matching conditions

If $V(x)$ is smooth, then solutions of the Schrödinger equation are smooth. If $V(x)$ has a finite discontinuity, integration of the equation across the discontinuity shows that both $\psi(x)$ and $\psi'(x)$ are continuous. Singular potentials, such as a delta function, require a modified derivative matching condition.

3.2.2 Examples

3.2.2.1 Free particle

For a free particle,

$$V(x) = 0. \quad (3.56)$$

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The plane-wave solutions are

$$\psi_k(x) \propto e^{ikx}, \quad E = \frac{\hbar^2 k^2}{2m}. \quad (3.57)$$

These are scattering states and are not square-normalizable on the entire real line. The two waves ψ_{+k} and ψ_{-k} have the same energy. Real eigenfunctions may be chosen as $\cos(kx)$ and $\sin(kx)$.

To properly normalize these plane waves, we have to utilize the Dirac delta function:

$$\langle k_1 | k_2 \rangle = \int dx \psi_{k_1}^*(x) \psi_{k_2}(x) = \delta(k_1 - k_2) \quad (3.58)$$

which means

$$\psi_k(x) = \frac{1}{\sqrt{2\pi}} e^{ikx} \quad (3.59)$$

Periodic boundary condition:

If we consider a plane wave on a periodic ring of length L , we need to enforce the *periodic boundary condition*

$$\psi_k(x) \equiv \psi_k(x + L) \quad (3.60)$$

which quantizes the wavevector k as

$$k = \frac{2\pi}{L} n_k, \quad n_k \in \mathbb{Z}. \quad (3.61)$$

3.2.2.2 Step potential

Consider

$$V(x) = \begin{cases} 0, & x < 0, \\ U, & x > 0, \end{cases} \quad U > 0. \quad (3.62)$$

Because $V(x) \geq 0$, there are no energy eigenstates with $E < 0$.

Case I: $E > U$

Define

$$k = \frac{\sqrt{2mE}}{\hbar}, \quad k' = \frac{\sqrt{2m(E-U)}}{\hbar}. \quad (3.63)$$

For a wave incident from the left, write

$$\psi(x) = \begin{cases} e^{ikx} + re^{-ikx}, & x < 0, \\ te^{ik'x}, & x > 0. \end{cases} \quad (3.64)$$

Continuity of ψ and ψ' at $x = 0$ gives

$$1 + r = t, \quad k(1 - r) = k't. \quad (3.65)$$

Therefore,

$$\boxed{r = \frac{k - k'}{k + k'}, \quad t = \frac{2k}{k + k'} = \frac{2}{1 + k'/k}.} \quad (3.66)$$

The reflection and transmission probabilities are determined by probability currents:

$$R = |r|^2, \quad T = \frac{k'}{k}|t|^2, \quad R + T = 1. \quad (3.67)$$

Case II: $0 < E < U$

Now define

$$k = \frac{\sqrt{2mE}}{\hbar}, \quad \kappa = \frac{\sqrt{2m(U-E)}}{\hbar}, \quad k' = i\kappa. \quad (3.68)$$

The physical solution is

$$\psi(x) = \begin{cases} e^{ikx} + re^{-ikx}, & x < 0, \\ te^{-\kappa x}, & x > 0. \end{cases} \quad (3.69)$$

The growing exponential for $x > 0$ is excluded. The matching formulas are the same as before after substituting $k' = i\kappa$:

$$t = \frac{2k}{k + i\kappa}, \quad r = \frac{k - i\kappa}{k + i\kappa}. \quad (3.70)$$

Thus $|r|^2 = 1$. There is no transmitted probability current, although the wave function penetrates a distance of order κ^{-1} into the classically forbidden region.

3.2.2.3 Delta-function potential

Consider

$$V(x) = D_0 \delta(x). \quad (3.71)$$

The wave function is continuous at the origin. Integrating the stationary Schrödinger equation from $-\epsilon$ to $+\epsilon$ and taking $\epsilon \rightarrow 0^+$ gives

$$\boxed{\psi'(0^+) - \psi'(0^-) = \frac{2mD_0}{\hbar^2}\psi(0).} \quad (3.72)$$

Scattering states: $E > 0$

Let

$$k = \frac{\sqrt{2mE}}{\hbar}, \quad (3.73)$$

and use

$$\psi(x) = \begin{cases} e^{ikx} + re^{-ikx}, & x < 0, \\ te^{ikx}, & x > 0. \end{cases} \quad (3.74)$$

Continuity gives

$$1 + r = t. \quad (3.75)$$

The derivative jump condition gives

$$ik(t - 1 + r) = \frac{2mD_0}{\hbar^2}t. \quad (3.76)$$

Define

$$\alpha = \frac{mD_0}{\hbar^2 k}. \quad (3.77)$$

Then

$$\boxed{t = \frac{1}{1 + i\alpha}, \quad r = \frac{-i\alpha}{1 + i\alpha}.} \quad (3.78)$$

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Equivalently,

$$t = \frac{1}{1 - mD_0/(ik\hbar^2)}, \quad r = \frac{mD_0/(ik\hbar^2)}{1 - mD_0/(ik\hbar^2)}. \quad (3.79)$$

The fluxes satisfy

$$|t|^2 + |r|^2 = 1. \quad (3.80)$$

The same scattering result applies to either sign of D_0 as long as $E > 0$.

Bound state for an attractive delta potential

Let $D_0 < 0$ and $E < 0$. Define

$$\kappa = \frac{\sqrt{2m|E|}}{\hbar}. \quad (3.81)$$

Normalizability requires exponential decay at both infinities:

$$\psi(x) = \begin{cases} Ae^{\kappa x}, & x < 0, \\ Be^{-\kappa x}, & x > 0. \end{cases} \quad (3.82)$$

The continuity gives $A = B$ and the derivative jump condition becomes

$$-2\kappa A = \frac{2mD_0}{\hbar^2} A, \quad (3.83)$$

so

$$\kappa = -\frac{mD_0}{\hbar^2} = \frac{m|D_0|}{\hbar^2}. \quad (3.84)$$

Therefore the attractive delta potential has one bound state with energy

$$\boxed{E = -\frac{mD_0^2}{2\hbar^2}}. \quad (3.85)$$

Normalization gives

$$1 = \int_{-\infty}^{\infty} |\psi(x)|^2 dx = \frac{|A|^2}{\kappa}, \quad (3.86)$$

so one may choose

$$\boxed{\psi(x) = \sqrt{\kappa} e^{-\kappa|x|}}. \quad (3.87)$$

3.2.3 Bound and Scattering Spectra

Bound states are normalizable and have discrete energies. For the attractive delta potential, there is a single bound-state energy. In contrast, scattering states are not square-normalizable in the full space and form a continuous energy spectrum. They are instead normalized using delta-function normalization.

3.2.4 Summary

- The stationary Schrödinger equation is a linear, second-order ODE.
- For real potentials, energy eigenfunctions can be chosen real.
- Finite potential discontinuities preserve continuity of both ψ and ψ' .
- A delta potential preserves ψ but produces a finite jump in ψ' .
- Step and delta potentials illustrate reflection, transmission, evanescent waves, and bound-state quantization.

3.2 Solving one-dimensional potentials (10/2)

- Bound states have discrete energies, whereas scattering states form a continuous spectrum.

3.3 Bound states in one dimension (10/14)

- Refer to Tong 2.3, 14.1.4, 7.5

A bound-state wave function is square-normalizable:

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx < \infty. \quad (3.88)$$

Consequently,

$$\lim_{|x| \rightarrow \infty} \psi(x) = 0. \quad (3.89)$$

For a potential with a well-defined asymptotic value, a bound-state energy lies below the asymptotic continuum threshold:

$$E < \lim_{|x| \rightarrow \infty} V(x). \quad (3.90)$$

Unlike scattering states characterized by a continuous wavevector, bound states have a discrete spectrum.

3.3.1 The Infinite Potential Well

3.3.1.1 Potential and boundary conditions

Consider a box of width L centered at the origin:

$$V(x) = \begin{cases} 0, & |x| < L/2, \\ +\infty, & |x| > L/2. \end{cases} \quad (3.91)$$

The wave function vanishes outside the box and satisfies

$$\psi(-L/2) = \psi(L/2) = 0. \quad (3.92)$$

Inside the box, a general energy eigenfunction is

$$\psi(x) = Ae^{ipx/\hbar} + Be^{-ipx/\hbar}. \quad (3.93)$$

The two boundary conditions give

$$Ae^{ipL/(2\hbar)} + Be^{-ipL/(2\hbar)} = 0, \quad (3.94)$$

$$Ae^{-ipL/(2\hbar)} + Be^{ipL/(2\hbar)} = 0. \quad (3.95)$$

A nontrivial solution requires

$$e^{2ipL/\hbar} = 1, \quad (3.96)$$

so the allowed momenta are quantized.

3.3.1.2 Eigenfunctions and energies

It is convenient to write the normalized eigenfunctions as

$$\boxed{\psi_n(x) = \sqrt{\frac{2}{L}} \sin \left[\frac{n\pi}{L} \left(x + \frac{L}{2} \right) \right], \quad n = 1, 2, 3, \dots,} \quad (3.97)$$

for $|x| < L/2$, with $\psi_n(x) = 0$ outside the well. The energies are

$$E_n = \frac{\pi^2 \hbar^2}{2mL^2} n^2, \quad n = 1, 2, 3, \dots \quad (3.98)$$

The ground state has no interior node, the first excited state has one node, and the state with quantum number n has $n - 1$ interior nodes.

3.3.2 General Theorems for One-Dimensional Bound States

3.3.2.1 Nondegeneracy

The bound-state spectrum of a one-dimensional Schrödinger Hamiltonian is nondegenerate. Suppose ψ_1 and ψ_2 are square-normalizable solutions with the same energy E :

$$-\frac{\hbar^2}{2m}\psi_1'' + V\psi_1 = E\psi_1, \quad (3.99)$$

$$-\frac{\hbar^2}{2m}\psi_2'' + V\psi_2 = E\psi_2. \quad (3.100)$$

Multiply the first equation by ψ_2 , the second by ψ_1 , and subtract. This gives

$$\frac{d}{dx} (\psi_2\psi_1' - \psi_1\psi_2') = 0. \quad (3.101)$$

Thus the Wronskian is constant. Bound-state boundary conditions make this constant zero, so

$$\psi_2\psi_1' = \psi_1\psi_2'. \quad (3.102)$$

Where the wave functions are nonzero,

$$\frac{\psi_1'}{\psi_1} = \frac{\psi_2'}{\psi_2}, \quad (3.103)$$

and hence

$$\psi_1(x) = C\psi_2(x). \quad (3.104)$$

The two solutions therefore represent the same physical state.

3.3.2.2 Node theorems

For one-dimensional bound states:

1. The ground-state wave function has no node at finite x .
2. The m th excited state has exactly m nodes.

Equivalently, if the eigenstates are indexed by $n = 1, 2, 3, \dots$, then the n th state has $n - 1$ nodes.

3.3.3 The Finite Square Well

Consider the symmetric finite well

$$V(x) = \begin{cases} -V_0, & |x| < a, \\ 0, & |x| > a, \end{cases} \quad V_0 > 0. \quad (3.105)$$

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3.3.3.1 Scattering states with $E > 0$

For scattering states with $E > 0$, define

$$k = \frac{\sqrt{2mE}}{\hbar}, \quad k' = \frac{\sqrt{2m(E + V_0)}}{\hbar}. \quad (3.106)$$

A left-incident scattering state has the form

$$\psi(x) = \begin{cases} e^{ikx} + re^{-ikx}, & x < -a, \\ Ae^{ik'x} + Be^{-ik'x}, & |x| < a, \\ te^{ikx}, & x > a. \end{cases} \quad (3.107)$$

Continuity of ψ and ψ' at $x = -a$ gives

$$e^{-ika} + re^{ika} = Ae^{-ik'a} + Be^{ik'a}, \quad (3.108)$$

$$k(e^{-ika} - re^{ika}) = k'(Ae^{-ik'a} - Be^{ik'a}). \quad (3.109)$$

At $x = a$,

$$Ae^{ik'a} + Be^{-ik'a} = te^{ika}, \quad (3.110)$$

$$k'(Ae^{ik'a} - Be^{-ik'a}) = kte^{ika}. \quad (3.111)$$

These matching equations determine the reflection and transmission amplitudes.

Solving the two matching equations at $x = \pm a$ for A and B , we find

$$A = \frac{e^{ik'a}}{2} \left[\left(1 + \frac{k}{k'}\right) e^{-ika} + r \left(1 - \frac{k}{k'}\right) e^{ika} \right] = \frac{e^{-ik'a}}{2} te^{ika} \left(1 + \frac{k}{k'}\right), \quad (3.112)$$

$$B = \frac{e^{-ik'a}}{2} \left[\left(1 - \frac{k}{k'}\right) e^{-ika} + r \left(1 + \frac{k}{k'}\right) e^{ika} \right] = \frac{e^{ik'a}}{2} te^{ika} \left(1 - \frac{k}{k'}\right). \quad (3.113)$$

3.3.3.2 Bound states with $-V_0 < E < 0$

For bound states,

$$-V_0 < E < 0. \quad (3.114)$$

Define

$$\kappa = -ik = \frac{\sqrt{2m|E|}}{\hbar}, \quad k' = \frac{\sqrt{2m(E + V_0)}}{\hbar}. \quad (3.115)$$

These obey

$$\kappa^2 + (k')^2 = \frac{2mV_0}{\hbar^2}. \quad (3.116)$$

Normalizability requires exponential decay outside the well:

$$\psi(x) = \begin{cases} re^{\kappa x}, & x < -a, \\ Ae^{ik'x} + Be^{-ik'x}, & |x| < a, \\ te^{-\kappa x}, & x > a. \end{cases} \quad (3.117)$$

Note that the matching conditions for bound states are exactly the same as for scattering states, except for replacing $k = i\kappa$. Comparing the form of wavefunctions, we see that in bound states

$$|r|, |t| \rightarrow \infty \quad (3.118)$$

and Eqs. (3.112)-(3.113) becomes

$$\frac{r}{t} = e^{2ik'a} \frac{k' - k}{k' + k} = e^{-2ik'a} \frac{k' + k}{k' - k} = \pm 1. \quad (3.119)$$

for bound states.

3.3.3.3 Inversion symmetry

For an even potential,

$$V(x) = V(-x), \quad (3.120)$$

define the parity operator

$$\hat{P} = \int_{-\infty}^{\infty} dx | -x \rangle \langle x|. \quad (3.121)$$

It satisfies

$$\hat{P}^2 = \mathbb{I}, \quad \hat{P}\hat{x}\hat{P}^{-1} = -\hat{x}, \quad \hat{P}\hat{p}\hat{P}^{-1} = -\hat{p}. \quad (3.122)$$

For

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}), \quad (3.123)$$

one has

$$[\hat{P}, \hat{H}] = 0. \quad (3.124)$$

Because the one-dimensional bound spectrum is nondegenerate, each bound energy eigenstate is also a parity eigenstate:

$$\hat{P}|\psi\rangle = \frac{r}{t}|\psi\rangle = \pm|\psi\rangle. \quad (3.125)$$

Thus the wave function is either even or odd under the parity operator, depending on the sign of

$$\frac{r}{t} = e^{-2ik'a} \frac{k' + i\kappa}{k' - i\kappa} = \pm 1, \quad \kappa = \sqrt{2mV_0/\hbar^2 - (k')^2} \quad (3.126)$$

3.3.3.4 Even-parity bound states

For an even state, choose

$$\psi(x) = \begin{cases} Ce^{\kappa x}, & x < -a, \\ D \cos(k'x), & |x| < a, \\ Ce^{-\kappa x}, & x > a. \end{cases} \quad (3.127)$$

Matching ψ and ψ' at $x = a$ gives

$$\boxed{k' \tan(k'a) = \kappa.} \quad (3.128)$$

Using $\kappa^2 + (k')^2 = 2mV_0/\hbar^2$,

$$\boxed{k' \tan(k'a) = \sqrt{\frac{2mV_0}{\hbar^2} - (k')^2}.} \quad (3.129)$$

3.3.3.5 Odd-parity bound states

For an odd state, choose

$$\psi(x) = \begin{cases} -Ce^{\kappa x}, & x < -a, \\ D \sin(k'x), & |x| < a, \\ Ce^{-\kappa x}, & x > a. \end{cases} \quad (3.130)$$

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Matching at $x = a$ gives

$$\boxed{-k' \cot(k'a) = \kappa,} \quad (3.131)$$

or

$$\boxed{-k' \cot(k'a) = \sqrt{\frac{2mV_0}{\hbar^2} - (k')^2}.} \quad (3.132)$$

Equivalently,

$$\tan\left(k'a + \frac{\pi}{2}\right) = \frac{\kappa}{k'}. \quad (3.133)$$

3.3.3.6 Graphical solution

The allowed bound-state energies are obtained by solving Eq. (3.126), which can be simplified as

$$k' \tan(k'a) = \sqrt{\frac{r}{t}} \cdot \sqrt{\frac{2mV_0}{\hbar^2} - (k')^2} \quad (3.134)$$

Graphically, as shown in Fig. 3.1, the allowed values of k' are the intersections of the red curve

$$\kappa(k') = \sqrt{\frac{2mV_0}{\hbar^2} - (k')^2} \quad (3.135)$$

with the green even-parity branches $k' \tan(k'a)$ and the blue odd-parity branches $-k' \cot(k'a)$. Intersections with $k' \tan(k'a)$ give even-parity states, while intersections with $-k' \cot(k'a)$ give odd-parity states. The vertical dashed lines indicate the poles at $k'a = \pi/2, \pi, \dots$, and the red curve meets the horizontal axis at

$$k'a = a\sqrt{\frac{2mV_0}{\hbar^2}}. \quad (3.136)$$

Once an allowed k' is found,

$$E = -V_0 + \frac{\hbar^2(k')^2}{2m}. \quad (3.137)$$

Only finitely many intersections occur for a well of finite depth and width, so a finite square well supports finitely many bound states.

Finally, note that in the limit of

$$a \rightarrow 0^+, \quad V_0 \rightarrow +\infty, \quad 2aV_0 \rightarrow |D_0| \quad (3.138)$$

the square potential well discussed here reduces to the delta function potential studied in the last section.

3.3.4 Summary

- Bound states are square-normalizable and decay at spatial infinity.
- One-dimensional bound-state energy levels are nondegenerate.
- The ground state has no nodes, and successive excited states acquire one additional node each.
- For an even potential, nondegenerate bound states have definite parity.
- Finite-well energies follow from separate even- and odd-parity transcendental equations.

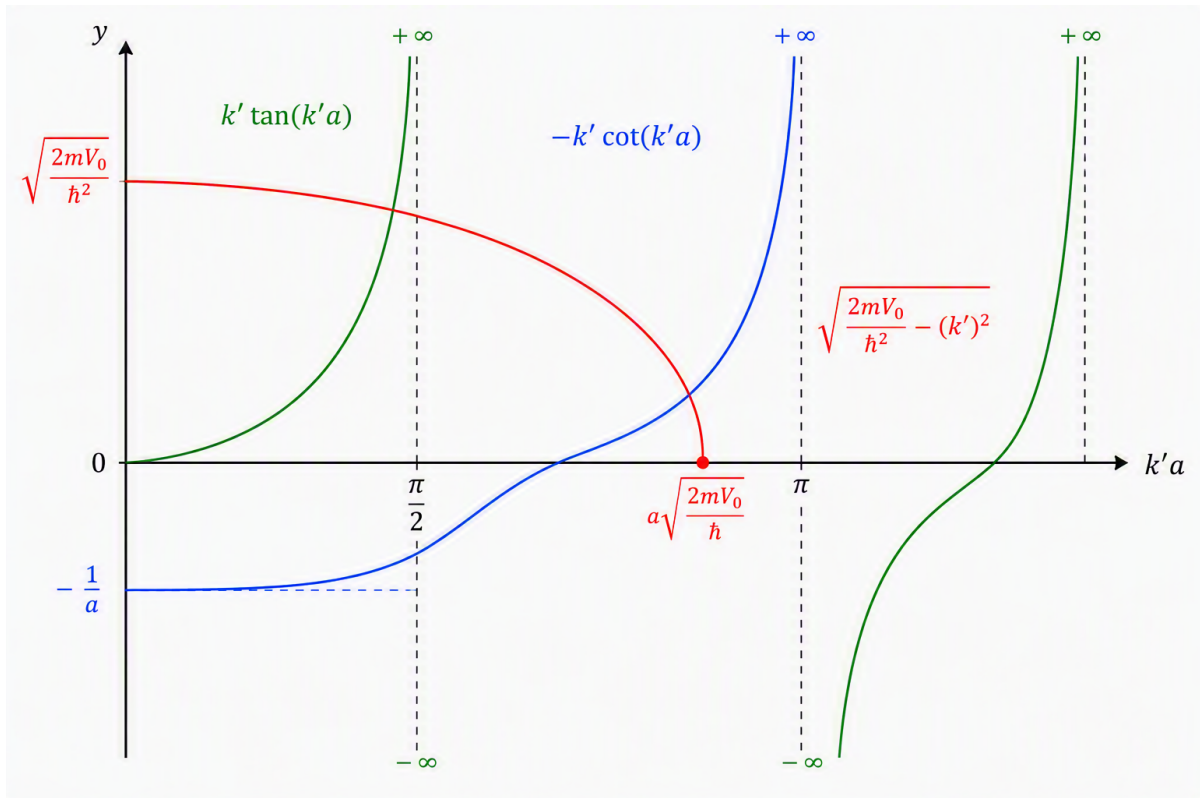


Figure 3.1: Graphical solution of the even- and odd-parity finite-well bound-state equations. Intersections with the red curve determine the allowed bound-state energies.

3.4 Scattering in one dimension (10/21)

- Refer to Tong 14.1.1-14.1.3, S&N 6.1

3.4.1 Scattering by a Delta Potential

We first review the case of a one-dimensional quantum particle scattered by a delta function potential. Consider

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}), \quad V(x) = V_0\delta(x). \quad (3.139)$$

If $V_0 > 0$, then $\hat{H} \geq 0$, so a negative-energy bound state is impossible. A bound state with $E < 0$ can occur only for an attractive delta potential.

3.4.1.1 Matching conditions at the singular point

The stationary Schrödinger equation is

$$-\frac{\hbar^2}{2m}\psi''(x) + V_0\delta(x)\psi(x) = E\psi(x). \quad (3.140)$$

The wave function remains continuous,

$$\psi(0^-) = \psi(0^+) = \psi(0), \quad (3.141)$$

but its derivative is discontinuous. Integrating across the origin gives

$$\psi'(0^+) - \psi'(0^-) = \frac{2mV_0}{\hbar^2}\psi(0). \quad (3.142)$$

These two conditions replace the usual continuity of both ψ and ψ' at a finite nonsingular potential discontinuity.

3.4.1.2 Scattering states

Consider a scattering state with $E > 0$, introduce

$$k = \frac{\sqrt{2mE}}{\hbar} > 0. \quad (3.143)$$

A general solution away from the origin is

$$\psi(x) = \begin{cases} \alpha e^{ikx} + \beta e^{-ikx}, & x < 0, \\ \gamma e^{ikx} + \delta e^{-ikx}, & x > 0. \end{cases} \quad (3.144)$$

Here we focus on the **in-state**, i.e. an incidental plane wave from the left

$$\psi_k^+(x) = \begin{cases} e^{ikx} + r e^{-ikx}, & x < 0, \\ t e^{ikx}, & x > 0. \end{cases} \quad (3.145)$$

where r and t correspond to the reflection and transmission amplitude after the incident wave was scattered by the delta function potential.

The continuity and derivative-jump equations are

$$1 + r = t, \quad ikt - ik(1 - r) = \frac{2mV_0}{\hbar^2}t. \quad (3.146)$$

Thus

$$\boxed{t = \frac{1}{1 - \frac{mV_0}{i\hbar^2 k}}, \quad r = \frac{\frac{mV_0}{i\hbar^2 k}}{1 - \frac{mV_0}{i\hbar^2 k}}.} \quad (3.147)$$

Equivalently, with $\eta = mV_0/(\hbar^2 k)$,

$$t = \frac{1}{1 + i\eta}, \quad r = \frac{-i\eta}{1 + i\eta}. \quad (3.148)$$

The reflection and transmission amplitudes obey

$$\boxed{|t|^2 + |r|^2 = 1.} \quad (3.149)$$

3.4.2 General properties of one-Dimensional scattering

3.4.2.1 In-state basis $\{|\psi_k^+\rangle\}$

Let $V(x)$ satisfy

$$\lim_{|x| \rightarrow \infty} V(x) = 0. \quad (3.150)$$

The stationary equation is

$$-\frac{\hbar^2}{2m}\psi''(x) + V(x)\psi(x) = E\psi(x). \quad (3.151)$$

For $E > 0$, its asymptotic solutions are plane waves:

$$\psi(x) \sim \alpha e^{ikx} + \beta e^{-ikx}, \quad k = \frac{\sqrt{2mE}}{\hbar}. \quad (3.152)$$

We consider a set of complete orthonormal basis for the scattering state, known as the **in-states** $\{|\psi_k^+\rangle\}$ which physically correspond to the incident plane waves

$$\psi_k^+(x) \sim \begin{cases} e^{ikx} + r e^{-ikx}, & x \rightarrow -\infty, \\ t e^{ikx}, & x \rightarrow +\infty, \end{cases} \quad (3.153)$$

$$\psi_{-k}^+(x) \sim \begin{cases} t' e^{-ikx}, & x \rightarrow -\infty, \\ e^{-ikx} + r' e^{ikx}, & x \rightarrow +\infty. \end{cases} \quad (3.154)$$

If bound states exist, they must also be added to in-states (or out-states) to complete the Hilbert space.

3.4.2.2 Probability flux

For a stationary wave function, define

$$\boxed{j(x) = \frac{i\hbar}{2m} (\psi^* \psi' - \psi (\psi^*)') = \frac{\hbar}{m} \text{Im}(\psi^* \psi').} \quad (3.155)$$

Subtracting the complex-conjugated Schrödinger equation from the original one gives the continuity equation for probability flux in a stationary state (eigenstate):

$$\frac{dj}{dx} = 0. \quad (3.156)$$

3 A quantum particle in one dimension

Thus the flux at $x \rightarrow -\infty$ equals the flux at $x \rightarrow +\infty$. Applied to $\psi_{\pm k}^+$, this yields

$$1 = |r|^2 + |t|^2 = |r'|^2 + |t'|^2 \quad (3.157)$$

3.4.2.3 Time-reversal symmetry

For a real scalar potential,

$$\hat{H}^* = \hat{H}. \quad (3.158)$$

Therefore, if $\psi(x)$ has energy E , then $\psi^*(x)$ has the same energy. This produces two independent scattering solutions at each energy $E > 0$.

Because ψ_k^+ and ψ_{-k}^+ have the same energy, form the complex conjugate of a linear combination that cancels one incoming component:

$$(t'\psi_k^+ - r\psi_{-k}^+)^* \propto \psi_{-k}(x) \quad (3.159)$$

This condition, combined with the conservation of probability flux in Eq. (3.157), yields the following relation:

$$t' = t, \quad r' = -\frac{r^*t}{t^*}. \quad (3.160)$$

Together with flux conservation, these relations are equivalent to the unitarity of the two-channel **scattering matrix** (also known as the S -matrix):

$$\hat{S}(k) = \begin{pmatrix} t & r \\ r' & t' \end{pmatrix} = \begin{pmatrix} t & r \\ -\frac{r^*t}{t^*} & t \end{pmatrix}, \quad \hat{S}^\dagger(k)\hat{S}(k) = \hat{1}. \quad (3.161)$$

3.4.2.4 Inversion symmetry in the case of an even potential $V(x) = V(-x)$

The inversion symmetry is implemented by a unitary operator

$$\hat{U} = \int_{-\infty}^{\infty} dx | -x \rangle \langle x |, \quad (3.162)$$

and satisfies

$$\hat{U}\hat{x}\hat{U}^\dagger = -\hat{x}, \quad \hat{U}\hat{p}\hat{U}^\dagger = -\hat{p}. \quad (3.163)$$

If $V(x) = V(-x)$, then $\hat{H} = \hat{U}\hat{H}\hat{U}^\dagger$ i.e. the Hamiltonian $\hat{H}$ preserves the inversion/parity symmetry. Hence, if $\psi(x)$ has energy E , so does $\psi(-x)$.

In the presence of inversion symmetry, the two in-states are related by the inversion operator $\hat{U}|\psi_k^+\rangle = |\psi_{-k}^+\rangle$:

$$\psi_{-k}^+(x) = \psi_k(-x) \quad (3.164)$$

and hence

$$r = r', \quad t = t'. \quad (3.165)$$

3.4.2.5 Out-state basis $\{|\psi_k^-\rangle\}$

The out-states are the time reversal partners of the in-states:

$$\psi_k^-(x) = [\psi_{-k}^+(x)]^*, \quad \psi_{-k}^-(x) = [\psi_k^+(x)]^*. \quad (3.166)$$

They are asymptotically outgoing plane waves. When the potential has no bound states, the out-states $\{|\psi_k^-\rangle\}$ also form a complete set of orthonormal basis of the Hilbert space, similar to

the in-state basis $\{|\psi_k^+\rangle\}$. If bound states exist, they must be added to complete the Hilbert space.

The unitary operator that relates these two basis sets $\{|\psi_k^+\rangle\}$ (in-states) and $\{|\psi_k^-\rangle\}$ (out-states) is exactly the scattering matrix:

$$S_{a,b}(k) = \langle \psi_{ak}^- | \psi_{bk}^+ \rangle, \quad a, b = \pm; \quad (3.167)$$

$$\hat{S}(k) = \begin{pmatrix} \langle \psi_k^- | \psi_k^+ \rangle & \langle \psi_k^- | \psi_{-k}^+ \rangle \\ \langle \psi_{-k}^- | \psi_k^+ \rangle & \langle \psi_{-k}^- | \psi_{-k}^+ \rangle \end{pmatrix} = \begin{pmatrix} t & r \\ -\frac{r^* t}{t^*} & t \end{pmatrix} \quad (3.168)$$

There is a deep reason behind this equation, as we explain below.

3.4.3 *Scattering theory in a nutshell

3.4.3.1 The Lippmann–Schwinger Equation

Consider a generic Hamiltonian

$$\hat{H} = \hat{H}_0 + \hat{V} \quad (3.169)$$

where $\hat{H}_0$ can be exactly solved (e.g. $\hat{H}_0 = \frac{\hat{p}^2}{2m}$ as we considered in one-dimensional scatterings), and $\hat{V}$ is a complicated term that makes it difficult to solve the full Hamiltonian $\hat{H}$. If

$$\hat{H}|\psi\rangle = E|\psi\rangle, \quad (3.170)$$

then

$$(E - \hat{H}_0)|\psi\rangle = \hat{V}|\psi\rangle. \quad (3.171)$$

Let $|a\rangle$ be a free eigenstate satisfying

$$\hat{H}_0|a\rangle = E_a|a\rangle. \quad (3.172)$$

The in- and out-states are eigenstates of full Hamiltonian $\hat{H}$ with the same energy E_a

$$(E_a - \hat{H} \pm i\epsilon)|\psi_a^\pm\rangle = 0 \quad (3.173)$$

This can be rewritten as (see HW7)

$$\boxed{|\psi_a^\pm\rangle = |a\rangle + \frac{1}{E_a - \hat{H}_0 \pm i\epsilon} \hat{V}|\psi_a^\pm\rangle = |a\rangle + \hat{G}_0^\pm(E_a) \hat{V}|\psi_a^\pm\rangle, \quad \epsilon \rightarrow 0^+} \quad (3.174)$$

where we defined the free retarded (+) and advanced (−) Green's function by

$$G_0^+(E) = \frac{1}{E - \hat{H}_0 + i\epsilon}, \quad G_0^-(E) = \frac{1}{E - \hat{H}_0 - i\epsilon}. \quad (3.175)$$

Iteration gives the so-called *Born series* and a formal solution to $|\psi_a^\pm\rangle$

$$|\psi_a^\pm\rangle = |a\rangle + G_0^\pm \hat{V}|a\rangle + (G_0^\pm \hat{V})^2 |a\rangle + \dots \quad (3.176)$$

$$= \sum_{j=0}^{\infty} (G_0^\pm \hat{V})^j |a\rangle = \frac{1}{1 - G_0^\pm \hat{V}} |a\rangle. \quad (3.177)$$

Equivalently,

$$|\psi_a^\pm\rangle = (E_a - \hat{H} \pm i\epsilon)^{-1} (E_a - \hat{H}_0 \pm i\epsilon) |a\rangle. \quad (3.178)$$

3.4.3.2 Adiabatic switching and the S matrix

Gell-Mann and Low introduced an adiabatically switched interaction

$$\hat{H}(t) = \hat{H}_0 + \hat{V}e^{-\epsilon|t|}, \quad \epsilon \rightarrow 0^+. \quad (3.179)$$

The time evolution operator in the interaction picture is

$$U_I^\epsilon(t, t_0) = e^{i\hat{H}_0 t/\hbar} \mathcal{T} \exp \left[-\frac{i}{\hbar} \int_{t_0}^t \hat{H}(t') dt' \right] e^{-i\hat{H}_0 t_0/\hbar}. \quad (3.180)$$

The Gell-Mann–Low construction gives the unnormalized scattering states

$$|\psi_a^+\rangle = \lim_{\substack{t_0 \rightarrow -\infty \\ \epsilon \rightarrow 0^+}} U_I^\epsilon(0, t_0) |a\rangle, \quad (3.181)$$

$$\langle \psi_a^- | = \lim_{\substack{t \rightarrow +\infty \\ \epsilon \rightarrow 0^+}} \langle a | U_I^\epsilon(t, 0). \quad (3.182)$$

Therefore,

$$S_{ab} = \langle \psi_a^- | \psi_b^+ \rangle = \lim_{\substack{t \rightarrow +\infty \\ t_0 \rightarrow -\infty \\ \epsilon \rightarrow 0^+}} \langle a | U_I^\epsilon(t, t_0) | b \rangle, \quad (3.183)$$

and

$$\hat{S} = \lim_{\epsilon \rightarrow 0^+} U_I^\epsilon(+\infty, -\infty). \quad (3.184)$$

Since time evolution is unitary, the scattering matrix is unitary.

3.4.3.3 The scattering matrix

At fixed energy $E = \hbar^2 k^2 / (2m)$, define

$$S_{ab} = \langle \psi_a^- | \psi_b^+ \rangle. \quad (3.185)$$

In the in- and out-state basis,

$$S(k) = \begin{pmatrix} t & r \\ r' & t \end{pmatrix}. \quad (3.186)$$

The scattering matrix is unitary:

$$S^\dagger S = \mathbb{I}. \quad (3.187)$$

3.4.3.4 Position-space Green function

With plane-wave normalization

$$\langle x | k \rangle = \frac{e^{ikx}}{\sqrt{2\pi}}, \quad \int_{-\infty}^{\infty} dk |k\rangle \langle k| = \mathbb{I}, \quad (3.188)$$

the Lippmann–Schwinger equation in position space is

$$\psi_k^+(x) = \frac{e^{ikx}}{\sqrt{2\pi}} + \int_{-\infty}^{\infty} dx' \langle x | G_0^+(E) | x' \rangle V(x') \psi_k^+(x'). \quad (3.189)$$

The Green-function kernel is

$$\langle x|G_0^\pm(E)|x'\rangle = \int_{-\infty}^{\infty} \frac{dq}{2\pi} \frac{e^{iq(x-x')}}{E - \hbar^2 q^2/(2m) \pm i\epsilon}. \quad (3.190)$$

The $+i\epsilon$ prescription produces the retarded Green function, while the $-i\epsilon$ prescription produces the advanced Green function.

Application to the delta potential:

For

$$V(x) = V_0 \delta(x), \quad (3.191)$$

the integral equation reduces to

$$\psi_k^+(x) = \frac{e^{ikx}}{\sqrt{2\pi}} + V_0 \langle x|G_0^+(E)|0\rangle \psi_k^+(0). \quad (3.192)$$

Evaluating the same equation at $x = 0$ gives an algebraic equation for $\psi_k^+(0)$, from which the reflection and transmission amplitudes of the delta potential follow.

3.4.4 Summary

- Delta-potential scattering follows from continuity of ψ and a finite jump in ψ' .
- Probability-flux conservation implies $|r|^2 + |t|^2 = 1$.
- Time reversal gives $t' = t$ and $r' = -r^* t/t^*$ for a general real one-dimensional potential.
- Inversion symmetry further constrains $r' = r$ for scattering by an even potential $V(x) = V(-x)$.
- The Lippmann–Schwinger equation encodes boundary conditions for in- and out-states through the $\pm i\epsilon$ prescriptions.
- The scattering matrix is unitary because the underlying time evolution is unitary.

4 Quantum optics

4.1 Simple harmonic oscillators (10/23)

- Refer to Tong 3.5.1, S&N 1.4, 2.3

4.1.1 Heisenberg's Uncertainty Principle

4.1.1.1 Expectation values and fluctuations

For a normalized pure state $|\psi\rangle$ and a Hermitian operator $\hat{A}$,

$$\langle \hat{A} \rangle = \langle \psi | \hat{A} | \psi \rangle. \quad (4.1)$$

Define the fluctuation operator

$$\Delta \hat{A} = \hat{A} - \langle \hat{A} \rangle. \quad (4.2)$$

The standard deviation of observable $\hat{A}$ is

$$\Delta A \equiv \sqrt{\langle (\Delta \hat{A})^2 \rangle} = \sqrt{\langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2}. \quad (4.3)$$

4.1.1.2 The uncertainty relation

The **Cauchy-Schwarz inequality** states

$$\langle \psi_1 | \psi_1 \rangle \langle \psi_2 | \psi_2 \rangle \geq |\langle \psi_1 | \psi_2 \rangle|^2, \quad (4.4)$$

with equality if and only if the two vectors are linearly dependent. Choose

$$|\psi_1\rangle = \Delta \hat{A} |\psi\rangle, \quad |\psi_2\rangle = \Delta \hat{B} |\psi\rangle. \quad (4.5)$$

Then

$$(\Delta A)^2 (\Delta B)^2 \geq \left| \langle \Delta \hat{A} \Delta \hat{B} \rangle \right|^2. \quad (4.6)$$

Decompose the product into commutator and anticommutator parts:

$$\Delta \hat{A} \Delta \hat{B} = \frac{1}{2} [\Delta \hat{A}, \Delta \hat{B}] + \frac{1}{2} \{\Delta \hat{A}, \Delta \hat{B}\}. \quad (4.7)$$

For Hermitian $\hat{A}$ and $\hat{B}$, the commutator expectation value is purely imaginary and the anticommutator expectation value is real. Therefore,

$$(\Delta A)^2 (\Delta B)^2 \geq \frac{1}{4} \left| \langle [\hat{A}, \hat{B}] \rangle \right|^2 + \frac{1}{4} \left| \langle \{\Delta \hat{A}, \Delta \hat{B}\} \rangle \right|^2 \quad (4.8)$$

$$\geq \frac{1}{4} \left| \langle [\hat{A}, \hat{B}] \rangle \right|^2. \quad (4.9)$$

Thus

$$\boxed{\Delta A \Delta B \geq \frac{1}{2} \left| \langle [\hat{A}, \hat{B}] \rangle \right|}. \quad (4.10)$$

4.1.1.3 Conditions for equality

Equality in the Schwarz inequality requires constants a and b , not both zero, such that

$$(a \Delta \hat{A} + b \Delta \hat{B}) |\psi\rangle = 0. \quad (4.11)$$

To saturate the commutator-only uncertainty relation, one also needs

$$\langle \{ \Delta \hat{A}, \Delta \hat{B} \} \rangle = 0. \quad (4.12)$$

The second condition follows from the first when $a^*b + ab^* = 0$ i.e.

$$\arg a = \arg b \pm \frac{\pi}{2} \quad (4.13)$$

4.1.1.4 Spin operators as an example

In this example we consider

$$\hat{A} = \hat{S}_x, \quad \hat{B} = \hat{S}_y. \quad (4.14)$$

Since

$$[\hat{S}_x, \hat{S}_y] = i\hbar \hat{S}_z, \quad \{\hat{S}_x, \hat{S}_y\} = 0 \quad (4.15)$$

for spin one-half, the uncertainty relation is

$$\Delta S_x \cdot \Delta S_y \geq \frac{\hbar}{2} |\langle \hat{S}_z \rangle| \quad (4.16)$$

For an S_x eigenstate, $\Delta S_x = 0$ and consistency requires $\langle \hat{S}_z \rangle = 0$.

4.1.1.5 Minimum-Uncertainty Wave Packets

For

$$\hat{A} = \hat{x}, \quad \hat{B} = \hat{p}, \quad [\hat{x}, \hat{p}] = i\hbar, \quad (4.17)$$

the uncertainty principle becomes

$$\boxed{\Delta x \Delta p \geq \frac{\hbar}{2}.} \quad (4.18)$$

A state saturating the bound obeys

$$\left(\Delta \hat{x} + iR^2 \frac{\Delta \hat{p}}{\hbar} \right) |\psi\rangle = 0. \quad (4.19)$$

Writing

$$x_0 = \langle \hat{x} \rangle, \quad p_0 = \langle \hat{p} \rangle, \quad (4.20)$$

the position-space equation is

$$\left[(x - x_0) + R^2 \left(\frac{d}{dx} - \frac{ip_0}{\hbar} \right) \right] \psi(x) = 0. \quad (4.21)$$

Its normalized solution is the Gaussian packet

$$\boxed{\psi(x) = \frac{1}{\pi^{1/4} \sqrt{R}} \exp \left(\frac{ip_0 x}{\hbar} - \frac{(x - x_0)^2}{2R^2} \right).} \quad (4.22)$$

The position variance is

$$(\Delta x)^2 = \langle (\hat{x} - x_0)^2 \rangle = \frac{R^2}{2}. \quad (4.23)$$

With

$$\langle x|p \rangle = \frac{e^{ipx/\hbar}}{\sqrt{2\pi\hbar}}, \quad (4.24)$$

the momentum-space wave function is

$$\tilde{\psi}(p) = \langle p | \psi \rangle = \frac{\sqrt{R}}{\pi^{1/4} \sqrt{\hbar}} \exp \left[-\frac{i(p - p_0)x_0}{\hbar} - \frac{R^2(p - p_0)^2}{2\hbar^2} \right], \quad (4.25)$$

up to an overall phase convention. Hence

$$(\Delta p)^2 = \frac{\hbar^2}{2R^2}, \quad \boxed{\Delta x \Delta p = \frac{\hbar}{2}}. \quad (4.26)$$

4.1.2 The Simple Harmonic Oscillator

The one-dimensional simple harmonic oscillator has the following Hamiltonian

$$\hat{H}_{\text{SHO}} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2 \hat{x}^2. \quad (4.27)$$

Near a stable minimum x_0 , a smooth potential has the expansion

$$V(x) = V(x_0) + \frac{1}{2}V''(x_0)(x - x_0)^2 + \dots. \quad (4.28)$$

Thus the harmonic oscillator is the leading approximation to motion near a stable equilibrium. It is also an exactly solvable model and a building block for systems such as quantized vibrational modes.

4.1.2.1 Ground state is the Gaussian wavepacket

Consider the Gaussian wavepacket $|\psi\rangle$ that saturates the Heisenberg uncertainty relation:

$$\left(\Delta \hat{x} + \frac{iR^2}{\hbar} \Delta \hat{p} \right) |\psi\rangle = 0. \quad (4.29)$$

Then we have

$$\begin{aligned} & \left(\Delta \hat{x} - \frac{iR^2}{\hbar} \Delta \hat{p} \right) \left(\Delta \hat{x} + \frac{iR^2}{\hbar} \Delta \hat{p} \right) |\psi\rangle \\ &= \left[(\Delta \hat{x})^2 + (\Delta \hat{p})^2 \left(\frac{R^2}{\hbar} \right)^2 + \frac{iR^2}{\hbar} [\Delta \hat{x}, \Delta \hat{p}] \right] |\psi\rangle \\ &= \left[(\Delta \hat{x})^2 + (\Delta \hat{p})^2 \left(\frac{R^2}{\hbar} \right)^2 - R^2 \right] |\psi\rangle = 0, \end{aligned} \quad (4.30)$$

where

$$[\Delta \hat{x}, \Delta \hat{p}] = i\hbar. \quad (4.31)$$

Multiplying by $\hbar^2/(2mR^4)$ gives

$$\left[\frac{(\Delta \hat{p})^2}{2m} + \frac{\hbar^2}{2mR^4} (\Delta \hat{x})^2 \right] |\psi\rangle = E |\psi\rangle, \quad (4.32)$$

with the eigen-energy given by

$$E = \frac{\hbar^2}{2mR^2} \quad (4.33)$$

Therefore the parent Hamiltonian for a Gaussian wavepacket is nothing but the simple harmonic

oscillator

$$\hat{H} = \frac{(\Delta\hat{p})^2}{2m} + \frac{m\omega^2}{2}(\Delta\hat{x})^2. \quad (4.34)$$

where the characteristic frequency ω is given by

$$\omega = \frac{\hbar}{mR^2} \quad (4.35)$$

and the ground state energy is

$$E_0 = \frac{\hbar^2}{2mR^2} = \frac{1}{2}\hbar\omega. \quad (4.36)$$

Note the only length scale R is proportional to the width of the wavepacket:

$$R = \sqrt{\frac{\hbar}{m\omega}} = \sqrt{2} \Delta x = \frac{\hbar}{\sqrt{2} \Delta p}. \quad (4.37)$$

4.1.2.2 Ground state wavefunction

In the simple harmonic oscillator Hamiltonian (4.27), we choose $\langle\hat{x}\rangle = \langle\hat{p}\rangle = 0$ and hence $\Delta\hat{x} = \hat{x}$ and $\Delta\hat{p} = \hat{p}$.

In position space,

$$\langle x|\hat{p}|\psi\rangle = -i\hbar \frac{d}{dx}\psi(x). \quad (4.38)$$

The condition (4.29) for ground state $|\psi_0\rangle$ therefore gives

$$\left(\sqrt{\frac{m\omega}{2\hbar}}x + \sqrt{\frac{\hbar}{2m\omega}} \frac{d}{dx} \right) \psi_0(x) = 0. \quad (4.39)$$

Hence we arrive at the ground state wavefunction

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} e^{-m\omega x^2/(2\hbar)} = \frac{1}{\pi^{1/4}\sqrt{R}} e^{-x^2/(2R^2)}. \quad (4.40)$$

4.1.2.3 Symmetries

The Hamiltonian (4.27) is obviously invariant under the inversion operation $\hat{\pi} \equiv \int dx |x\rangle\langle x|$:

$$\hat{x} \mapsto -\hat{x}, \quad \hat{p} \mapsto -\hat{p}, \quad \hat{H}_{\text{SHO}} \mapsto \hat{H}_{\text{SHO}} \quad (4.41)$$

In addition, the SHO Hamiltonian also preserves another $SO(2) \simeq U(1)$ symmetry:

$$\begin{pmatrix} \hat{x} \\ \hat{p}/m\omega \end{pmatrix} \mapsto \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \hat{x} \\ \hat{p}/m\omega \end{pmatrix} \quad (4.42)$$

where $\theta \in [0, 2\pi)$ is an arbitrary rotation angle.

4.1.3 Equation of motion in the Heisenberg picture

Recall that in the Heisenberg picture, states remain constants while the operators evolve with time:

$$|\psi_H(t)\rangle = |\psi_0\rangle, \quad \hat{O}_H(t) = \hat{U}^\dagger(t,0)\hat{O}\hat{U}(t,0) = e^{i\hat{H}t/\hbar}\hat{O}e^{-i\hat{H}t/\hbar} \quad (4.43)$$

Therefore the operators in the Heisenberg picture obey the following equation of motion:

$$\mathrm{i}\hbar \frac{\mathrm{d}}{\mathrm{d}t} \hat{O}_H(t) = [\hat{O}_H(t), \hat{H}] = e^{\mathrm{i}\hat{H}t/\hbar} [\hat{O}, \hat{H}] e^{-\mathrm{i}\hat{H}t/\hbar} \quad (4.44)$$

for a time-independent Hamiltonian $\hat{H}$. If we intend to compute the time dependence of expectation values of observable $\hat{O}$, it is more convenient to use the Heisenberg picture.

For example, in the case of a simple harmonic oscillator (4.27), using the canonical commutation relation between position and momentum operators:

$$[\hat{x}, \hat{p}] = \mathrm{i}\hbar \quad (4.45)$$

it is straightforward to show that

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle \hat{x}(t) \rangle = \frac{\langle \hat{p}(t) \rangle}{m}, \quad (4.46)$$

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle \hat{p}(t) \rangle = -\langle V'(\hat{x}(t)) \rangle = -m\omega^2 \langle \hat{x}(t) \rangle \quad (4.47)$$

where $V'(x)$ is the derivative of potential $V(x)$ in the Hamiltonian. Solving these two equations of motion produces the harmonic oscillation that is well known in classical mechanics.

4.2 From ladder operators to coherent states (10/28)

- Refer to Tong 5.1, 5.2; S&N 2.3; S&Z 2.1-2.4

4.2.1 Algebraic solution to the simple harmonic oscillator

For the centered oscillator, the minimum-energy Gaussian wavepacket has a width of

$$R = \sqrt{\frac{\hbar}{m\omega}}. \quad (4.48)$$

Label the ground state by $|0\rangle$. The minimum-uncertainty condition is

$$\left(\hat{x} + iR^2\frac{\hat{p}}{\hbar}\right)|0\rangle = 0. \quad (4.49)$$

The commutator of the two conjugate combinations is

$$[\hat{x} + iR^2\hat{p}/\hbar, \hat{x} - iR^2\hat{p}/\hbar] = 2R^2. \quad (4.50)$$

4.2.1.1 Ladder operators

Define

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\hat{x} + \frac{i}{\sqrt{2\hbar m\omega}}\hat{p}, \quad (4.51)$$

$$\hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\hat{x} - \frac{i}{\sqrt{2\hbar m\omega}}\hat{p}. \quad (4.52)$$

They obey

$$\boxed{[\hat{a}, \hat{a}^\dagger] = 1.} \quad (4.53)$$

The inverse relations are

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a} + \hat{a}^\dagger), \quad (4.54)$$

$$\hat{p} = -i\sqrt{\frac{\hbar m\omega}{2}}(\hat{a} - \hat{a}^\dagger). \quad (4.55)$$

Substitution into the Hamiltonian gives

$$\boxed{\hat{H}_{\text{SHO}} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right).} \quad (4.56)$$

The ground state satisfies

$$\hat{a}|0\rangle = 0, \quad \hat{H}|0\rangle = \frac{1}{2}\hbar\omega|0\rangle. \quad (4.57)$$

The commutators between the Hamiltonian and ladder operators are

$$[\hat{H}, \hat{a}^\dagger] = \hbar\omega\hat{a}^\dagger, \quad [\hat{H}, \hat{a}] = -\hbar\omega\hat{a}. \quad (4.58)$$

If

$$\hat{H}|\psi\rangle = E|\psi\rangle, \quad (4.59)$$

then

$$\hat{H}\hat{a}^\dagger|\psi\rangle = (E + \hbar\omega)\hat{a}^\dagger|\psi\rangle, \quad (4.60)$$

$$\hat{H}\hat{a}|\psi\rangle = (E - \hbar\omega)\hat{a}|\psi\rangle. \quad (4.61)$$

Thus $\hat{a}^\dagger$ is the creation or raising operator, because it raises the energy by $\hbar\omega$. Meanwhile $\hat{a}$ is the annihilation or lowering operator, since it lowers the energy by $\hbar\omega$.

4.2.1.2 Occupation number basis

The normalized energy eigenstates are

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle, \quad n = 0, 1, 2, \dots \quad (4.62)$$

They satisfy

$$\hat{H}|n\rangle = \left(n + \frac{1}{2}\right)\hbar\omega|n\rangle. \quad (4.63)$$

The ladder actions are

$$\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle, \quad (4.64)$$

$$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle. \quad (4.65)$$

Equivalently,

$$\hat{a}^\dagger = \sum_{n=0}^{\infty} \sqrt{n+1}|n+1\rangle\langle n|, \quad (4.66)$$

$$\hat{a} = \sum_{n=0}^{\infty} \sqrt{n+1}|n\rangle\langle n+1|. \quad (4.67)$$

4.2.1.3 Oscillator Wave Functions

Ground state:

In position space,

$$\langle x|\hat{p}|\psi\rangle = -i\hbar\frac{d}{dx}\psi(x). \quad (4.68)$$

The condition $\hat{a}|0\rangle = 0$ gives

$$\left(\sqrt{\frac{m\omega}{2\hbar}}x + \sqrt{\frac{\hbar}{2m\omega}}\frac{d}{dx}\right)\psi_0(x) = 0. \quad (4.69)$$

Hence

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-m\omega x^2/(2\hbar)} = \frac{1}{\pi^{1/4}\sqrt{R}} e^{-x^2/(2R^2)}. \quad (4.70)$$

First excited state:

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Using $|1\rangle = \hat{a}^\dagger|0\rangle$,

$$\psi_1(x) = \langle x|\hat{a}^\dagger|0\rangle \quad (4.71)$$

$$= \left(\sqrt{\frac{m\omega}{2\hbar}} x - \sqrt{\frac{\hbar}{2m\omega}} \frac{d}{dx} \right) \psi_0(x) \quad (4.72)$$

$$= \sqrt{\frac{2m\omega}{\hbar}} x \psi_0(x). \quad (4.73)$$

Thus

$$\psi_1(x) \propto x e^{-x^2/(2R^2)}. \quad (4.74)$$

Second excited state:

The second excited state is

$$|2\rangle = \frac{(\hat{a}^\dagger)^2}{\sqrt{2!}}|0\rangle = \frac{\hat{a}^\dagger}{\sqrt{2}}|1\rangle. \quad (4.75)$$

Applying the creation operator again gives

$$\boxed{\psi_2(x) = \frac{1}{\sqrt{2}} \left(\frac{2x^2}{R^2} - 1 \right) \psi_0(x).} \quad (4.76)$$

4.2.2 Coherent States

A coherent state is an eigenstate of the annihilation operator:

$$\boxed{\hat{a}|z\rangle = z|z\rangle}, \quad z \in \mathbb{C}. \quad (4.77)$$

Using

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a} + \hat{a}^\dagger), \quad \hat{p} = i\sqrt{\frac{m\hbar\omega}{2}}(\hat{a}^\dagger - \hat{a}), \quad (4.78)$$

one obtains

$$\langle z|\hat{x}|z\rangle = \sqrt{\frac{2\hbar}{m\omega}} \operatorname{Re} z, \quad (4.79)$$

$$\langle z|\hat{p}|z\rangle = \sqrt{2m\hbar\omega} \operatorname{Im} z. \quad (4.80)$$

4.2.2.1 Time evolution

$$|z(t)\rangle = e^{-i\hat{H}t/\hbar}|z\rangle. \quad (4.81)$$

Since

$$\hat{a}|z(t)\rangle = e^{-i\omega t} z|z(t)\rangle, \quad (4.82)$$

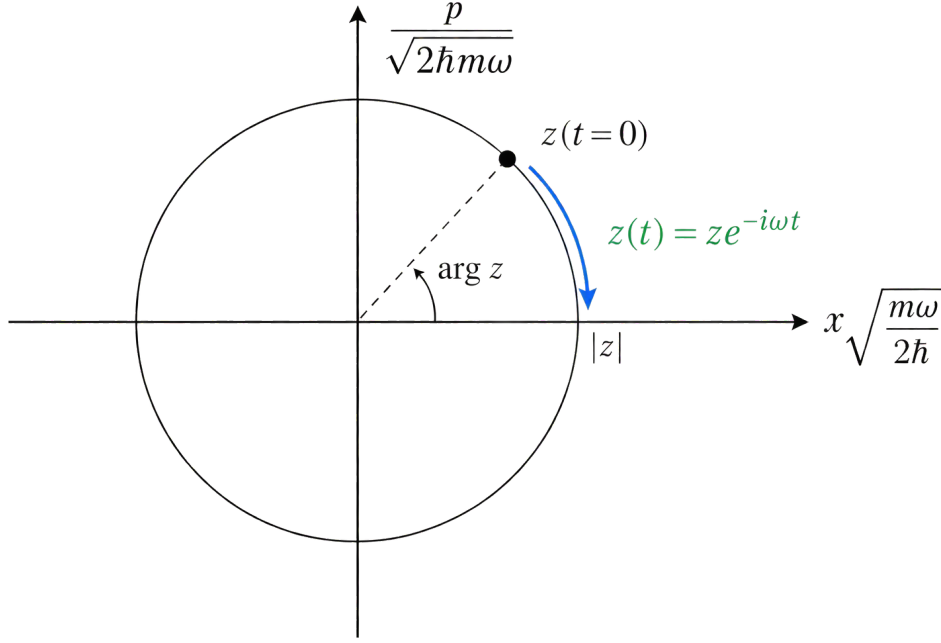
we have

$$\boxed{z(t) = z e^{-i\omega t}}. \quad (4.83)$$

Thus $\langle \hat{x}(t) \rangle$ and $\langle \hat{p}(t) \rangle$ follow the classical oscillator trajectory (see Fig. 4.1):

$$\langle \hat{x}(t) \rangle = \sqrt{\frac{2\hbar}{m\omega}} \operatorname{Re}(z e^{-i\omega t}), \quad (4.84)$$

$$\langle \hat{p}(t) \rangle = \sqrt{2m\hbar\omega} \operatorname{Im}(z e^{-i\omega t}) \quad (4.85)$$


 Figure 4.1: Phase-space evolution of a coherent state, $z(t) = ze^{-i\omega t}$.

4.2.2.2 Number-state expansion

Write

$$|z\rangle = \sum_{n=0}^{\infty} C_n |n\rangle. \quad (4.86)$$

The eigenvalue equation gives

$$C_{n+1} = \frac{z}{\sqrt{n+1}} C_n \implies C_n = \frac{z^n}{\sqrt{n!}} C_0. \quad (4.87)$$

The normalized coherent state is

$$|z\rangle = e^{-|z|^2/2} \sum_{n=0}^{\infty} \frac{z^n}{\sqrt{n!}} |n\rangle = e^{-|z|^2/2} e^{z\hat{a}^\dagger} |0\rangle. \quad (4.88)$$

Equivalently,

$$|z\rangle = \hat{D}(z) |0\rangle \quad (4.89)$$

where $D(z)$ is known as the displacement operator:

$$\hat{D}(z) = e^{z\hat{a}^\dagger - z^*\hat{a}}. \quad (4.90)$$

It shifts the center of the Gaussian wavepacket from $\langle \hat{x} \rangle = \langle \hat{p} \rangle = 0$ to

$$z = \sqrt{\frac{m\omega}{2\hbar}} x_0 + \frac{i}{\sqrt{2m\hbar\omega}} p_0 \quad (4.91)$$

4.2.2.3 Overlap and completeness

$$\langle z|z'\rangle = \exp \left[-\frac{|z|^2}{2} - \frac{|z'|^2}{2} + z^* z' \right], \quad (4.92)$$

$$\int_{\mathbb{C}} \frac{d^2 z}{\pi} |z\rangle \langle z| = \mathbb{I}. \quad (4.93)$$

Coherent states form a nonorthogonal overcomplete basis.

4.2.2.4 Gaussian wavepacket as its wavefunction

Let

$$R = \sqrt{\frac{\hbar}{m\omega}}, \quad x_0 = \sqrt{\frac{2\hbar}{m\omega}} \operatorname{Re} z, \quad p_0 = \sqrt{2m\hbar\omega} \operatorname{Im} z. \quad (4.94)$$

Then

$$\left[(\hat{x} - x_0) + \frac{iR^2}{\hbar} (\hat{p} - p_0) \right] |z\rangle = 0. \quad (4.95)$$

Hence we obtain the wavefunction of coherent states as a Gaussian wavepacket:

$$\langle x|z\rangle \propto \exp \left(\frac{ip_0 x}{\hbar} - \frac{(x - x_0)^2}{2R^2} \right), \quad (4.96)$$

$$\langle p|z\rangle \propto \exp \left(-\frac{ix_0 p}{\hbar} - \frac{R^2(p - p_0)^2}{2\hbar^2} \right). \quad (4.97)$$

Since all coherent states saturate the Heisenberg uncertainty relation, among all quantum states, they are closest to the classical trajectory $x(t), p(t)$ of a simple harmonic oscillator.

4.3 Quantization of electromagnetic fields (10/30)

- Refer to S&N 7.6, Tong 5.3, 13.4.1, S&Z 1.1

4.3.1 Classical Electromagnetic Field

Let

$$x^a = (ct, \mathbf{r}), \quad A^a = (\phi/c, \mathbf{A}), \quad (4.98)$$

and

$$F_{ab} = \frac{\partial A_b}{\partial x^a} - \frac{\partial A_a}{\partial x^b}. \quad (4.99)$$

The electromagnetic Lagrangian density is

$$\mathcal{L} = -\frac{1}{4\mu} F^{ab} F_{ab} = \frac{1}{2} \left(\epsilon \mathbf{E}^2 - \frac{\mathbf{B}^2}{\mu} \right), \quad c = \frac{1}{\sqrt{\mu\epsilon}}. \quad (4.100)$$

The electric and magnetic fields are

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t} - \nabla \phi, \quad \mathbf{B} = \nabla \times \mathbf{A}. \quad (4.101)$$

They are invariant under the following gauge transformations

$$\mathbf{A} \mapsto \mathbf{A} + \nabla \Theta, \quad \phi \mapsto \phi - \frac{\partial \Theta}{\partial t}. \quad (4.102)$$

Below we choose the Weyl and Coulomb gauges:

$$\phi = 0, \quad \nabla \cdot \mathbf{A} = 0. \quad (4.103)$$

4.3.2 One-dimensional optical cavity

In a cavity which is confined along x, z direction but has length L along y direction, we consider one linearly polarized mode propagating along y ,

$$\mathbf{A} = (A(y, t), 0, 0), \quad \mathbf{E} = (E_x, 0, 0), \quad \mathbf{B} = (0, 0, B_z), \quad (4.104)$$

with

$$E_x = -\frac{\partial A}{\partial t}, \quad B_z = -\frac{\partial A}{\partial y}. \quad (4.105)$$

The cavity Lagrangian is

$$L = \int_0^L dy \left[\frac{\epsilon}{2} \left(\frac{\partial A}{\partial t} \right)^2 - \frac{1}{2\mu} \left(\frac{\partial A}{\partial y} \right)^2 \right]. \quad (4.106)$$

4.3.2.1 Standing wave boundary conditions

Impose perfectly reflecting standing-wave boundary conditions

$$\boxed{A(0, t) = A(L, t) = 0.} \quad (4.107)$$

The normalized spatial modes are

$$u_n(y) = \sqrt{\frac{2}{L}} \sin(k_n y), \quad k_n = \frac{n\pi}{L}, \quad n = 1, 2, 3, \dots, \quad (4.108)$$

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and satisfy

$$\int_0^L dy u_n(y) u_m(y) = \delta_{nm}. \quad (4.109)$$

Expand the field as

$$A(y, t) = \sum_{n=1}^{\infty} q_n(t) u_n(y) = \sqrt{\frac{2}{L}} \sum_{n=1}^{\infty} q_n(t) \sin(k_n y). \quad (4.110)$$

Then

$$E_x(y, t) = -\sqrt{\frac{2}{L}} \sum_{n=1}^{\infty} \dot{q}_n(t) \sin(k_n y), \quad (4.111)$$

$$B_z(y, t) = -\sqrt{\frac{2}{L}} \sum_{n=1}^{\infty} k_n q_n(t) \cos(k_n y). \quad (4.112)$$

4.3.2.2 Mode Lagrangian

Using

$$\int_0^L dy \sin(k_n y) \sin(k_m y) = \frac{L}{2} \delta_{nm}, \quad (4.113)$$

$$\int_0^L dy \cos(k_n y) \cos(k_m y) = \frac{L}{2} \delta_{nm}, \quad (4.114)$$

one obtains the following Lagrangian

$$L = \sum_{n=1}^{\infty} \left(\frac{\epsilon}{2} \dot{q}_n^2 - \frac{k_n^2}{2\mu} q_n^2 \right). \quad (4.115)$$

Thus there is one real harmonic oscillator for each standing-wave mode.

4.3.3 Canonical Quantization

The momentum conjugate to q_n is

$$p_n = \frac{\partial L}{\partial \dot{q}_n} = \epsilon \dot{q}_n. \quad (4.116)$$

Canonical quantization gives

$$[\hat{q}_n, \hat{p}_m] = i\hbar \delta_{nm}, \quad [\hat{q}_n, \hat{q}_m] = 0, \quad [\hat{p}_n, \hat{p}_m] = 0. \quad (4.117)$$

The Hamiltonian is

$$\hat{H} = \sum_{n=1}^{\infty} \left(\hat{p}_n \dot{\hat{q}}_n - L_n \right) \quad (4.118)$$

$$= \sum_{n=1}^{\infty} \left(\frac{\hat{p}_n^2}{2\epsilon} + \frac{k_n^2}{2\mu} \hat{q}_n^2 \right). \quad (4.119)$$

Identifying each term with

$$\frac{\hat{p}_n^2}{2M} + \frac{1}{2}M\omega_n^2\hat{q}_n^2 \quad (4.120)$$

gives

$$M = \epsilon, \quad \omega_n^2 = \frac{k_n^2}{\mu\epsilon} = c^2 k_n^2. \quad (4.121)$$

Therefore,

$$\boxed{\omega_n = ck_n = \frac{n\pi c}{L}.} \quad (4.122)$$

4.3.3.1 Creation and annihilation operators

Define

$$\hat{a}_n = \sqrt{\frac{\epsilon\omega_n}{2\hbar}}\hat{q}_n + \frac{i}{\sqrt{2\hbar\epsilon\omega_n}}\hat{p}_n, \quad (4.123)$$

$$\hat{a}_n^\dagger = \sqrt{\frac{\epsilon\omega_n}{2\hbar}}\hat{q}_n - \frac{i}{\sqrt{2\hbar\epsilon\omega_n}}\hat{p}_n. \quad (4.124)$$

Then

$$[\hat{a}_n, \hat{a}_m^\dagger] = \delta_{nm}. \quad (4.125)$$

The inverse relations are

$$\hat{q}_n = \sqrt{\frac{\hbar}{2\epsilon\omega_n}}(\hat{a}_n + \hat{a}_n^\dagger), \quad (4.126)$$

$$\hat{p}_n = -i\sqrt{\frac{\hbar\epsilon\omega_n}{2}}(\hat{a}_n - \hat{a}_n^\dagger). \quad (4.127)$$

Thus

$$\boxed{\hat{H} = \sum_{n=1}^{\infty} \hbar\omega_n \left(\hat{a}_n^\dagger \hat{a}_n + \frac{1}{2} \right).} \quad (4.128)$$

The total zero-point energy is

$$E_0 = \frac{1}{2} \sum_{n=1}^{\infty} \hbar\omega_n. \quad (4.129)$$

4.3.3.2 Photon number states

For one mode,

$$\hat{N}_n = \hat{a}_n^\dagger \hat{a}_n, \quad \hat{N}_n |N_n\rangle = N_n |N_n\rangle. \quad (4.130)$$

Each excitation carries energy $\hbar\omega_n$, and

$$E_n(N_n) = \hbar\omega_n \left(N_n + \frac{1}{2} \right). \quad (4.131)$$

4.3.4 Quantized Standing-Wave Fields

4.3.4.1 Vector potential

In the Schrödinger picture,

$$\hat{A}(y) = \sum_{n=1}^{\infty} \sqrt{\frac{\hbar}{\epsilon L \omega_n}} \sin(k_n y) (\hat{a}_n + \hat{a}_n^\dagger). \quad (4.132)$$

In the Heisenberg picture,

$$\hat{A}(y, t) = \sum_{n=1}^{\infty} \sqrt{\frac{\hbar}{\epsilon L \omega_n}} \sin(k_n y) \left(\hat{a}_n e^{-i\omega_n t} + \hat{a}_n^\dagger e^{i\omega_n t} \right). \quad (4.133)$$

The boundary conditions are automatically satisfied because

$$\sin(k_n 0) = \sin(k_n L) = 0. \quad (4.134)$$

4.3.4.2 Electric and magnetic fields

Using $\hat{E}_x = -\partial_t \hat{A}$,

$$\hat{E}_x(y, t) = i \sum_{n=1}^{\infty} \sqrt{\frac{\hbar \omega_n}{\epsilon L}} \sin(k_n y) \left(\hat{a}_n e^{-i\omega_n t} - \hat{a}_n^\dagger e^{i\omega_n t} \right). \quad (4.135)$$

Using $\hat{B}_z = -\partial_y \hat{A}$,

$$\hat{B}_z(y, t) = - \sum_{n=1}^{\infty} k_n \sqrt{\frac{\hbar}{\epsilon L \omega_n}} \cos(k_n y) \left(\hat{a}_n e^{-i\omega_n t} + \hat{a}_n^\dagger e^{i\omega_n t} \right). \quad (4.136)$$

4.4 *Squeezed states of light (11/4)

- Refer to S&Z 2.5-2.7

4.4.1 Squeezed states

4.4.1.1 Field quadratures

For a single mode, define the field quadratures:

$$\hat{X} = \frac{\hat{a} + \hat{a}^\dagger}{\sqrt{2}}, \quad \hat{Y} = \frac{i(\hat{a}^\dagger - \hat{a})}{\sqrt{2}}. \quad (4.137)$$

Then

$$[\hat{X}, \hat{Y}] = i, \quad \Delta X \Delta Y \geq \frac{1}{2}. \quad (4.138)$$

For coherent states,

$$\Delta X = \Delta Y = \frac{1}{\sqrt{2}}. \quad (4.139)$$

4.4.1.2 Squeezing

A minimum-uncertainty squeezed state satisfies

$$(\Delta \hat{X} + iR^2 \Delta \hat{Y}) |\psi\rangle = 0. \quad (4.140)$$

Equivalently,

$$\left[(1 - R^2) \hat{a}^\dagger + (1 + R^2) \hat{a} - (X_0 + iR^2 Y_0) \right] |\psi\rangle = 0. \quad (4.141)$$

Define the squeeze parameter by

$$\frac{1 - R^2}{1 + R^2} = \tanh r, \quad R = e^{-r}. \quad (4.142)$$

Then

$$\Delta X = \frac{e^{-r}}{\sqrt{2}}, \quad \Delta Y = \frac{e^r}{\sqrt{2}}, \quad \Delta X \Delta Y = \frac{1}{2}. \quad (4.143)$$

4.4.1.3 Electric-field noise

For phase

$$\Phi = \omega t - \mathbf{k} \cdot \mathbf{r}, \quad (4.144)$$

the electric field of a propagating (non-standing) wave

$$\hat{E}(\mathbf{r}, t) = \sqrt{\frac{\hbar\omega}{\epsilon L}} \left(\hat{a} e^{-i(\omega t - \mathbf{k} \cdot \mathbf{r})} - \hat{a}^\dagger e^{i(\omega t - \mathbf{k} \cdot \mathbf{r})} \right). \quad (4.145)$$

can be rewritten as a superposition of quadratures:

$$\hat{E}(\mathbf{r}, t) = \sqrt{\frac{2\hbar\omega}{\epsilon V}} \left(\hat{X} \cos \Phi + \hat{Y} \sin \Phi \right). \quad (4.146)$$

For a squeezed state with vanishing quadrature covariance,

$$(\Delta E)^2 = \frac{\hbar\omega}{\epsilon V} \left(e^{-2r} \cos^2 \Phi + e^{2r} \sin^2 \Phi \right). \quad (4.147)$$

For $r > 0$,

$$\Delta E_{\min} = \Delta X = \frac{e^{-r}}{\sqrt{2}}, \quad \Phi = 0 \pmod{\pi}, \quad (4.148)$$

$$\Delta E_{\max} = \Delta Y = \frac{e^r}{\sqrt{2}}, \quad \Phi = \frac{\pi}{2} \pmod{\pi}, \quad (4.149)$$

$$\Delta E_{\min} \Delta E_{\max} = \Delta X \Delta Y = \frac{1}{2}. \quad (4.150)$$

4.4.2 Generation of coherent and squeezed states in quantum optics

4.4.2.1 Generating coherent states

With a classical current,

$$\hat{H} = \hat{H}_{\text{EM}} - \int \mathbf{j}(\mathbf{r}, t) \cdot \hat{\mathbf{A}}(\mathbf{r}) d^3\mathbf{r}. \quad (4.151)$$

The evolved vacuum is a coherent state:

$$e^{-i\hat{H}t/\hbar}|0\rangle = \exp \left[\sum_k (\alpha_k \hat{a}_k^\dagger - \alpha_k^* \hat{a}_k) \right] |0\rangle, \quad (4.152)$$

where

$$\alpha_k = - \int d^3\mathbf{r} \int_0^t dt' e^{i(\omega t' - \mathbf{k} \cdot \mathbf{r})} \mathbf{j}(\mathbf{r}, t') \cdot \mathbf{e}_k \sqrt{\frac{\hbar}{\omega \epsilon V}}. \quad (4.153)$$

The displacement operator is

$$\boxed{\hat{D}(\alpha) = e^{\alpha \hat{a}^\dagger - \alpha^* \hat{a}}}. \quad (4.154)$$

4.4.2.2 Generating squeezed states

Degenerate parametric amplification, achieved by placing a nonlinear crystal nonlinear inside an optical cavity, produces the effective Hamiltonian

$$\hat{H}_{\text{dpa}} = \hbar \chi \beta_0 \left[(\hat{a}^\dagger)^2 e^{-i\phi} + \hat{a}^2 e^{i\phi} \right]. \quad (4.155)$$

Define the squeeze operator:

$$\boxed{\hat{S}(\xi) = \exp \left[\frac{1}{2} \left(\xi^* \hat{a}^2 - \xi (\hat{a}^\dagger)^2 \right) \right]}, \quad \xi = r e^{i\theta}. \quad (4.156)$$

It satisfies

$$\hat{S}^\dagger(\xi)(\hat{X} + i\hat{Y})\hat{S}(\xi) = e^{-r}\hat{X} + ie^r\hat{Y} \quad (4.157)$$

for a suitable squeezing phase. A squeezed coherent state is

$$\boxed{|\xi, z\rangle = \hat{S}(\xi)\hat{D}(z)|0\rangle \sim \hat{S}(\xi)|z\rangle}. \quad (4.158)$$

Its quadrature variances are

$$\langle \xi, z | (\Delta \hat{X})^2 | \xi, z \rangle = \frac{e^{-2r}}{2}, \quad \langle \xi, z | (\Delta \hat{Y})^2 | \xi, z \rangle = \frac{e^{2r}}{2}. \quad (4.159)$$

4.4.3 Application: squeezed light in LIGO

4.4.3.1 Gravitational-wave measurement

LIGO is a Michelson interferometer with long optical cavities in its two perpendicular arms. A passing gravitational wave changes the arm lengths in opposite directions. If the unperturbed arm length is L , define the gravitational-wave strain by

$$h(t) = \frac{\Delta L_x(t) - \Delta L_y(t)}{L}. \quad (4.160)$$

The differential change in arm length produces a phase shift in the light returning from the two arms. The output photodetector measures this phase shift.

Ideally, the interferometer is operated near a dark fringe, so little classical carrier light exits through the antisymmetric port. Quantum vacuum fluctuations can nevertheless enter through this port and interfere with the carrier field. These fluctuations contribute quantum noise to the measured signal.

4.4.3.2 Two sources of quantum noise

Introduce the amplitude and phase quadratures

$$\hat{X} = \frac{\hat{a} + \hat{a}^\dagger}{\sqrt{2}}, \quad \hat{Y} = \frac{i(\hat{a}^\dagger - \hat{a})}{\sqrt{2}}, \quad (4.161)$$

with

$$[\hat{X}, \hat{Y}] = i, \quad \Delta X \Delta Y \geq \frac{1}{2}. \quad (4.162)$$

The two quadratures affect LIGO in different ways:

1. The phase quadrature $\hat{Y}$ produces uncertainty in the optical phase measured at the output. This is called shot noise.
2. The amplitude quadrature $\hat{X}$ produces fluctuations in the number of photons incident on the mirrors. The corresponding fluctuating radiation pressure moves the mirrors. This is called quantum radiation-pressure noise.

The approximate frequency dependence of the quadrature variance is

$$S_h^{\text{shot}}(\Omega) \sim \Delta Y^2 \propto \frac{1}{P}, \quad S_h^{\text{rad}}(\Omega) \sim \Delta X^2 \propto \frac{P}{\Omega^4}, \quad (4.163)$$

where P is the circulating optical power and Ω is the gravitational-wave angular frequency. Shot noise is most important at high frequencies, while radiation-pressure noise becomes important at low frequencies.

Increasing the laser power reduces shot noise but increases radiation-pressure noise. This tradeoff leads to the standard quantum limit.

4.4.3.3 Injection of squeezed vacuum

For ordinary vacuum,

$$(\Delta X)^2 = (\Delta Y)^2 = \frac{1}{2}. \quad (4.164)$$

A squeezed vacuum state redistributes these uncertainties. For phase squeezing,

$$(\Delta X)^2 = \frac{1}{2}e^{2r}, \quad (\Delta Y)^2 = \frac{1}{2}e^{-2r}, \quad (4.165)$$

where $r > 0$ is the squeeze parameter. Therefore,

$$\Delta X \Delta Y = \frac{1}{2}. \quad (4.166)$$

Injecting this squeezed vacuum into the antisymmetric port reduces the phase uncertainty:

$$\Delta Y_{\text{sq}} = e^{-r} \Delta Y_{\text{vac}}. \quad (4.167)$$

The shot-noise contribution is consequently reduced by

$$S_{h,\text{sq}}^{\text{shot}} = e^{-2r} S_{h,\text{vac}}^{\text{shot}}. \quad (4.168)$$

The uncertainty principle is not violated. The reduction of phase noise is accompanied by an increase in amplitude noise:

$$\Delta X_{\text{sq}} = e^r \Delta X_{\text{vac}}, \quad S_{h,\text{sq}}^{\text{rad}} = e^{2r} S_{h,\text{vac}}^{\text{rad}}. \quad (4.169)$$

The increased amplitude noise can increase radiation-pressure noise at low frequencies.

4.4.3.4 Frequency-dependent squeezing

A fixed squeezing angle cannot reduce both forms of quantum noise over the entire detection band. LIGO therefore uses frequency-dependent squeezing.

Let the quadrature vector be

$$\hat{\mathbf{q}} = \begin{pmatrix} \hat{X} \\ \hat{Y} \end{pmatrix}. \quad (4.170)$$

A rotation of the squeezing ellipse through an angle θ is represented by

$$\hat{\mathbf{q}}_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \hat{\mathbf{q}}. \quad (4.171)$$

Before the squeezed vacuum enters the main interferometer, it is reflected from an auxiliary optical filter cavity. The cavity produces a frequency-dependent quadrature rotation,

$$\theta = \theta(\Omega). \quad (4.172)$$

The squeezing ellipse is therefore oriented differently for different gravitational-wave frequencies.

At high frequencies, the squeezed quadrature is aligned to reduce phase noise:

$$(\Delta Y)_{\text{high frequency}}^2 = \frac{1}{2} e^{-2r}. \quad (4.173)$$

This reduces shot noise.

At low frequencies, the filter cavity rotates the ellipse so that the amplitude quadrature is squeezed:

$$(\Delta X)_{\text{low frequency}}^2 = \frac{1}{2} e^{-2r}. \quad (4.174)$$

This reduces radiation-pressure fluctuations acting on the mirrors.

Schematically,

low frequency: amplitude squeezing $\implies$ less radiation-pressure noise, high frequency: phase squeezing $\implies$ less shot noise.	(4.175)
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4.4.3.5 Optical loss

Optical loss replaces part of the squeezed state with ordinary vacuum. If η is the total optical efficiency, the observed squeezed variance is approximately

$$(\Delta Q)_{obs}^2 = \eta \frac{e^{-2r}}{2} + (1 - \eta) \frac{1}{2}, \quad (4.176)$$

where $\hat{Q}$ denotes the intended squeezed quadrature.

Even if the generated squeezing is very strong, the measured noise reduction is limited by optical loss and fluctuations of the squeezing angle. High optical efficiency and accurate control of the squeezing phase are therefore essential.

4.4.3.6 Summary

LIGO does not evade the uncertainty principle. Instead, squeezed light places less quantum uncertainty in the quadrature that limits a particular part of the measurement and more uncertainty in the conjugate quadrature.

Frequency-dependent squeezing extends this idea by rotating the squeezing ellipse as a function of frequency. It reduces the relevant quantum noise at both low and high frequencies, allowing weaker gravitational-wave signals to be distinguished from the quantum fluctuations of the light.

This section summarizes LIGO's use of vacuum fluctuations entering the unused interferometer port, squeezed states to reduce shot noise, and frequency-dependent squeezing to control the tradeoff between high-frequency phase noise and low-frequency radiation-pressure noise.

LIGO reports that frequency-dependent squeezing has been operating since the observing run began in May 2023 and improves sensitivity across its gravitational-wave frequency range.

5 Quantum dynamics

5.1 Sudden and Adiabatic Limits (11/6)

In this section we study the time evolution of a quantum state under the time-dependent Schrodinger equation

$$|\psi(t)\rangle = \mathcal{T} \exp \left[-\frac{i}{\hbar} \int_0^t \hat{H}(\tau) d\tau \right] |\psi(0)\rangle, \quad i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H}(t) |\psi(t)\rangle. \quad (5.1)$$

Specifically we focus on two limit cases: the extremely fast limit, described by the sudden approximation; and the extremely slow limit, described by the adiabatic evolution.

5.1.1 Sudden approximation

In a very short time t

$$t \ll \frac{\hbar}{\Delta E}, \quad (5.2)$$

where ΔE is the smallest level spacing between any two energy levels in the spectrum of the Hamiltonian, the unitary time evolution operator is approximately a global phase factor

$$\hat{U}(t, 0) = \mathcal{T} \exp \left[-\frac{i}{\hbar} \int_0^t \hat{H}(\tau) d\tau \right] \simeq e^{i\phi(t)} \mathbb{I}, \quad (5.3)$$

and hence the state is approximately the same up to a phase factor

$$\boxed{|\psi(t)\rangle \simeq |\psi(0)\rangle}. \quad (5.4)$$

For example,

$$\hat{H}(t) = \hat{H}_0 + \delta \hat{H} \Theta(t) \quad (5.5)$$

is a toy model that describes a sudden change from $\hat{H}_0$ to $\hat{H}_0 + \delta \hat{H}$. In this toy model, it takes zero time for the Hamiltonian to jump from $\hat{H}_0$ at $t = -\epsilon$ to $\hat{H}$ at $t = +\epsilon$, and we have

$$|\psi(t = -\epsilon)\rangle \approx |\psi(t = +\epsilon)\rangle \quad (5.6)$$

5.1.1.1 Sudden quench of a harmonic oscillator

Consider

$$\hat{H}(t) = \hbar\omega(t) \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right), \quad \omega(t) = \begin{cases} \omega_0, & t < 0, \\ \omega, & t > 0. \end{cases} \quad (5.7)$$

Let

$$R_0 = \sqrt{\frac{\hbar}{m\omega_0}}, \quad R = \sqrt{\frac{\hbar}{m\omega}}. \quad (5.8)$$

The old annihilation operator is

$$\begin{aligned} \hat{a}_0 &= \frac{\hat{x}}{\sqrt{2}R_0} + \frac{iR_0\hat{p}}{\sqrt{2}\hbar} \\ &= \frac{1}{2} \left(\sqrt{\frac{\omega}{\omega_0}} + \sqrt{\frac{\omega_0}{\omega}} \right) \hat{a} + \frac{1}{2} \left(\sqrt{\frac{\omega}{\omega_0}} - \sqrt{\frac{\omega_0}{\omega}} \right) \hat{a}^\dagger. \end{aligned} \quad (5.9)$$

Define the squeezing parameter r as

$$\frac{\omega_0}{\omega} = e^{2r}. \quad (5.10)$$

Then

$$\hat{a}_0 = \cosh r \hat{a} + \sinh r \hat{a}^\dagger. \quad (5.11)$$

Since

$$\hat{a}_0|0\rangle_{\omega_0} = 0, \quad (5.12)$$

one has

$$\left(\hat{a} + \tanh r \hat{a}^\dagger\right) |0\rangle_{\omega_0} = 0. \quad (5.13)$$

Thus the old ground state $|0\rangle_{\omega_0}$ of a SHO with frequency ω_0 is a squeezed state of the new oscillator with frequency ω .

5.1.2 Schrodinger equation in the instantaneous eigenstate basis

Define the instantaneous eigenstates $\{|n, t\rangle\}$ of a time-dependent Hamiltonian $\hat{H}(t)$:

$$\hat{H}(t)|n, t\rangle = E_n(t)|n, t\rangle, \quad \langle m, t|n, t\rangle = \delta_{m,n} \quad (5.14)$$

Expand

$$|\psi(t)\rangle = \sum_n C_n(t) \exp\left[-\frac{i}{\hbar} \int_0^t E_n(\tau) d\tau\right] |n, t\rangle. \quad (5.15)$$

Then the Schrodinger equation becomes

$$\dot{C}_m = - \sum_n \exp\left[-i \int_0^t \omega_{nm}(\tau) d\tau\right] C_n \langle m, t | \frac{d}{dt} |n, t\rangle, \quad (5.16)$$

where

$$\omega_{nm}(t) = \frac{E_n(t) - E_m(t)}{\hbar}. \quad (5.17)$$

Differentiating the instantaneous eigenvalue equation (5.14) gives, for $m \neq n$,

$$\langle m, t | \frac{d}{dt} |n, t\rangle = \frac{\langle m, t | \dot{\hat{H}}(t) |n, t\rangle}{E_n(t) - E_m(t)}. \quad (5.18)$$

Thus

$$\begin{aligned} \dot{C}_m = & - C_m \langle m, t | \frac{d}{dt} |m, t\rangle \\ & - \sum_{n \neq m} \exp\left[-i \int_0^t \omega_{nm}(\tau) d\tau\right] C_n \frac{\langle m, t | \dot{\hat{H}}(t) |n, t\rangle}{E_n(t) - E_m(t)}. \end{aligned} \quad (5.19)$$

5.1.3 Adiabatic theorem and Berry phase

Typically the time-dependence of a Hamiltonian comes from one (or more) time-dependent parameter(s):

$$\hat{H}(t) = \hat{H}(\lambda(t)), \quad \dot{\hat{H}} = \frac{\partial \hat{H}}{\partial \lambda} \dot{\lambda}. \quad (5.20)$$

A sufficient adiabatic condition is

$$\left| \frac{\hbar \dot{\lambda} \langle m | \partial_\lambda \hat{H} | n \rangle}{(E_n - E_m)^2} \right| \ll 1, \quad m \neq n. \quad (5.21)$$

Neglecting transitions between distinct instantaneous levels,

$$\dot{C}_m = -C_m \dot{\lambda} \langle m | \partial_\lambda | m \rangle. \quad (5.22)$$

Therefore, in the adiabatic limit we have

$$C_m(t) = C_m(0) \exp \left[- \int_{\lambda(0)}^{\lambda(t)} \langle m | \partial_\lambda | m \rangle d\lambda \right]. \quad (5.23)$$

Define the Berry connection

$$\boxed{\mathcal{A}_m(\lambda) = -i \langle m | \partial_\lambda | m \rangle.} \quad (5.24)$$

It is real because

$$\partial_\lambda \langle m | m \rangle = \langle \partial_\lambda m | m \rangle + \langle m | \partial_\lambda m \rangle = 0. \quad (5.25)$$

The geometric phase is

$$\boxed{e^{i\gamma_m} = \exp \left[-i \int \mathcal{A}_m(\lambda) d\lambda \right].} \quad (5.26)$$

For example, for a qubit in a slowly varying magnetic field,

$$\hat{H} = -\hbar\omega \hat{\mathbf{n}}(t) \cdot \boldsymbol{\sigma}, \quad |\hat{\mathbf{n}}| = 1, \quad (5.27)$$

the Berry phase around a closed loop is proportional to the solid angle subtended on the Bloch sphere.

5.2 A quantum particle in electromagnetic fields (11/13)

5.2.1 Probability conservation

For

$$\mathrm{i}\hbar\partial_t\psi = -\frac{\hbar^2}{2m}\nabla^2\psi + V\psi, \quad (5.28)$$

the probability density and current are

$$\rho = |\psi|^2, \quad \mathbf{j} = \frac{\mathrm{i}\hbar}{2m}(\psi\nabla\psi^* - \psi^*\nabla\psi) = \frac{\hbar}{m}\mathrm{Im}(\psi^*\nabla\psi). \quad (5.29)$$

They satisfy

$$\boxed{\partial_t\rho + \nabla \cdot \mathbf{j} = 0.} \quad (5.30)$$

Hence

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_R \rho \mathrm{d}^3r = - \oint_{\partial R} \mathbf{j} \cdot \mathrm{d}\mathbf{S}. \quad (5.31)$$

For charge e particles, the electric charge density and electric current are given by

$$\rho_e = e\rho, \quad \mathbf{j}_e = e\mathbf{j}. \quad (5.32)$$

5.2.2 Gauge fields and minimal coupling

The electromagnetic fields are

$$\mathbf{B} = \nabla \times \mathbf{A}, \quad \mathbf{E} = -\nabla\phi - \partial_t\mathbf{A}. \quad (5.33)$$

Gauge transformations are

$$\phi \mapsto \phi + \partial_t f, \quad \mathbf{A} \mapsto \mathbf{A} - \nabla f. \quad (5.34)$$

The Hamiltonian is

$$\boxed{\hat{H} = \frac{[\hat{\mathbf{p}} - e\mathbf{A}(\hat{\mathbf{r}}, t)]^2}{2m} + e\phi(\hat{\mathbf{r}}, t).} \quad (5.35)$$

Gauge invariance requires

$$\psi \mapsto e^{-\mathrm{i}ef/\hbar}\psi, \quad \psi^* \mapsto e^{\mathrm{i}ef/\hbar}\psi^*. \quad (5.36)$$

The canonical momentum is not gauge invariant:

$$\hat{\mathbf{p}} = -\mathrm{i}\hbar\nabla \mapsto \hat{\mathbf{p}} - e\nabla f. \quad (5.37)$$

Instead the kinetic momentum (also known as mechanical momentum)

$$\boxed{\hat{\boldsymbol{\pi}} = \hat{\mathbf{p}} - e\mathbf{A}} \quad (5.38)$$

is gauge invariant, and different components of the kinetic momenta generally don't commute:

$$\boxed{[\hat{\pi}_a, \hat{\pi}_b] = \mathrm{i}\hbar e\epsilon_{abc}B_c.} \quad (5.39)$$

5.2.3 Ehrenfest equations

In the Heisenberg picture,

$$\mathrm{i}\hbar\frac{\mathrm{d}\hat{O}}{\mathrm{d}t} = [\hat{O}, \hat{H}]. \quad (5.40)$$

For the position operator,

$$\frac{d\langle \hat{\mathbf{r}} \rangle}{dt} = \frac{\langle \hat{\boldsymbol{\pi}} \rangle}{m}. \quad (5.41)$$

For the kinetic momentum,

$$\frac{d\langle \hat{\boldsymbol{\pi}} \rangle}{dt} = e\langle \mathbf{E} \rangle + \frac{e}{2m} \langle \hat{\boldsymbol{\pi}} \times \mathbf{B} - \mathbf{B} \times \hat{\boldsymbol{\pi}} \rangle. \quad (5.42)$$

For a uniform and static electromagnetic field, we therefore have the following equation of motion:

$$\boxed{m \frac{d^2 \langle \hat{\mathbf{r}} \rangle}{dt^2} = e\mathbf{E} + e \frac{d\langle \hat{\mathbf{r}} \rangle}{dt} \times \mathbf{B}.} \quad (5.43)$$

5.2.4 Solving Schrodinger equation in a uniform magnetic field

5.2.4.1 Landau gauge

Let

$$\mathbf{B} = (0, 0, B), \quad \mathbf{A} = (0, Bx, 0). \quad (5.44)$$

Neglecting free motion along z ,

$$\hat{H} = \frac{\hat{p}_x^2 + (\hat{p}_y - eB\hat{x})^2}{2m}. \quad (5.45)$$

Since

$$[\hat{p}_y, \hat{H}] = 0, \quad (5.46)$$

write

$$\hat{p}_y = \hbar k_y. \quad (5.47)$$

Then

$$\hat{H} = \frac{\hat{p}_x^2}{2m} + \frac{(eB)^2}{2m} \left(\hat{x} - \frac{\hbar k_y}{eB} \right)^2. \quad (5.48)$$

Define

$$\omega_c = \frac{|eB|}{m}, \quad \ell_B = \sqrt{\frac{\hbar}{|eB|}}. \quad (5.49)$$

The Landau levels are

$$\boxed{E_n = \hbar\omega_c \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, \dots} \quad (5.50)$$

The eigenstates are

$$\psi_{n,k_y}(x, y) \propto e^{ik_y y} \varphi_n \left(x - \frac{\hbar k_y}{eB} \right). \quad (5.51)$$

For $n = 0$,

$$\psi_{0,k_y}(x, y) \propto e^{ik_y y} \exp \left[-\frac{(x - \hbar k_y / eB)^2}{2\ell_B^2} \right]. \quad (5.52)$$

5.2.4.2 Landau-level degeneracy

With periodic boundary conditions along y ,

$$k_y = \frac{2\pi j}{L_y}. \quad (5.53)$$

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The guiding-center spacing is

$$\Delta X = \frac{\hbar \Delta k_y}{|eB|} = \frac{2\pi\hbar}{|eB|L_y}. \quad (5.54)$$

The degeneracy in area $L_x L_y$ is

$$D = \frac{|eB|L_x L_y}{2\pi\hbar} = \frac{\Phi}{\Phi_0}, \quad (5.55)$$

where

$$\Phi_0 = \frac{2\pi\hbar}{|e|} \quad (5.56)$$

is the flux quantum.

5.2.4.3 Filling factor

The filling factor is defined as the ratio of total electron number and Landau level degeneracy:

$$\nu = \frac{N_e}{D} = \frac{\rho_e \Phi_0}{B}. \quad (5.57)$$

Physical observables such as conductivity and magnetization oscillate with the magnetic field, known as the quantum oscillations. The quantum oscillations are periodic in $1/B$:

$$\Delta \left(\frac{1}{B} \right) = \frac{1}{\rho_e \Phi_0}. \quad (5.58)$$

5.3 Nonintegrable phase factors (11/18)

5.3.1 A review of adiabatic evolution and Berry's phase

Let

$$\hat{H}(\lambda)|n, \lambda\rangle = E_n(\lambda)|n, \lambda\rangle. \quad (5.59)$$

In the adiabatic limit,

$$|\psi(t)\rangle = \sum_n C_n(t) \exp \left[-\frac{i}{\hbar} \int_0^t E_n(t') dt' \right] |n, t\rangle, \quad (5.60)$$

where

$$\dot{C}_n(t) = -C_n(t) \langle n, t | \frac{d}{dt} | n, t \rangle. \quad (5.61)$$

Thus

$$C_n(t) = C_n(0) \exp \left[-i \int_{\lambda(0)}^{\lambda(t)} \mathcal{A}_n(\lambda) d\lambda \right], \quad (5.62)$$

with the *Berry connection* defined as

$$\boxed{\mathcal{A}_n(\lambda) = -i \langle n, \lambda | \partial_\lambda | n, \lambda \rangle.} \quad (5.63)$$

Since $\langle n, \lambda | n, \lambda \rangle = 1$,

$$\mathcal{A}_n^*(\lambda) = \mathcal{A}_n(\lambda). \quad (5.64)$$

For several slowly-varying parameters $\{\lambda_j\}$, there can be more than one component of Berry connection

$$\mathcal{A}_{n,j} = -i \langle n | \partial_{\lambda_j} | n \rangle. \quad (5.65)$$

and

$$C_n(t) = C_n(0) \exp \left[-i \sum_j \int_{\lambda(0)}^{\lambda(t)} \mathcal{A}_{n,j}(\lambda) d\lambda_j \right], \quad (5.66)$$

Notice that the phase factor in $C_n(t)$ depends not only on the endpoints $\lambda(0), \lambda(t)$, but also on the full trajectory in parameter space. Therefore it is a non-integrable phase factor.

5.3.2 Gauge redundancy

Under

$$|n, \lambda\rangle \mapsto |\tilde{n}, \lambda\rangle = e^{if(\lambda)} |n, \lambda\rangle, \quad (5.67)$$

the Berry connection transforms as

$$\boxed{\tilde{\mathcal{A}}_n(\lambda) = \mathcal{A}_n(\lambda) + \frac{df}{d\lambda}.} \quad (5.68)$$

For an open path,

$$e^{i\gamma_n} = \exp \left[-i \int_{\lambda_0}^{\lambda_1} \mathcal{A}_n(\lambda) d\lambda \right] \quad (5.69)$$

changes by an endpoint phase difference $f(\lambda_1) - f(\lambda_0)$. For a closed contour C ,

$$\boxed{e^{i\gamma_n(C)} = \exp \left[-i \oint_C \mathcal{A}_n(\lambda) d\lambda \right]} \quad (5.70)$$

is gauge invariant.

5.3.3 Aharonov–Bohm phase

Consider

$$\hat{H}(\boldsymbol{\lambda}) = \frac{(\hat{\mathbf{p}} - e\mathbf{A})^2}{2m} + e\phi(\hat{\mathbf{r}}) + V(\hat{\mathbf{r}} - \boldsymbol{\lambda}), \quad (5.71)$$

where V traps the particle near $\boldsymbol{\lambda}$. For a slow displacement $\boldsymbol{\lambda} \rightarrow \boldsymbol{\lambda} + \boldsymbol{\delta}$,

$$|n, \boldsymbol{\lambda} + \boldsymbol{\delta}\rangle \simeq e^{-i\hat{\boldsymbol{\pi}} \cdot \boldsymbol{\delta}/\hbar} |n, \boldsymbol{\lambda}\rangle, \quad \hat{\boldsymbol{\pi}} = \hat{\mathbf{p}} - e\mathbf{A}. \quad (5.72)$$

Hence

$$-i\langle n, \boldsymbol{\lambda} | \frac{\partial}{\partial \lambda_j} | n, \boldsymbol{\lambda} \rangle = \frac{1}{\hbar} \langle n | \hat{\pi}_j | n \rangle \simeq -\frac{e}{\hbar} A_j(\boldsymbol{\lambda}). \quad (5.73)$$

The geometric phase is therefore

$$e^{i\gamma} = \exp \left[\frac{ie}{\hbar} \int_{\boldsymbol{\lambda}(0)}^{\boldsymbol{\lambda}(t)} \mathbf{A}(\mathbf{r}) \cdot d\mathbf{r} \right]. \quad (5.74)$$

Under

$$\mathbf{A} \mapsto \mathbf{A} - \nabla f, \quad \psi \mapsto e^{-ief/\hbar} \psi, \quad (5.75)$$

the induced endpoint phase agrees with the gauge transformation of the wave function.

For a closed contour C ,

$$e^{i\gamma_{AB}} = \exp \left[\frac{ie}{\hbar} \oint_C \mathbf{A} \cdot d\mathbf{r} \right] = \exp \left(2\pi i \frac{\Phi}{\Phi_0} \right), \quad (5.76)$$

where

$$\Phi_0 = \frac{2\pi\hbar}{e} = \frac{h}{e} \quad (5.77)$$

is the flux quantum.

5.3.4 Particle on a ring

For a particle on a ring of radius R threaded by flux Φ ,

$$\hat{H} = \frac{(\hat{p} - eA)^2}{2m} = \frac{\hbar^2}{2mR^2} (-i\partial_\theta - \phi)^2, \quad (5.78)$$

where

$$\phi = \frac{eAR}{\hbar} = \frac{\Phi}{\Phi_0}. \quad (5.79)$$

Periodic boundary conditions give

$$\psi_n(\theta) = \frac{e^{in\theta}}{\sqrt{2\pi}}, \quad n \in \mathbb{Z}, \quad (5.80)$$

and

$$E_n(\phi) = \frac{\hbar^2}{2mR^2} (n - \phi)^2. \quad (5.81)$$

Therefore,

$$E_{n+1}(\phi + 1) = E_n(\phi), \quad (5.82)$$

so the spectrum is periodic in Φ with period Φ_0 .

For two spinless electrons and $0 \leq \phi \leq 1$, the total energy is given by

$$U(\phi) = E_0(\phi) + E_1(\phi) = \frac{\hbar^2}{2mR^2} [\phi^2 + (\phi - 1)^2]. \quad (5.83)$$

The magnetization is

$$M = -\frac{\partial U}{\partial \Phi} = \frac{\hbar^2}{mR^2\Phi_0}(1 - 2\phi). \quad (5.84)$$

It is a periodic function of ϕ with a periodicity of 1.

5.3.5 Dirac quantization

Cover a sphere surrounding a magnetic monopole by northern and southern patches R and $\bar{R}$, with common boundary C . Stokes' theorem gives

$$\oint_C \mathbf{A}_R \cdot d\mathbf{r} = \int_R \mathbf{B} \cdot d\mathbf{a}, \quad (5.85)$$

and

$$\oint_C \mathbf{A}_{\bar{R}} \cdot d\mathbf{r} = - \int_{\bar{R}} \mathbf{B} \cdot d\mathbf{a}. \quad (5.86)$$

Single-valuedness of the phase requires

$$\frac{e}{\hbar} (\Phi_R + \Phi_{\bar{R}}) = 2\pi N, \quad N \in \mathbb{Z}. \quad (5.87)$$

Thus the total monopole flux is quantized:

$$\Phi_{S^2} = N\Phi_0, \quad N \in \mathbb{Z}. \quad (5.88)$$

5.4 Interaction Picture and Rabi Oscillation (11/20)

5.4.1 The interaction picture

Let

$$\hat{H}(t) = \hat{H}_0 + \hat{V}(t), \quad \hat{H}_0|a\rangle = E_a|a\rangle. \quad (5.89)$$

The Schrodinger-picture evolution operator satisfies

$$i\hbar \frac{\partial}{\partial t} \hat{U}(t, t_0) = \hat{H}(t) \hat{U}(t, t_0). \quad (5.90)$$

Define

$$|\psi_I(t)\rangle = e^{i\hat{H}_0 t/\hbar} |\psi_S(t)\rangle, \quad (5.91)$$

$$\hat{U}_I(t, t_0) = e^{i\hat{H}_0 t/\hbar} \hat{U}(t, t_0) e^{-i\hat{H}_0 t_0/\hbar}, \quad (5.92)$$

$$\hat{V}_I(t) = e^{i\hat{H}_0 t/\hbar} \hat{V}(t) e^{-i\hat{H}_0 t/\hbar}. \quad (5.93)$$

Then

$$i\hbar \frac{d}{dt} |\psi_I(t)\rangle = \hat{V}_I(t) |\psi_I(t)\rangle, \quad (5.94)$$

$$i\hbar \frac{\partial}{\partial t} \hat{U}_I(t, t_0) = \hat{V}_I(t) \hat{U}_I(t, t_0). \quad (5.95)$$

An interaction-picture operator is

$$\hat{O}_I(t) = e^{i\hat{H}_0 t/\hbar} \hat{O}_S e^{-i\hat{H}_0 t/\hbar}, \quad (5.96)$$

with

$$i\hbar \frac{d\hat{O}_I}{dt} = [\hat{O}_I, \hat{H}_0]. \quad (5.97)$$

5.4.1.1 Dyson equation and Dyson series

Integrating the evolution equation gives the Dyson equation

$$\boxed{\hat{U}_I(t, t_0) = \mathbb{I} - \frac{i}{\hbar} \int_{t_0}^t \hat{V}_I(t') \hat{U}_I(t', t_0) dt'.} \quad (5.98)$$

Iteration yields the Dyson series:

$$\hat{U}_I(t, t_0) = \mathbb{I} - \frac{i}{\hbar} \int_{t_0}^t \hat{V}_I(t_1) dt_1 + \left(-\frac{i}{\hbar}\right)^2 \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \hat{V}_I(t_1) \hat{V}_I(t_2) + \dots \quad (5.99)$$

$$= \mathcal{T} \exp \left[-\frac{i}{\hbar} \int_{t_0}^t \hat{V}_I(t') dt' \right]. \quad (5.100)$$

Finally we mention that the Gellmann-Low theorem introduced in section 3.4.3.2 can be derived from Dyson's equation and the Lippmann-Schwinger equation.

5.4.2 Rabi Oscillations

5.4.2.1 Spin one-half in a time-dependent magnetic field

Let

$$\hat{H}(t) = -\boldsymbol{\mu} \cdot \mathbf{B}(t), \quad \boldsymbol{\mu} = \frac{ge}{2m_e} \mathbf{S}, \quad (5.101)$$

and

$$\mathbf{B}(t) = B_0 \hat{\mathbf{z}} + B_1 (\hat{\mathbf{x}} \cos \omega t - \hat{\mathbf{y}} \sin \omega t). \quad (5.102)$$

Write

$$\hat{H}(t) = \hat{H}_0 + \hat{V}(t), \quad \hat{H}_0 = -\frac{\hbar \omega_{12}}{2} \sigma_z, \quad (5.103)$$

where

$$\omega_{12} = \frac{geB_0}{2m_e}, \quad \gamma = \frac{geB_1}{4m_e} \quad (5.104)$$

Let

$$\hat{V}(t) = \hbar \gamma (\sigma_x \cos \omega t - \sigma_y \sin \omega t) = \hbar \gamma \begin{pmatrix} 0 & e^{i\omega t} \\ e^{-i\omega t} & 0 \end{pmatrix}. \quad (5.105)$$

In the interaction picture,

$$\hat{V}_I(t) = -\hbar \gamma \begin{pmatrix} 0 & e^{i(\omega - \omega_{12})t} \\ e^{-i(\omega - \omega_{12})t} & 0 \end{pmatrix}. \quad (5.106)$$

Our goal is to solve the time dependence of the 2-component wavefunction:

$$|\psi_I(t)\rangle = C_1(t)|\uparrow\rangle + C_2(t)|\downarrow\rangle. \quad (5.107)$$

Then

$$i\dot{C}_1 = -\gamma e^{i(\omega - \omega_{12})t} C_2, \quad (5.108)$$

$$i\dot{C}_2 = -\gamma e^{-i(\omega - \omega_{12})t} C_1. \quad (5.109)$$

With detuning

$$\Delta = \omega - \omega_{12}, \quad (5.110)$$

one obtains

$$\ddot{C}_1 - i\Delta \dot{C}_1 + \gamma^2 C_1 = 0. \quad (5.111)$$

The characteristic frequencies are

$$\Omega_{\pm} = \frac{\Delta}{2} \pm \sqrt{\gamma^2 + \frac{\Delta^2}{4}}. \quad (5.112)$$

Define the generalized Rabi frequency

$$\boxed{\Omega_R = \sqrt{\gamma^2 + \frac{\Delta^2}{4}}}. \quad (5.113)$$

For

$$C_1(0) = 1, \quad C_2(0) = 0, \quad (5.114)$$

the transition probability is

$$\boxed{|C_2(t)|^2 = \frac{\gamma^2}{\gamma^2 + \Delta^2/4} \sin^2(\Omega_R t)}. \quad (5.115)$$

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The survival probability is

$$|C_1(t)|^2 = 1 - |C_2(t)|^2. \quad (5.116)$$

At resonance,

$$\omega = \omega_{12}, \quad \Delta = 0 \quad (5.117)$$

we have

$$\boxed{|C_2(t)|^2 = \sin^2(\gamma t), \quad |C_1(t)|^2 = \cos^2(\gamma t).} \quad (5.118)$$

5.4.2.2 Nuclear magnetic resonance

The resonance condition is

$$\boxed{\omega_{\text{res}} = \omega_{12} = \mu B_0.} \quad (5.119)$$

In an NMR experiment,

$$B_0 = H + M = H + \chi H. \quad (5.120)$$

The magnetization and susceptibility are

$$M = \frac{\omega_{\text{res}}}{\mu} - H, \quad \chi = \frac{M}{H} = \frac{\omega_{\text{res}}}{\mu H} - 1. \quad (5.121)$$

5.5 *Electrons in a uniform electric field (12/2)

5.5.1 Solving Schrodinger equation in a uniform electric field

Let the electric field be $\mathcal{E}\hat{x}$ and define the electric force

$$F = e\mathcal{E}. \quad (5.122)$$

for an electron of charge e . In one dimension, the electric field $\mathcal{E}$ is related to scalar and vector potentials by

$$\mathcal{E} = -\nabla\phi - \partial_t A \quad (5.123)$$

5.5.1.1 Gauge 1: time-dependent vector-potential gauge

In one dimension,

$$\hat{H} = \frac{[-i\hbar\partial_x - eA(x,t)]^2}{2m} + e\phi(x,t). \quad (5.124)$$

Choose the following gauge:

$$\phi_1 = 0, \quad A_1 = -\mathcal{E}t. \quad (5.125)$$

Then

$$i\hbar \frac{d}{dt}|\psi\rangle = \frac{(\hat{p} + e\mathcal{E}t)^2}{2m}|\psi\rangle. \quad (5.126)$$

For initial state

$$\psi(x, 0) = e^{ikx}, \quad (5.127)$$

one obtains

$$\boxed{\psi(x, t) = \exp \left[ikx - \frac{i}{6me\mathcal{E}\hbar} ((\hbar k + e\mathcal{E}t)^3 - (\hbar k)^3) \right]}. \quad (5.128)$$

5.5.1.2 Gauge 2: scalar-potential gauge and Airy functions

Alternatively we can choose

$$\phi_2 = -\mathcal{E}x, \quad A_2 = 0. \quad (5.129)$$

Then

$$\hat{H} = \frac{\hat{p}^2}{2m} - e\mathcal{E}\hat{x}. \quad (5.130)$$

The stationary equation is

$$-\frac{\hbar^2}{2m}\psi''(x) - e\mathcal{E}x\psi(x) = E\psi(x). \quad (5.131)$$

With

$$u = \left(\frac{2m|e|\mathcal{E}}{\hbar^2} \right)^{1/3} \left(x - \frac{E}{e\mathcal{E}} \right), \quad (5.132)$$

one obtains the Airy equation

$$\frac{d^2\psi}{du^2} = u\psi. \quad (5.133)$$

The decaying solution is

$$\boxed{\text{Ai}(u) = \frac{1}{\pi} \int_0^\infty \cos \left(\frac{t^3}{3} + ut \right) dt.} \quad (5.134)$$

Its asymptotic forms are

$$\text{Ai}(u) \sim \frac{1}{2\sqrt{\pi}} u^{-1/4} e^{-2u^{3/2}/3}, \quad u \gg 1, \quad (5.135)$$

$$\text{Ai}(u) \sim \frac{1}{\sqrt{\pi}} |u|^{-1/4} \cos\left(\frac{2}{3}|u|^{3/2} - \frac{\pi}{4}\right), \quad u \ll -1. \quad (5.136)$$

The spectrum is continuous.

5.5.1.3 Relation between the two gauges

Use the gauge convention

$$A' = A - \partial_x f, \quad \phi' = \phi + \partial_t f, \quad \psi' = e^{-iqf/\hbar} \psi. \quad (5.137)$$

The choice

$$\boxed{f(x, t) = -\mathcal{E}tx} \quad (5.138)$$

transforms Gauge 1 into Gauge 2:

$$A_2 = A_1 - \partial_x f = 0, \quad \phi_2 = \phi_1 + \partial_t f = -\mathcal{E}x. \quad (5.139)$$

Therefore,

$$\boxed{\psi_2(x, t) = e^{iFtx/\hbar} \psi_1(x, t).} \quad (5.140)$$

The Hamiltonians satisfy

$$\hat{H}_2 = \hat{U}_f \hat{H}_1 \hat{U}_f^\dagger + i\hbar \dot{\hat{U}}_f \hat{U}_f^\dagger, \quad \hat{U}_f = e^{iFt\hat{x}/\hbar}. \quad (5.141)$$

The transformed plane-wave solution in gauge 2 is

$$\boxed{\psi_2(x, t) = C \exp\left[i\left(k + \frac{Ft}{\hbar}\right)x - \frac{i}{2m\hbar} \int_0^t (\hbar k + F\tau)^2 d\tau\right]}. \quad (5.142)$$

It is a superposition of stationary Airy eigenstates $\{|\varphi_E\rangle\}$:

$$\psi_2(x, t) = \int_{-\infty}^{\infty} c(E) e^{-iEt/\hbar} \varphi_E(x) dE. \quad (5.143)$$

5.5.1.4 Density and electric current in the two gauges

For the plane-wave solution in Gauge 1, the charge density and electric currents are

$$\rho_{q,1} = e|C|^2, \quad (5.144)$$

$$j_{q,1} = \frac{e|C|^2}{m} (\hbar k + Ft). \quad (5.145)$$

For the gauge-transformed solution in Gauge 2, we also have

$$\rho_{q,2} = e|C|^2, \quad (5.146)$$

$$j_{q,2} = \frac{e|C|^2}{m} (\hbar k + Ft). \quad (5.147)$$

Hence

$$\boxed{\rho_{q,1} = \rho_{q,2}, \quad j_{q,1} = j_{q,2}.} \quad (5.148)$$

A real stationary Airy state $|\varphi_E\rangle$ in Gauge 2 is

$$\psi_{2,E}(x, t) = e^{-iEt/\hbar} \varphi_E(x), \quad \varphi_E(x) \in \mathbb{R}. \quad (5.149)$$

Its charge density and current are

$$\rho_{q,2}^{(E)} = e|\varphi_E(x)|^2, \quad \boxed{j_{q,2}^{(E)} = 0.} \quad (5.150)$$

The corresponding Gauge-1 state is

$$\boxed{\psi_{1,E}(x, t) = e^{-iFtx/\hbar} e^{-iEt/\hbar} \varphi_E(x).} \quad (5.151)$$

For this state,

$$\rho_{q,1}^{(E)} = e|\varphi_E(x)|^2, \quad \boxed{j_{q,1}^{(E)} = 0.} \quad (5.152)$$

The phase gradient contributes canonical momentum $-Ft$, while the vector potential contributes $+Ft$ to the kinetic momentum, so the physical current vanishes in both gauges.

5.5.2 Landau–Zener Tunneling in solids

5.5.2.1 Band crossing

Consider

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}) + |e|\mathcal{E}\hat{x}, \quad V(x+a) = V(x). \quad (5.153)$$

Near a valence-conduction avoided crossing, the conduction and valence bands can be modeled by a two-level system, where the energy of conduction band $|c\rangle$ is higher than the valence band $|v\rangle$ by the band gap $E_g > 0$:

$$\hat{H} \simeq \begin{pmatrix} E_g/2 + \hbar^2 k^2/(2m_v) & \Delta \\ \Delta^* & -E_g/2 - \hbar^2 k^2/(2m_c) \end{pmatrix}. \quad (5.154)$$

Δ models the tunneling term between the two bands.

The electric field gives

$$\hbar \frac{dk}{dt} \simeq e\mathcal{E}, \quad k(t) = k_0 + \frac{e\mathcal{E}}{\hbar} t. \quad (5.155)$$

5.5.2.2 Landau–Zener model

In a semiconductor, we expand the wavevector k around the smallest direct gap location k_0 , and we arrive at the Landau-Zener model

$$\hat{H}_{\text{LZ}}(t) = \begin{pmatrix} \alpha t & \Delta \\ \Delta & -\alpha t \end{pmatrix}, \quad |\psi(t)\rangle = \begin{pmatrix} C_1(t) \\ C_2(t) \end{pmatrix}. \quad (5.156)$$

where

$$\alpha = |e|\mathcal{E}v_F/\hbar, \quad v_F = \frac{\hbar k_0}{m^*}, \quad \frac{1}{m^*} = \frac{1}{m_c} + \frac{1}{m_v}. \quad (5.157)$$

The 2-component Schrodinger equation is

$$i\hbar \dot{C}_1 = \alpha t C_1 + \Delta C_2, \quad (5.158)$$

$$i\hbar \dot{C}_2 = \Delta C_1 - \alpha t C_2. \quad (5.159)$$

Eliminating C_1 gives

$$-\hbar^2 \ddot{C}_2 = (\alpha^2 t^2 + \Delta^2 - i\hbar\alpha) C_2. \quad (5.160)$$

5.5.2.3 Computing the tunneling probability

Define

$$z = e^{-i\pi/4} \sqrt{\frac{2\alpha}{\hbar}} t, \quad \delta = \frac{\Delta^2}{2\hbar\alpha}. \quad (5.161)$$

Then

$$\boxed{\frac{d^2 C_2}{dz^2} + \left(\nu + \frac{1}{2} - \frac{z^2}{4} \right) C_2 = 0, \quad \nu = i\delta.} \quad (5.162)$$

This is Weber's equation. Its solutions are parabolic-cylinder functions $D_\nu(z)$, also called Weber functions:

$$\boxed{C_2(z) = AD_{i\delta}(z) + BD_{i\delta}(-z).} \quad (5.163)$$

The remaining amplitude follows from

$$\boxed{C_1(t) = \frac{i\hbar\dot{C}_2 + \alpha t C_2}{\Delta}.} \quad (5.164)$$

If the initial state is $|\psi(t = -\infty)\rangle = (0, 1)^T$

$$C_2(-\infty) = 1, \quad C_1(-\infty) = 0, \quad (5.165)$$

then

$$|C_2(+\infty)|^2 = e^{-2\pi\Delta^2/(\hbar\alpha)} \equiv P_{\text{diabatic}}. \quad (5.166)$$

Hence the transition probability is

$$\boxed{P_{1 \rightarrow 2} = 1 - e^{-2\pi\Delta^2/(\hbar\alpha)}.} \quad (5.167)$$

The diabatic probability (Zener excitation probability) P_{diabatic} increases, as parameter α increases.

In other words, as the electric field $\mathcal{E}$ (and hence α) increases, there is a larger chance for electrons to diabatically transfer from the lower-energy valence band to higher-energy conduction band.

A Homework 1 (due 9/4)

A.1 Franck–Hertz experiment

In the Franck-Hertz experiment (see Fig. 1.2 in lecture notes), the sharp drop of electric current through the vacuum tube at a voltage of 4.9 V corresponds to the absorption of electron energy by the mercury atoms in the vacuum tube, where the atomic energy level of mercury atoms go through a transition. Compute the wavelength of light associated with this atomic energy level transition, show that it is approximately 254 nm, as observed in optical emission of the same vacuum tube.

A.2 Davisson–Germer experiment

(1) *Bragg's law* Consider the diffraction of a wave by a lattice of atoms with a lattice constant of d between layers. Derive the Bragg's law

$$n\lambda = 2d \cos \frac{\theta}{2}, \quad n = 0, 1, 2, 3, \dots \quad (\text{A.1})$$

for the diffraction pattern.

(2) Consider the Davisson-Germer experiment illustrated in Fig. 1.5 of overleaf lecture notes. For $\theta = 50^\circ$, $d = 0.091$ nm for nickel crystals, and an electron beam accelerated by a $V = 54$ volts voltage, show that it satisfies the Bragg's law with $n = 1$.

In part (1) and (2), we have dealt with a simplified version of the actual Davisson-Germer experiment[3], assuming the electron beam was scattered by a three-dimensional simple cubic lattice. Below we will seriously consider the actual situation of Davisson-Germer experiment in history, where an electron beam was scattered by the (111) surface of a nickel crystal in which Ni atoms form a face-centered cubic (FCC) lattice. In this case, we need to first derive the general Bragg condition for two-dimensional diffraction peaks and apply it to the setup of Davisson-Germer experiment. Below we carry out this calculation step by step.

(3) **Bonus** For electrons of energy 54 eV in the Davisson-Germer experiment, their penetration depth into the nickel crystal is less than 1 nanometer, comparable to the FCC lattice constant $a = 3.52$ Å. This means as a zeroth order approximation, we can model the Davisson-Germer experiment as electrons scattered by the top layer of Ni atoms parallel to the crystal surface. In Davisson-Germer experiment the incident beam of electrons is perpendicular to the cleaved (111) surface of the nickel crystal. Can you draw the two-dimensional lattice formed on the (111) surface of a nickel crystal? Given the FCC lattice constant $a = 3.52$ Å in nickel crystals, what is the distance $\tilde{a}$ between two nearest Ni atoms on its (111) surface?

(4) **Bonus** For the two-dimensional (2d) plane parallel to the (111) surface, we can choose the following two orthonormal vectors as two basis vectors for the 2d plane:

$$\hat{e}_1 = (-1, 1, 0)/\sqrt{2}, \quad \hat{e}_2 = (-1, -1, 2)/\sqrt{6}, \quad \hat{e}_1 \times \hat{e}_2 = (1, 1, 1)/\sqrt{3}. \quad (\text{A.2})$$

In this two-dimensional lattice on (111) surface, the periodic pattern of Ni atoms are described

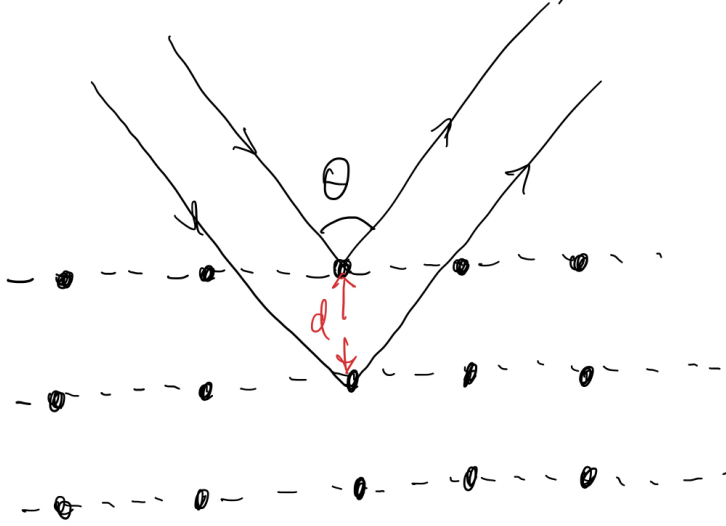


Figure A.1: Scattering of a wave by a lattice of atoms with a lattice constant d .

by two primitive vectors

$$\vec{a}_1 = \tilde{a}\hat{e}_1, \quad \vec{a}_2 = \tilde{a}\frac{-\hat{e}_1 + \sqrt{3}\hat{e}_2}{2}. \quad (\text{A.3})$$

where $\tilde{a}$ is the distance $\tilde{a}$ between two nearest Ni atoms on its (111) surface. They are called primitive vectors in the sense that any lattice site position $\vec{R}$ in the 2d lattice is a summation of integer multiples of $\vec{a}_i$

$$\vec{R} = \sum_{j=1}^2 m_j \vec{a}_j, \quad m_j \in \mathbb{Z}. \quad (\text{A.4})$$

One can also define *reciprocal lattice primitive vectors* $\vec{b}_{1,2}$ by the following conditions

$$\vec{a}_i \cdot \vec{b}_j = 2\pi\delta_{i,j} \quad (\text{A.5})$$

Note that $\vec{b}_{1,2}$ are both perpendicular to (1,1,1) as well.

For $\vec{a}_{1,2}$ defined in (A.3), express $\vec{b}_{1,2}$ using $\tilde{a}$ and $\hat{e}_{1,2}$.

Most generally, the Bragg condition for diffraction peaks is given by

$$\sum_{\vec{R}} e^{i\vec{q} \cdot \vec{R}} \neq 0 \iff \vec{q} \cdot \vec{a}_j = 0 \pmod{2\pi}, \quad \forall j = 1, 2. \quad (\text{A.6})$$

where $\vec{q}$ is the in-plane component of electron momentum transfer before and after scattered by the nickel crystal:

$$\vec{q} \equiv (\vec{k}_{\text{out}} - \vec{k}_{\text{in}}) - \frac{(1, 1, 1)}{3}(\vec{k}_{\text{out}} - \vec{k}_{\text{in}}) \cdot (1, 1, 1). \quad (\text{A.7})$$

In later lectures we will come back to (A.6) when discussing Fourier transform as a change of

basis (i.e. a unitary transformation). **From (A.6), show that**

$$\vec{q} = \sum_{j=1}^2 n_j \vec{b}_j, \quad n_j \in \mathbb{Z} \quad (\text{A.8})$$

(5) **Bonus** In the Davisson-Germer experiment, the incident momentum $\vec{k}_{\text{in}}$ is perpendicular to the (111) surface i.e. $\vec{k}_i = \frac{2\pi}{\lambda}(1, 1, 1)/\sqrt{3}$ where λ is the wavelength of electrons with a kinetic energy of 54 eV. For elastic scatterings responsible for the diffraction pattern, we have

$$|\vec{k}_i| = |\vec{k}_f| = \frac{2\pi}{\lambda} \quad (\text{A.9})$$

In the experimental setup of Fig. 1.5 (left), using results (A.7)-(A.8) of the previous part, derive the following Bragg's law for Davisson-Germer experiment:

$$\lambda \sqrt{n_1^2 + n_1 n_2 + n_2^2} = \sqrt{\frac{3}{8}} a \sin \theta, \quad n_{1,2} \in \mathbb{Z} \quad (\text{A.10})$$

Comment: specifically, if we choose $(n_1, n_2) = n(1, 0)$ or $n(0, 1)$ or $n(1, -1)$, Eq. (A.10) reduces to the more familiar one

$$n\lambda = d_{(111)} \sin \theta, \quad n \in \mathbb{Z}, \quad d_{(111)} = a\sqrt{3/8} \approx 2.16 \text{ \AA}. \quad (\text{A.11})$$

(6) **Bonus** Note that the in-plane momentum transfer $\vec{q}$ can be written as

$$\vec{q} = |q|(\cos \phi \hat{e}_1 + \sin \phi \hat{e}_2) \quad (\text{A.12})$$

where ϕ is the azimuth angle in Fig. 1.5 of the lecture notes. Use relations (A.7)-(A.9) to explain the ϕ -dependent peaks in the right panel of Fig. 1.5.

(7) **Bonus** Can you qualitatively explain in Fig. 1.5 right panel, why are the peak heights at $\phi = \frac{\pi}{2} \bmod 2\pi/3$ are different from those at $\phi = \frac{\pi}{6} \bmod 2\pi/3$?

A.3 Hilbert space of linear operators

(1) Consider linear operators on complex vector space $\mathbb{V} = \mathbb{C}^n$, i.e. $n \times n$ complex matrices. Prove that they form a vector space, which we label as $\mathbb{L}_{\mathbb{V}}$. For the definition of a vector space, refer to Chapter 1.1 of Shankar.

(2) Prove that the following binary operation $(\cdot, \cdot)$ on $\mathbb{L}_{\mathbb{V}} \times \mathbb{L}_{\mathbb{V}}$ defined by

$$(A, B) \equiv \text{tr}(A^\dagger B), \quad A, B \in \mathbb{L}_{\mathbb{V}} \quad (\text{A.13})$$

is an inner product. For the definition of an inner product, refer to Shankar 1.2 or N&C 2.1.4. This is known as the Hilbert-Schmidt inner product on operators.

(3) Using the above Hilbert-Schmidt inner product, show that 3 Pauli matrices $\sigma_x, \sigma_y, \sigma_z$ defined below

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (\text{A.14})$$

and the 2×2 identity matrix $\hat{1}$ together can form a complete orthonormal basis for the vector space $\mathbb{L}_{\mathbb{C}^2}$ of all 2×2 complex matrices.

(4) **Bonus** Prove the following *completeness relation of Pauli matrices*:

$$\sum_{i=x,y,z} (\sigma_i)_{a,b} (\sigma_i)_{c,d} = \vec{\sigma}_{a,b} \cdot \vec{\sigma}_{c,d} = 2\delta_{a,d}\delta_{b,c} - \delta_{a,b}\delta_{c,d} \quad (\text{A.15})$$

Hint: use the conclusion of part (3), i.e. any 2×2 Hermitian matrix $\hat{M}$ can be written as

$$\hat{M} = \frac{\text{Tr}(\hat{M})}{2} \hat{1}_{2 \times 2} + \sum_{a=x,y,z} \frac{\text{Tr}(\hat{M}\sigma_a)}{2} \sigma_a \quad (\text{A.16})$$

A.4 Compton shift

Consider the Compton effect discussed in lecture notes (Fig. 1.1), where an X -ray photon was scattered by a stationary electron.

- (1) Write down the momenta conservation and energy conservation formula.
- (2) Use energy and momentum conservations to derive the Compton shift formula:

$$\lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \theta) \quad (\text{A.17})$$

A.5 Quantization of angular momentum

(1) Consider the circular motion of an electron around the nucleus of hydrogen atom. Using Bohr's quantization condition of the angular momentum

$$mvr = n\hbar \quad (\text{A.18})$$

derive the formula of quantized energy

$$E_n = -\frac{mk^2 e^4}{2\hbar^2} \frac{1}{n^2} \quad (\text{A.19})$$

- (2) Compute the energy

$$|E_1| = \frac{mk^2 e^4}{2\hbar^2} \quad (\text{A.20})$$

in unit of electron volt (eV), and then in unit of joule (J).

A.6 Mode counting in blackbody radiation

Consider light waves in a cubic box of size $L \times L \times L$.

- (1) Using the standing wave condition, derive the condition for the wavevector $\vec{k} = (k_x, k_y, k_z)$:

$$k_i = n_i \frac{\pi}{L}, \quad i = x, y, z, \quad n_x, n_y, n_z = 1, 2, 3, \dots \quad (\text{A.21})$$

(2) **Bonus** Show that in the large box limit of $L \rightarrow \infty$, the total number of standing wave modes in the box is given by

$$N_{\text{modes}} = \sum_{n_x, n_y, n_z} = \frac{L^3}{2\pi^2} \int |k|^2 d|k| \quad (\text{A.22})$$

B Homework 2 (due 9/11)

B.0.1 Pauli matrices

(1) Diagonalize the 3 Pauli matrices and find out their eigenvalues and eigenkets:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (\text{B.1})$$

(2) For an arbitrary function $f(\cdot)$ from complex numbers to complex numbers, if $\vec{n}$ is a normalized vector in three dimensions and θ is a real number, prove the following relation

$$f(\theta \vec{n} \cdot \vec{\sigma}) = \frac{f(\theta) + f(-\theta)}{2} \hat{1} + \frac{f(\theta) - f(-\theta)}{2} \vec{n} \cdot \vec{\sigma} \quad (\text{B.2})$$

where $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are three Pauli matrices.

Hint: a function $f(\hat{M})$ of a matrix (operator) $\hat{M}$ can be defined by the Taylor expansion:

$$f(\hat{M}) = \sum_{n=0}^{\infty} \frac{\hat{M}^n}{n!} \left. \frac{d^n f}{dx^n} \right|_{x=0} \quad (\text{B.3})$$

B.0.2 Linear algebra

(1) Use the Cauchy-Schwarz inequality to prove the triangle inequality

$$||u\rangle + |v\rangle| \leq ||u\rangle| + ||v\rangle|, \quad \forall |u\rangle, |v\rangle \in V \quad (\text{B.4})$$

(2) A linear operator $\hat{P}$ is a projection operator (or a projector) if $\hat{P}^2 = \hat{P}$. Prove that the eigenvalues of a projector are all either 0 or 1.

(3) **Bonus** A projector has the following form:

$$\hat{P} = \alpha |1\rangle\langle 1| + \beta |2\rangle\langle 2| \quad (\text{B.5})$$

where $|1\rangle$ and $|2\rangle$ are two normalized vectors in the Hilbert space that are linearly independent. What conditions should coefficients α and β satisfy in order for $\hat{P}$ defined above to be a projector (i.e. to satisfy $\hat{P}^2 = \hat{P}$)?

(4) **Bonus** Assume Hamiltonian H commutes with two linear operators, A_1 and A_2 , which do not commute with each other:

$$[H, A_1] = [H, A_2] = 0, \quad [A_1, A_2] \neq 0 \quad (\text{B.6})$$

Show that the eigenstates of H are in general degenerate.

(5) **Bonus** Assume two observables A_1 and A_2 anticommute with each other, i.e.

$$\{A_1, A_2\} = A_1 A_2 + A_2 A_1 = 0 \quad (\text{B.7})$$

Show that for any eigenstate $|\psi_1\rangle$ of observable A_1 with a nonzero eigenvalue, its expectation

B Homework 2 (due 9/11)

value w.r.t. A_2 must vanish

$$\langle \psi_1 | A_2 | \psi_1 \rangle = 0 \quad (\text{B.8})$$

Comment: one such example is Pauli matrices $A_1 = \sigma_x$ and $A_2 = \sigma_y$.

(6) Consider two observables $A_{1,2}$ that anticommute with each other. Is it possible to have simultaneous eigenket of A_1 and A_2 ? Explain.

(7) **Bonus** For two linear operators satisfying $[\hat{A}, \hat{B}] = \alpha \hat{1}$ where $\hat{1}$ is the identity operator, prove the following identity:

$$[\hat{A}, e^{\hat{B}}] = \alpha e^{\hat{B}} \quad (\text{B.9})$$

Hint: first show that

$$\hat{f}(x) = e^{-x\hat{B}} \hat{A} e^{x\hat{B}} = \hat{A} + x\alpha \hat{1} \quad (\text{B.10})$$

(8) For a quantum-mechanical particle in one dimension, its position $\hat{x}$ and momentum $\hat{p}$ operators satisfy the canonical commutation relation:

$$[\hat{x}, \hat{p}] = i\hbar = ih/2\pi \quad (\text{B.11})$$

Use the results of part (7) to show that for every $\hat{x}$ eigenstate $|x_0\rangle$ with eigenvalue x_0

$$\hat{x}|x_0\rangle = x_0|x_0\rangle \quad (\text{B.12})$$

the following state

$$|x_0 + a\rangle \equiv e^{-i\hat{p}a/\hbar}|x_0\rangle \quad (\text{B.13})$$

is also an eigenstate of $\hat{x}$ with eigenvalue $x_0 + a$.

Comment: physically this means the linear operator $\hat{T}(a) \equiv e^{-i\hat{p}a/\hbar}$ translate the position by a , from x_0 to $x_0 + a$.

B.0.3 Consecutive measurements of incompatible operators

Two consecutive projective measurements, for Pauli matrices $\hat{\sigma}_x$ and $\hat{\sigma}_z$ respectively, are carried out one after another on a quantum state of a single qubit (i.e. a Hilbert space $\mathbb{V} = \mathbb{C}^2$). In the first procedure, we first measured $\hat{\sigma}_x$ with an outcome $+1$ and probability $p(X = +1)$, and then measured $\hat{\sigma}_z$ with an outcome $+1$ with probability $p(Z = +1|X = +1)$. In the second procedure, we switched the order of the two measurements, i.e. we first measured $\hat{\sigma}_z$ with an outcome $+1$ and probability $p(Z = +1)$, and then measured $\hat{\sigma}_x$ with an outcome $+1$ with probability $p(X = +1|Z = +1)$. It is found that the probabilities do not depend on the order of the two incompatible measurements, i.e.

$$p(X = +1) = p(X = +1|Z = +1), \quad p(Z = +1|X = +1) = p(Z = +1) \quad (\text{B.14})$$

(1) Find a single-qubit state $|\psi\rangle$ that satisfies the condition (B.14).

(2) **Bonus** If the quantum state is a single-qubit mixed state, it is generally described by a density matrix of the following form:

$$\hat{\rho} = \frac{\hat{1} + \vec{r} \cdot \vec{\sigma}}{2}, \quad \vec{r} = (x, y, z) \in \mathbb{R}^3, \quad |\vec{r}| \leq 1. \quad (\text{B.15})$$

Find those mixed states that satisfy the same condition (B.14).

Comment: you will need to learn about mixed states and measurements in N&C 2.4.1.

B.0.4 Quantum Zeno effect

Consider a single qubit evolving under the following Hamiltonian:

$$\hat{H} = -\mu_B B \hat{\sigma}_z \quad (\text{B.16})$$

where $\mu_B = \hbar e / 2m_e$ is the Bohr magneton. We define time T as follows:

$$T = \frac{\pi m_e}{eB} \quad (\text{B.17})$$

Starting from the initial state

$$|\psi_0\rangle = |+\rangle = \frac{|\uparrow\rangle + |\downarrow\rangle}{\sqrt{2}} \quad (\text{B.18})$$

at time $t = 0$, we evolve the system by time T/n , followed by a measurement of $\hat{\sigma}_x$, and then repeat the same 2-step procedure (evolution by $\frac{T}{n} + \hat{\sigma}_x$ measurement) n times.

(1) Starting from the initial state $|+\rangle$, after unitary time evolution of time T/n , obtain the new state

$$|\psi_+\rangle = e^{-i\hat{H}T/n\hbar}|+\rangle \quad (\text{B.19})$$

Hint: use the results of problem 1 on Pauli matrices.

(2) After a measurement of $\hat{\sigma}_x$ on the above state $|\psi_+\rangle$, what is the probability of measurement outcome $\text{Pr}(\sigma_x = +1)$? What does the state collapse into after reading out the measurement outcome $\sigma_x = +1$?

(3) Consider the same measurement of $\hat{\sigma}_x$ on the above state $|\psi_+\rangle$, what is the probability of measurement outcome $\text{Pr}(\sigma_x = -1)$? What does the state collapse into after reading out the measurement outcome $\sigma_x = -1$?

(4) **Bonus** What is the probability at time T (after n such 2-step procedures) to produce the measurement outcome of $\sigma_x = +1$?

C Homework 3 (due 9/18)

C.0.1 Spin precession

(1) For a unit vector $\vec{n} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$, diagonalize the matrix $\vec{n} \cdot \vec{\sigma} = \sum_{i=x,y,z} n_i \sigma_i$. Prove that it has two eigenvalues ± 1 , and find out the two eigenvectors.

(2) We denote $|\vec{n}\rangle$ as the eigenvector of $\vec{n} \cdot \vec{\sigma}$ with eigenvalue $+1$. Starting from time zero, the state vector $|\psi(0)\rangle \equiv |\vec{n}\rangle$ evolves under the following Hamiltonian:

$$\hat{H} = -B\sigma_z \quad (\text{C.1})$$

Prove that the following expectation value of $S_z = \hbar\sigma_z/2$

$$\langle S_z(t) \rangle \equiv \langle \psi(t) | S_z | \psi(t) \rangle \quad (\text{C.2})$$

does not change with time.

(3) Compute the time-dependent expectation values of S_x and S_y , and show that they oscillate with time.

(4) Compute the transition probability that the state vector $|\vec{n}\rangle$ returns to itself after time t :

$$P(\vec{n} \rightarrow \vec{n} : t) \equiv |\langle \psi(t) | \psi(0) \rangle|^2 \quad (\text{C.3})$$

C.0.2 Neutrino oscillation

Read the neutrino oscillation in S&N 2.1. Using the energy eigenvalues

$$E_i = \sqrt{p^2 c^2 + m_i^2 c^4} \approx pc \left(1 + \frac{m_i^2 c^2}{2p^2}\right) \quad (\text{C.4})$$

prove the probability

$$P(\nu_e \rightarrow \nu_e) = 1 - \sin^2(2\theta) \sin^2 \left[\frac{(m_1^2 - m_2^2) c^4 t}{4E\hbar} \right], \quad E \approx pc. \quad (\text{C.5})$$

C.0.3 Propagator

(1) Consider the following propagator for Hamiltonian $\hat{H}$:

$$G(i, f; t) \equiv -i \langle f | e^{-i\hat{H}t/\hbar} | i \rangle \quad (\text{C.6})$$

Show that its Fourier transform

$$G(i, f : \omega + i0^+) \equiv \lim_{\epsilon \rightarrow 0^+} \int_0^\infty e^{i(\omega + i\epsilon)t} G(i, f : t) \quad (\text{C.7})$$

is equal to

$$G(i, f : \omega + i0^+) = \langle f | \frac{1}{\omega + i0^+ - \hat{H}} | i \rangle \quad (\text{C.8})$$

where we have set $\hbar = 1$.

(2) **Bonus** Now consider a quantum particle in one dimension, by choosing i, f to be position eigenstates $|i\rangle = |x\rangle, |f\rangle = |y\rangle$. Show that

$$G(x, y; t) = -i \sum_n \psi_n^*(x) \psi_n(y) e^{-iE_n t/\hbar} \quad (\text{C.9})$$

where E_n and $\psi_n(x)$ are the eigenvalues and eigenstates of Hamiltonian $\hat{H}$.

(3) **Bonus** Show that the imaginary part of the frequency-domain propagator is given by

$$\Im G(x, x; \omega + i0^+) = -\pi \sum_n |\psi_n(x)|^2 \delta(\omega - E_n/\hbar) \quad (\text{C.10})$$

where $\delta(x)$ is a Dirac delta function.

C.0.4 2-qubit Hamiltonian

Consider a closed quantum system consisting of two spin-1/2 particles $\{\vec{S}_i = \frac{\hbar}{2}\vec{\sigma}_i | i = 1, 2\}$, i.e. two qubits. The Hamiltonian of the system is given by the *Heisenberg model*

$$\hat{H} = J\vec{S}_1 \cdot \vec{S}_2 = \sum_{a=x,y,z} JS_{1,a} \otimes S_{2,a} \quad (\text{C.11})$$

(1) Using the matrix representation of Pauli matrices, show the commutation relation for each qubit:

$$[S_{j,a}, S_{j,b}] = i\hbar \epsilon_{abc} S_{j,c}, \quad j = 1, 2; \quad a, b, c = x, y, z \quad (\text{C.12})$$

where ϵ_{abc} is the Levi-Civita symbol.

(2) Prove that each component of the total spin

$$S_a^{\text{tot}} \equiv S_{1,a} \otimes \hat{1}_2 + \hat{1}_1 \otimes S_{2,a} \quad (\text{C.13})$$

commutes with the Hamiltonian:

$$[S_a^{\text{tot}}, \hat{H}] = 0, \quad a = x, y, z. \quad (\text{C.14})$$

The total spin of the system is defined as

$$S_{\text{tot}}^2 = \vec{S}^{\text{tot}} \cdot \vec{S}^{\text{tot}} = \sum_a S_a^{\text{tot}} S_a^{\text{tot}} \quad (\text{C.15})$$

Show that it commutes with both the Hamiltonian $\hat{H}$, and S_a^{tot} , $a = x, y, z$.

(3) Show that if the Hamiltonian $\hat{H}$ commutes with another Hermitian operator $\hat{O}$ (known as a conserved quantity or a constant of motion), for any two eigenstates $|\psi_1\rangle, |\psi_2\rangle$ of operator $\hat{O}$ with different eigenvalues $\lambda_1 \neq \lambda_2$, the following matrix must vanish

$$\langle \psi_1 | \hat{H} | \psi_2 \rangle = 0 \quad (\text{C.16})$$

In other words, Hamiltonian $\hat{H}$ is a block diagonal matrix in the eigenbasis of the conserved quantity $\hat{O}$. This is known as the selection rule in quantum mechanics.

(4) Diagonalize the Hamiltonian $\hat{H}$ and find its energy levels. Find the simultaneous eigenstates of $\hat{H}$, S_z^{tot} and S_{tot}^2 , in the orthonormal basis of $\{|\sigma_{1,z} = \pm 1\rangle \otimes |\sigma_{2,z} = \pm 1\rangle\}$. Note: these 4 eigenstates can be properly chosen to form the *Bell basis* of 2-qubit Hilbert space $\mathbb{C}^4$. (You are allowed to use computer programs, but please show your codes if you do so.)

C Homework 3 (due 9/18)

(5) **Bonus** Can you explain the reason for the degenerate and non-degenerate energy levels, using what you have learnt from Problem 2.(4) in Homework 2?

D Homework 4 (due 9/25)

D.0.1 Bloch ball for single-qubit mixed states

(1) Show that an arbitrary density matrix for a single-qubit mixed state can be written as

$$\hat{\rho} = \frac{\hat{1} + \vec{r} \cdot \vec{\sigma}}{2} \quad (\text{D.1})$$

where $\vec{\sigma}$ are 3 Pauli matrices, and $\vec{r}$ is a 3-dimensional vector such that $|\vec{r}| \leq 1$. This vector $\vec{r}$ is known as the *Bloch vector* for the state $\hat{\rho}$.

Hint: a density matrix $\hat{\rho}$ is defined to satisfy two properties: $\hat{\rho}$ is positive (i.e. it can be diagonalized and all eigenvalues are non-negative) and $\text{Tr} \rho = 1$.

(2) Show that a state ρ is pure if and only if $|\vec{r}| = 1$ i.e. $\vec{r}$ is a unit vector.

Hint: a density matrix $\hat{\rho}$ is a pure state if and only if $\hat{\rho}^2 = \hat{\rho}$.

(3) Consider the eigenstate $|\vec{r}\rangle$ of operator $\vec{r} \cdot \vec{\sigma}$ with eigenvalue +1, where $\vec{r}$ is a unit vector. Show that in this case, its density matrix $\rho = |\vec{r}\rangle\langle\vec{r}|$ is exactly given by Eq. (D.1) with $\vec{r}$ being a unit vector. This is known as the Bloch sphere.

(4) Compute the von Neumann entropy $S(\rho) = -\text{Tr} \rho \log \rho$ for the mixed state (D.1). Plot it as a function of $|\vec{r}|$.

D.0.2 Entanglement entropy

(1) If ρ_1 and ρ_2 are two density matrices, show that

$$\hat{\rho}(x) \equiv x\hat{\rho}_1 + (1-x)\hat{\rho}_2, \quad 0 \leq x \leq 1. \quad (\text{D.2})$$

is also a density matrix.

(2) **Bonus** Assume that $\hat{\rho}(x)$ in (D.2) is always invertible for any $0 \leq x \leq 1$, i.e. its inverse matrix $[\hat{\rho}(x)]^{-1}$ always exists. Prove that its von Neumann entropy is a concave function of x , i.e.

$$\frac{d^2 S(\hat{\rho}(x))}{dx^2} \leq 0 \quad (\text{D.3})$$

(3) **Bonus** Use the concave property (D.3) to prove the concavity of von Neumann entropy

$$S\left(\sum_j p_j \hat{\rho}_j\right) \geq \sum_j p_j S(\hat{\rho}_j) \quad (\text{D.4})$$

where $\{p_j\}$ is a probability distribution and $\{\hat{\rho}_j\}$ is a set of density matrices.

Hint: for a concave function $f(x)$ satisfying $f'' \leq 0$, we always have $f(px_1 + (1-p)x_2) \geq pf(x_1) + (1-p)f(x_2)$ for $0 \leq p \leq 1$.

(4) Consider a pure state $|\psi\rangle_{AB}$ in a composite quantum system with Hilbert space $A \otimes B$. Show that for the reduced density matrices

$$\rho_A = \text{Tr}_B |\psi_{AB}\rangle\langle\psi_{AB}|, \quad \rho_B = \text{Tr}_A |\psi_{AB}\rangle\langle\psi_{AB}| \quad (\text{D.5})$$

they share the same von Neumann entropy

$$S(\rho_A) = S(\rho_B) \quad (\text{D.6})$$

Therefore either $S(\rho_A)$ or $S(\rho_B)$ can be defined as the entanglement entropy.

(5) **Bonus** Consider a composite system of two qubits A and B . Can you write down the generic form of a maximally entangled state $|\psi_{AB}\rangle$, satisfying that the reduced density matrix has a maximal entanglement entropy?

Hint: for a maximally entangled 2-qubit state, we must have

$$\hat{\rho}_A = \hat{\rho}_B = \frac{1}{2}\hat{1}. \quad (\text{D.7})$$

so that $S(\rho_{A,B}) = \log 2$.

(6) For a pure state in a composite system of 2 qubits A and B :

$$|\psi_{AB}\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle \quad (\text{D.8})$$

show that this state is (bipartite) entangled if and only if

$$\alpha\delta - \beta\gamma \neq 0. \quad (\text{D.9})$$

Hint: a pure state is (bipartite) entangled if and only if its bipartite entangled entropy is nonzero:

$$S(\rho_A) = S(\rho_B) > 0 \quad (\text{D.10})$$

(7) **Bonus** This problem is about *tripartite entanglement*. Show that in a 3-qubit system, there are pure states that cannot be written in the following form:

$$|\psi\rangle = \sum_{j=1}^2 \lambda_j |\alpha_j\rangle_1 \otimes |\beta_j\rangle_2 \otimes |\gamma_j\rangle_3. \quad (\text{D.11})$$

where $\{\alpha_j|j=1,2\}$ is a set of complete orthonormal basis for $\mathbb{C}^2$, and so are $\{\beta_j|j=1,2\}$ and $\{\gamma_j|j=1,2\}$. Find one such state.

D.0.3 Witnessing entanglement

Consider the singlet Bell state

$$|\text{EPR}\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \quad (\text{D.12})$$

where we have used shorthand notation $|10\rangle \equiv |1\rangle_1 \otimes |0\rangle_2$ for two qubits 1 and 2. From the lecture, we know that this state is invariant under spin rotations, i.e.

$$|\text{EPR}\rangle = \frac{1}{\sqrt{2}}(|+\vec{n}, -\vec{n}\rangle - |-\vec{n}, +\vec{n}\rangle) \quad (\text{D.13})$$

where $\vec{n}$ is any unit vector, and $|\vec{n}\rangle$ is the eigenstate of $\vec{n} \cdot \vec{\sigma}$:

$$\vec{n} \cdot \hat{\sigma} |\vec{n}\rangle = +|\vec{n}\rangle \quad (\text{D.14})$$

15.7 Two photons are initially prepared at a source and move along the y -direction in opposite directions toward polarisers at A and B. They are initially in the state

$$|\Psi\rangle = \frac{1}{\sqrt{2}}[|V_A V_B\rangle + |H_A H_B\rangle],$$

where $|V\rangle$ and $|H\rangle$ are the two linear polarisation states, vertical and horizontal, of the photon moving to the polariser at $I = A$ or B . To be explicit, we will fix the vertical polarisation direction here as the $\hat{z}$ -direction. In particular, the V state is in the direction of $\hat{z}$ and the H state is in the $\hat{x}$ -direction, thus perpendicular to both the $\hat{z}$ -direction and the direction of motion for the photon. Let $\underline{n}_I$ be other directions at polarisers $I = A$ or B that are perpendicular to the direction of motion of the photons, so in the $\hat{x}$ - $\hat{z}$ plane, and at an angle θ_{Az} to the $\hat{z}$ -direction. If photon I is measured through a polariser oriented with respect to the $\underline{n}_I$ -direction:

$$|V_{I1}\rangle = \cos \theta_{Iz} |V\rangle + \sin \theta_{Iz} |H\rangle,$$

$$|H_{I1}\rangle = \sin \theta_{Iz} |V\rangle + \cos \theta_{Iz} |H\rangle.$$

The two photons initially in the state $|\Psi\rangle$ are then measured respectively at A and B, with polarisers oriented along the $\underline{n}_A$ and $\underline{n}_B$ -directions, respectively. Let $P_{VV}(\underline{n}_A, \underline{n}_B)$ be the probability that the vertical polarisation of the photon at A is measured along direction $\underline{n}_A$ for polarisers rotated by angle θ_{Az} and for the photon at B the vertical polarisation is measured along direction $\underline{n}_B$ for polarisers rotated by angle θ_{Bz} . Show that

$$P_{VV}(\underline{n}_A, \underline{n}_B) = |\langle V_{A1} V_{B1} | \Psi \rangle|^2 = \frac{1}{2} \cos^2(\theta_{B1} - \theta_{A1}).$$

Also compute $P_{HH}(\underline{n}_A, \underline{n}_B)$, $P_{VH}(\underline{n}_A, \underline{n}_B)$ and $P_{HV}(\underline{n}_A, \underline{n}_B)$.

Show that

$$E(\underline{n}_A, \underline{n}_B) \equiv P_{VV}(\underline{n}_A, \underline{n}_B) + P_{HH}(\underline{n}_A, \underline{n}_B) - P_{VH}(\underline{n}_A, \underline{n}_B) - P_{HV}(\underline{n}_A, \underline{n}_B) = \cos[2(\theta_{Az} - \theta_{Bz})].$$

For the expression

$$S = E(\underline{n}_A, \underline{n}_B) - E(\underline{n}_A, \underline{n}'_B) + E(\underline{n}'_A, \underline{n}_B) + E(\underline{n}'_A, \underline{n}'_B),$$

a realistic local theory gives the Bell inequality $-2 \leq S \leq 2$. Show that this inequality is violated for the quantum-mechanical expressions for photons derived here. For this choose the orientations

$$\theta_{Bz} - \theta_{Az} = \theta_{A'z} - \theta_{Bz} = \theta_{B'z} - \theta_{A'z} = \phi \quad \text{and} \quad \theta_{B'z} - \theta_{Az} = 3\phi,$$

$$\underline{n}_i \cdot \underline{n}_j = \underline{n}'_i \cdot \underline{n}_j = \underline{n}'_i \cdot \underline{n}'_j = \cos \phi \quad \text{and} \quad \underline{n}_i \cdot \underline{n}'_j = \cos 3\phi$$

and plot as a function of ϕ . Determine the range of angles for ϕ for which Bell's inequality is violated.

Figure D.1: Problem 3.(2)

(1) Show that

$$\langle \vec{m} | \vec{n} \cdot \hat{\vec{\sigma}} | \vec{m} \rangle = \vec{m} \cdot \vec{n} \quad (\text{D.15})$$

where $\vec{m}, \vec{n}$ are both unit vectors.

(2) For two unit vectors $\vec{m}$ and $\vec{n}$, show that

$$\langle \text{EPR} | (\vec{m} \cdot \vec{\sigma}_1) \otimes (\vec{n} \cdot \vec{\sigma}_2) | \text{EPR} \rangle = -\vec{m} \cdot \vec{n} \quad (\text{D.16})$$

(3) Show that for simultaneous measurements of two compatible observables $\vec{m} \cdot \hat{\vec{\sigma}}_1$ and $\vec{n} \cdot \hat{\vec{\sigma}}_2$ in the singlet Bell state $|\text{EPR}\rangle$, the probability for the measurement outcome $\vec{m} \cdot \vec{\sigma}_1 = +1$ and $\vec{n} \cdot \vec{\sigma}_2 = +1$ is given by

$$P(\vec{m}, \vec{n}) = \frac{1}{2} |\langle -\vec{m} | \vec{n} \rangle|^2 \quad (\text{D.17})$$

where $\vec{m}, \vec{n}$ are unit vectors.

For $\vec{m} = (\cos \theta_1, \sin \theta_1, 0)$ and $\vec{n} = (\cos \theta_2, \sin \theta_2, 0)$ in x - y plane, further show that

$$P(\vec{m}, \vec{n}) = \frac{1}{2} \sin^2 \frac{\theta_1 - \theta_2}{2} \quad (\text{D.18})$$

(4) **Bonus** Tsirelson's inequality: N&C Problem 2.3

(5) **Bonus** As shown in Fig. D.1, to understand Aspect's experiment to demonstrate a violation to CHSH inequality in 1982 (Nobel Prize 2022), solve B&DD problem 15.7 (see Fig. D.1).

E Homework 5 (due 10/2)

E.0.1 Revisiting the quantum Zeno effect

We will now revisit the quantum Zeno effect in the lens of generalized measurements. Consider an initial state of a single qubit

$$\hat{\rho}_0 = |+\rangle\langle+|, \quad |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad (\text{E.1})$$

evolving under the following Hamiltonian

$$\hat{H} = -\mu_B B \hat{Z}, \quad T = \frac{\pi m_e}{eB} \quad (\text{E.2})$$

where $\mu_B = \hbar e/2m_e$ is the Bohr magneton. After a small period $\Delta t = T/n$ of unitary time evolution, we carry out a projective measurement on Pauli matrix $\hat{X}$. Recall that while $|0\rangle, |1\rangle$ are eigenstates of Pauli $\hat{Z}$ operator, $|\pm\rangle$ are eigenstates of $\hat{X}$.

(1) As we have learnt in the lecture, the 2-step procedure, of unitary time evolution for $\Delta t = T/n$ followed by a projective measurement on $\hat{X}$, can be described by a generalized measurement. Compute the measurement operators $\{\hat{M}_m|m = \pm\}$ as 2×2 matrices, and show that they satisfy the completeness relation

$$\sum_{m=\pm} \hat{M}_m^\dagger \hat{M}_m = \hat{1}. \quad (\text{E.3})$$

(2) In the generalized measurement discussed above, if we do not know the measurement outcomes, the density matrix after the measurement is given by the action of a quantum channel $\mathcal{E}$, which is characterized by measurement operators $\{\hat{M}_m\}$:

$$\mathcal{E}[\hat{\rho}] \equiv \sum_m \hat{M}_m \hat{\rho} \hat{M}_m^\dagger \quad (\text{E.4})$$

We have previously learnt that any single-qubit density matrix can be expressed in terms of the Bloch ball:

$$\hat{\rho} = \frac{\hat{1} + \vec{r} \cdot \vec{\sigma}}{2}, \quad |\vec{r}| \leq 1. \quad (\text{E.5})$$

How does the vector $\vec{r}$ transform under the quantum channel (E.4)?

(3) In the discussion of quantum Zeno effect, such a generalized measurement (without knowing measurement outcomes) was performed n times. Starting from the initial density matrix $\hat{\rho}_0$ in (E.1), what is the final density matrix $\mathcal{E}^n[\rho_0]$ after n such operation?

Hint: use the results of part (2).

(4) Using the above results, compute the probability at time T (after n such 2-step procedures) to produce the measurement outcome of $\sigma_x = +1$.

Comment: recall that we have obtained this probability in HW2. You should compare your results here to HW2.

E.0.2 Measurements

(1) **Bonus: distinguishing states by POVM measurements** N&C Exercise 2.64.

(2) Let $|\psi\rangle = a|0\rangle + b|1\rangle$ be any single-qubit quantum state. Suppose we receive one of the four states $|\psi\rangle, \hat{X}|\psi\rangle, \hat{Y}|\psi\rangle, \hat{Z}|\psi\rangle$ with equal prior probabilities of $1/4$, where $(\hat{X}, \hat{Y}, \hat{Z})$ are the three Pauli matrices. This is known as Pauli-twirling. Show that the density matrix received by us is exactly $\hat{\rho} = \frac{1}{2}\hat{1}$.

Comment: for such a single-qubit mixed state with $\hat{\rho} = \frac{1}{2}\hat{1}$, the outcome for any projective measurement $\hat{P}$ has a probability of one half, thus the received state contains no information at all about the identity of $|\psi\rangle$. This is Problem 4 in Chapter 16 exercises of Tong.

E.0.3 Quantum channels

(1) **Bonus** Consider an open quantum system A interacting with its environment B . In the lectures, we have shown that for an initial state of $\rho_A \otimes |0\rangle\langle 0|_B$ where $|0\rangle$ is a pure state in B , the time evolution for system A is given by a quantum channel ε , which is specified by a set of Kraus operators $\{\hat{M}_k\}$ satisfying the completeness relation

$$\sum_k \hat{M}_k^\dagger \hat{M}_k = \hat{1}_A. \quad (\text{E.6})$$

Show that this is still true if we consider the initial state to be a product state $\rho_A \otimes \rho_B$, where ρ_B is an arbitrary density matrix in B .

Hint: you will need to find the set of Kraus operators satisfying the completeness relation in this case. It is convenient to work in the eigenbasis of ρ_B .

(2) N&C Exercise 8.15

(3) **Bonus: generalized depolarizing channel** N&C Exercise 8.19

E.0.4 The master equation

E.0.4.1 Phase damping

Time evolution of a single-qubit open system under the phase damping quantum channel ε_z is given by:

$$\hat{\rho} = \begin{pmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{pmatrix} \rightarrow \varepsilon_z[\hat{\rho}] = (1-p)\hat{\rho} + p\hat{Z}\hat{\rho}\hat{Z} = \begin{pmatrix} \rho_{11} & (1-2p)\rho_{12} \\ (1-2p)\rho_{21} & \rho_{22} \end{pmatrix} \quad (\text{E.7})$$

where p is the probability of error. For the time evolution by a small time Δt , the error probability is given by

$$2p = \Gamma \Delta t \quad (\text{E.8})$$

where Γ is the rate of error.

(1) Derive the differential equation

$$\frac{d\rho}{dt} = \lim_{\Delta t \rightarrow 0^+} \frac{\varepsilon_z[\rho] - \rho(t=0)}{\Delta t} \quad (\text{E.9})$$

and solve it. Obtain the form of $\rho(t)$ for a given initial state $\hat{\rho}(t=0) = \begin{pmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{pmatrix}$.

(2) Write the differential equation in part (1) into the form of a master equation. Find the Hamiltonian $\hat{H}$ and the Lindblad operators $\{\hat{L}_k\}$ in the master equation.

E.0.4.2 Amplitude damping

(1) Solve the following master equation for a 1-qubit density matrix $\rho(t)$, with a single Lindblad operator $L = \sqrt{\lambda}\sigma_- \equiv \sqrt{\lambda}(\hat{X} - i\hat{Y})/2 = \sqrt{\lambda} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$:

$$\frac{d\rho}{dt} = \lambda\sigma_-\rho\sigma_+ - \frac{\lambda}{2}\{\rho, \sigma_+\sigma_-\} \quad (\text{E.10})$$

where $\sigma_+ = (\hat{X} + i\hat{Y})/2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Express $\rho(t)$ as a function of the initial density matrix

$$\rho(t=0) = \begin{pmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{pmatrix} \quad (\text{E.11})$$

Comment: this master equation describes the amplitude damping of a 1-qubit system.

(2) Consider the time evolution of a 1-qubit system by an amplitude-damping quantum channel ε_p with the following set of Kraus operators:

$$\hat{M}_0 = \begin{pmatrix} \sqrt{1-p} & 0 \\ 0 & 1 \end{pmatrix}, \quad \hat{M}_1 = \begin{pmatrix} 0 & 0 \\ \sqrt{p} & 0 \end{pmatrix} \quad (\text{E.12})$$

For the same initial state (E.11), obtain the final density matrix $\rho(p) = \varepsilon_p[\rho(t=0)]$ after the action of the quantum channel, as a function of the amplitude-damping probability p . Compare it with the previous problem, and obtain the relation between p in this problem, and t in the previous problem.

E.0.4.3 Amplitude + phase dampings

Bonus N&C Exercise 8.30. There is a typo in N&C for this problem: you need to show that $T_2 = 2T_1$ for amplitude damping only, and $T_2 \leq 2T_1$ for both amplitude and phase dampings.

F Homework 6 (Due 10/14)

F.0.1 (Bonus) Position and momentum operators on a discrete periodic chain

During the lecture, we showed that the position-space basis $\{|x\rangle\}$ and momentum-space basis $\{|p\rangle\}$ in the continuum can be approached as the limit of a one-dimensional periodic ring of L lattice sites $\{|n\rangle \equiv |n+L\rangle | 1 \leq n \leq L\}$, where two neighboring sites are separated by a lattice constant a . The unitary translation operator (it's easy to check $\hat{T}_x^\dagger \hat{T}_x = \hat{1}$)

$$\hat{T}_x = \sum_{n=1}^L |(n \bmod L) + 1\rangle \langle n| = e^{-ia\hat{p}/\hbar} \quad (\text{F.1})$$

allows us to define the unitary momentum operator $\hat{p}$, and the position operator is given by

$$\hat{x} = \sum_{n=1}^L (n - \frac{L}{2})a |n\rangle \langle n| \quad (\text{F.2})$$

where a labels the lattice constant. In lectures we claimed that in the simultaneous limits of $a \rightarrow 0^+$ and $La \rightarrow \infty$, they correspond to the position and momentum operators in the continuum. In this problem, we consider them in a finite chain for a finite positive integer L .

(1) **Bonus** Find all eigenvalues and their associated eigenstates of the translation operator $\hat{T}_x$ for a finite $L \in \mathbb{Z}_+$.

Hint: there should be L eigenstates for a $L \times L$ unitary operator. These are also eigenstates of momentum operator $\hat{p}$.

(2) **Bonus** In lectures, we used the fact that $[\hat{x}, \hat{T}_x] = a\hat{T}_x$ to derive the commutation relation $[\hat{x}, \hat{p}] = i\hbar\hat{1}$. But we know that in a finite-dimensional Hilbert space, it is impossible for the commutator of two linear operators to be a nonzero constant matrix. Here we will investigate what really happens in a finite-dimensional space, which is $\mathbb{C}^L$ here.

Compute the commutator $[\hat{x}, \hat{T}_x]$ of the position and translation operators defined in (F.2) and (F.1), in the complete orthonormal basis of $\{|n\rangle\}$. Show that it is almost the same as $a\hat{T}_x$, except for one matrix element.

(3) **Bonus** For an arbitrary momentum eigenstate (you can choose any one of the L eigenstates of $\hat{T}_x$), compute the variance of the position operator:

$$(\Delta x)^2 \equiv \langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2 \quad (\text{F.3})$$

How does the variance increase with L in the limit of $L \rightarrow \infty$?

F.0.2 Dirac delta function

(1) In lectures we learnt that the Dirac delta function is given by

$$\delta(x) = \frac{1}{2\pi} \int dp e^{ipx} \quad (\text{F.4})$$

However, this integral does not converge since the integrand e^{ipx} does not vanish at infinity. To regulate this integral, and to define Dirac delta function more rigorously, we define

$$\delta(x) = \lim_{\lambda \rightarrow 0^+} \int \frac{dp}{2\pi} e^{ipx - \lambda p^2} \quad (\text{F.5})$$

Carry out this integral to show that

$$\delta(x) = \lim_{\lambda \rightarrow 0^+} \frac{e^{-x^2/4\lambda}}{\sqrt{4\pi\lambda}} \quad (\text{F.6})$$

(2) Show that the above definition (F.6) satisfies the properties of Dirac delta functions:

$$\delta(x) = \delta(-x), \quad (\text{F.7})$$

$$\delta(x \neq 0) = 0, \quad (\text{F.8})$$

$$\int dx \delta(x) = 1. \quad (\text{F.9})$$

(3) Prove the following property of the delta function:

$$\delta(ax) = \delta(x)/|a| \quad (\text{F.10})$$

Hint: consider the property (F.9) of Dirac delta functions.

(4) Prove another property of the delta function:

$$\delta[f(x)] = \sum_i \frac{\delta(x - x_i)}{|df/dx|_{x=x_i}} \quad (\text{F.11})$$

where x_i are the zeros of $f(x)$.

F.0.3 Inversion symmetry in one dimension

(1) For a one-dimensional quantum mechanics problem, consider the following unitary operator $\hat{U}$:

$$\hat{U} = \int dx |-x\rangle\langle x| \quad (\text{F.12})$$

Show that $\hat{U}$ is a unitary operator first.

(2) How does operator $\hat{U}$ act on the momentum-space basis? Using Fourier transform to show that

$$\hat{U}|p\rangle = |-p\rangle \quad (\text{F.13})$$

$\hat{U}$ is known as the inversion symmetry operator.

(3) Consider a non-degenerate eigenstate $|\psi\rangle$ of a Hamiltonian

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}) \quad (\text{F.14})$$

satisfying $V(x) = V(-x)$. Show that

$$\langle\psi|\hat{x}|\psi\rangle = \langle\psi|\hat{p}|\psi\rangle = 0 \quad (\text{F.15})$$

F.0.4 Double delta function potential

Because the inversion symmetry operator $\hat{U}$ commutes with the Hamiltonian, each energy eigenstate must also be an eigenstate of inversion operator $\hat{U}$. Consider the Hamiltonian (F.14) with the following attractive potential

$$V(x) = -V_0 \delta(x - x_0) - V_0 \delta(x + x_0) \quad (\text{F.16})$$

where both V_0 and x_0 are positive constants.

(1) Write down the form of an $E < 0$ bound state wavefunction in the three regions:

$$(i) \ x < -x_0, \quad (ii) \ -x_0 < x < x_0 \quad (iii) \ x > x_0$$

(2) Write down the matching conditions at $x = \pm x_0$, to continuously connect the wavefunctions of the 3 regions.

(3) Solve for the wavefunction of an energy eigenstate with a negative energy $E < 0$. What is the inversion eigenvalue of this state?

Hint: recall that the energies of bound states are discrete. It won't be surprising if your results depend on the energy E and parameters V_0, x_0 .

G Homework 7 (due 10/28)

G.0.1 (Bonus) Lippmann-Schwinger equation

In this problem, we delve into the derivation of Lippmann-Schwinger equation, which was briefly mentioned in our lectures. Consider a Hamiltonian

$$\hat{H} = \hat{H}_0 + \hat{V} \quad (\text{G.1})$$

where $\hat{H}_0$ is the non-interacting Hamiltonian which is exactly solvable, e.g. the free particle Hamiltonian $\hat{p}^2/2m$ in lectures. $\hat{V}$ represents an interaction term, e.g. the potential term $V(\hat{x})$ for a quantum particle in free space, whose presence makes the full Hamiltonian $\hat{H}$ hard to solve.

Next we assume that $\hat{H}$ and $\hat{H}_0$ have the same energy spectrum. This is exactly the case for a localized potential

$$\lim_{|x| \rightarrow \infty} V(x) = 0. \quad (\text{G.2})$$

that we have discussed in lectures: it is obvious from the asymptotic behavior of the Schrodinger equation at $|x| \rightarrow \infty$.

Now we label the orthonormal diagonal basis of non-interacting Hamiltonian $\hat{H}_0$ to be $\{|a\rangle\}$, satisfying

$$\hat{H}_0|a\rangle = E_a|a\rangle, \quad \langle a|b\rangle = \delta_{a,b}. \quad (\text{G.3})$$

We further label the eigenstate of full Hamiltonian $\hat{H}$ with energy E_a as $|\psi_a^\pm\rangle$, satisfying the stationary Schrodinger equation

$$\hat{H}|\psi_a^\pm\rangle = (E_a \pm i\epsilon)|\psi_a^\pm\rangle \quad (\text{G.4})$$

where $\epsilon \rightarrow 0^+$ is an infinitely small positive number.

(1) **Bonus** Show that

$$|\psi_a^\pm\rangle = \alpha|a\rangle + (E_a - \hat{H}_0 \pm i\epsilon)^{-1}\hat{V}|\psi_a^\pm\rangle \quad (\text{G.5})$$

where α is a complex coefficient.

Comment: we can always absorb the coefficient α into the un-normalized eigenstate $|\psi_a^\pm\rangle$, which leads to the familiar form of Lippmann-Schwinger equation:

$$|\psi_a^\pm\rangle = |a\rangle + \frac{1}{E_a - \hat{H}_0 \pm i\epsilon}\hat{V}|\psi_a^\pm\rangle \quad (\text{G.6})$$

But it is sometimes important to keep the coefficient α in mind, as we will see in the next problem when we are dealing with piecewise continuous wavefunctions.

(2) **Bonus** Show that $\{|\psi_a^+\rangle\}$ forms another set of complete orthonormal basis of the Hilbert space. In other words, show that

$$\langle \psi_a^+ | \psi_b^+ \rangle = \delta_{a,b} = \langle a | b \rangle \quad (\text{G.7})$$

Comment: we need to use the Gellmann-Low theorem for this problem, which we mentioned during lectures. Similarly one can show that $\{|\psi_a^-\rangle\}$ is also a set of complete orthonormal basis.

G.0.2 (Bonus) Solving Lippmann-Schwinger eq. for a delta function potential

Next, we will use the Lippmann-Schwinger equation to solve the eigenstate wavefunctions for scattering states with a delta function potential:

$$\hat{H} = \hat{H}_0 + \hat{V}, \quad \hat{H}_0 = \frac{\hat{p}^2}{2m}, \quad \hat{V} = V(\hat{x}) \quad (\text{G.8})$$

Recall the Lippmann-Schwinger equation:

$$\psi_k^\pm(x) = \alpha \langle x|k \rangle + \int dy G_0^\pm(E|x, y) V(y) \psi_k^\pm(y) \quad (\text{G.9})$$

where α is an arbitrary coefficient to be determined, and $|k\rangle$ is a plane wave

$$\langle x|k \rangle = \frac{1}{\sqrt{2\pi}} e^{ikx} \quad (\text{G.10})$$

with wavevector $k = \sqrt{2mE}/\hbar > 0$, with a positive energy $E > 0$ for the free particle Hamiltonian $\hat{H}_0$. The retarded (G^+) and advanced (G^-) Green's functions are defined as

$$G_0^\pm(E|x, y) = \int dk \langle x|k \rangle \frac{1}{E - \hbar^2 k^2 / 2m \pm i\epsilon} \langle k|y \rangle = \int \frac{dk}{2\pi} \frac{e^{ik(x-y)}}{E - \hbar^2 k^2 / 2m \pm i\epsilon} \quad (\text{G.11})$$

where $\epsilon \rightarrow 0^+$ is an infinitely small positive number.

(1) **Bonus** Show the following relation holds for an arbitrary potential $V(\hat{x})$

$$[\psi_k^+(x)]^* = \psi_{-k}^-(x) \quad (\text{G.12})$$

using the Lippmann-Schwinger equation (G.9). This relation is a consequence of the time reversal symmetry $\hat{H}^* = \hat{H}$.

(2) **Bonus** Using the contour integral in complex analysis, compute the retarded Green's function $G_0^+(E|x, y)$ in (G.11). Explain why it corresponds to an outgoing wave.

Hint: it should have different behaviors in $x < y$ and $x > y$ regions.

(3) **Bonus** Similarly compute the advanced Green's function $G_0^-(E|x, y)$ in (G.11). Explain why it corresponds to an incoming wave.

(4) **Bonus** Now we turn to the delta function potential $V(x) = V_0 \delta(x)$. We will focus on only the outgoing states $|\psi_k^+\rangle$. Write down the Lippmann-Schwinger equation (G.9) for $x > 0$ and $x < 0$ separately. Note that the coefficient α can be different for the two regions. Solve the Lippmann-Schwinger equation to obtain the transmission and reflection coefficients t and r . They should recover the exact solution we derived in lectures.

G.0.3 S matrix and the phase shift

In lectures we have mentioned that the S matrix defined as the change of basis between two sets of complete orthonormal basis $\{|\psi_a^+\rangle\}$ and $\{|\psi_a^-\rangle\}$ (see problem 1.(2))

$$S_{a,b} = \langle \psi_a^- | \psi_b^+ \rangle = \langle a | U_I(+\infty, -\infty) | b \rangle \quad (\text{G.13})$$

is a unitary matrix, satisfying $\hat{S}^\dagger \hat{S} = \hat{1}$. A unitary matrix can always be diagonalized, and its eigenvalues are phase factors $e^{2i\delta}$ known as phase shifts in scattering problems. We will learn more about the phase shift in this problem.

In our lectures, we have derived the asymptotic form of exact eigenstates for an arbitrary localized potential $V(x)$ satisfying $\lim_{|x| \rightarrow \infty} V(x) = 0$:

$$\psi_k^+(x) = \begin{cases} te^{ikx}, & x \rightarrow +\infty, \\ e^{ikx} + re^{-ikx}, & x \rightarrow -\infty. \end{cases} \quad (\text{G.14})$$

$$\psi_{-k}^+(x) = \begin{cases} e^{-ikx} + r'e^{ikx}, & x \rightarrow +\infty, \\ t'e^{-ikx}, & x \rightarrow -\infty. \end{cases} \quad (\text{G.15})$$

and

$$\psi_k^-(x) = [\psi_{-k}^+(x)]^*, \quad \psi_{-k}^-(x) = [\psi_k^+(x)]^* \quad (\text{G.16})$$

(1) Using the continuity of probability flux, show that

$$|t|^2 + |r|^2 = 1 = |t'|^2 + |r'|^2. \quad (\text{G.17})$$

(2) Using the fact that $(t\psi_{-k}^+ - r'\psi_k^+)^* \propto \psi_k^+$ and relation (G.17), show that

$$t' = t, \quad r' = -r^*t/t^*. \quad (\text{G.18})$$

(3) Using the relations proved in parts (1) and (2), show that

$$\langle \psi_k^+ | \psi_{-k}^+ \rangle = \langle \psi_k^- | \psi_{-k}^- \rangle = 0. \quad (\text{G.19})$$

Comment: this verifies that either $\{|\psi_k^+\rangle, |\psi_{-k}^+\rangle\}$ or $\{|\psi_k^-\rangle, |\psi_{-k}^-\rangle\}$ forms a complete orthogonal basis of the Hilbert space.

(4) **Bonus** In the subspace with energy $E = \hbar^2 k^2 / 2m > 0$, the $\hat{S}$ matrix is given by

$$\hat{S}(k) = \begin{pmatrix} \langle \psi_k^- | \psi_k^+ \rangle & \langle \psi_k^- | \psi_{-k}^+ \rangle \\ \langle \psi_{-k}^- | \psi_k^+ \rangle & \langle \psi_{-k}^- | \psi_{-k}^+ \rangle \end{pmatrix} \quad (\text{G.20})$$

Diagonalize it to obtain its eigenvalues $e^{2i\delta_+}$ and $e^{2i\delta_-}$ as a function of r and t .

Hint: since $|t|^2 + |r|^2 = 1$ we can parameterize them as

$$t = \cos \theta e^{i\phi_t}, \quad r = \sin \theta e^{i\phi_r}, \quad 0 \leq \theta \leq \pi/2. \quad (\text{G.21})$$

Comment: the diagonal basis of $\hat{S}(k)$ are inversion eigenstates with inversion eigenvalue $+1$ and -1 respectively, if the potential is inversion symmetric i.e. $V(-x) = V(x)$.

G.0.4 Tunneling through a finite potential barrier

Consider a finite potential barrier

$$V(x) = \begin{cases} 0, & |x| > a/2, \\ U > 0, & |x| < a/2. \end{cases} \quad (\text{G.22})$$

where $a > 0$ is the width of the potential barrier.

(1) Consider an eigenstate, the in $|\psi_k^+\rangle$ with energy $0 < E < U$. Write down the form of its eigenstate wavefunction $\psi_k^+(x)$, without computing the coefficients.

(2) Write down the boundary conditions satisfied by $\psi_k^+(x)$ at $x = \pm a/2$.

(3) Obtain the reflection and transmission coefficients r and t as functions of E, U, a .

(4) Show that in the low energy limit of

$$U - E \gg \frac{\hbar^2}{ma^2} \quad (\text{G.23})$$

the transmission probability decays exponentially with $\sqrt{U - E}$

$$T \equiv |t|^2 \sim e^{-C\sqrt{U-E}} \quad (\text{G.24})$$

Compute the coefficient C .

(5) Now we consider $U < 0$ case, i.e. a finite potential well which we discussed in details in our lectures. In this case, the even-parity bound state solution is determined by the following equation

$$k' \tan(k'a/2) = \sqrt{\frac{2m|U|}{\hbar^2} - (k')^2}, \quad E = \frac{(\hbar k')^2}{2m} - |U| < 0. \quad (\text{G.25})$$

Show that in the limits of

$$a \rightarrow 0^+, \quad |U| \rightarrow \infty, \quad a|U| = |D_0| = \text{const.} \quad (\text{G.26})$$

the finite potential well is reduced to an attractive delta function $V(x) = -|D_0|\delta(x)$, and show the bound state energy is given by

$$E = -\frac{m|D_0|^2}{2\hbar^2} \quad (\text{G.27})$$

G.0.5 A step potential

Consider the following step potential

$$V(x) = \begin{cases} 0, & x < 0, \\ U > 0, & x > 0 \end{cases} \quad (\text{G.28})$$

- (1) Solve for the eigenstate wavefunction with energy $E > U$.
- (2) Solve for the eigenstate wavefunction with energy $0 < E < U$.

H Homework 8 (due 11/4)

H.0.1 The simple harmonic oscillator

(1) **Bonus** When we learnt about Heisenberg's uncertainty relation, we know that the equality $\Delta x \Delta p = \hbar/2$ holds when the following two conditions hold simultaneously:

$$(\Delta \hat{x} + C \Delta \hat{p})|\psi\rangle = 0, \quad (\text{H.1})$$

$$\langle\psi|\{\Delta \hat{x}, \Delta \hat{p}\}|\psi\rangle = 0 \quad (\text{H.2})$$

where C is a constant, and

$$\Delta \hat{x} \equiv \hat{x} - \langle\hat{x}\rangle, \quad \Delta \hat{p} \equiv \hat{p} - \langle\hat{p}\rangle. \quad (\text{H.3})$$

Using the fact that $\Delta \hat{x}$ and $\Delta \hat{p}$ are both Hermitian operators, show that the constant C must be an imaginary number (e.g. we can choose $C = iR^2/\hbar$), and the two conditions (H.1)-(H.2) turn out to be just one condition

$$(\Delta \hat{x} + \frac{iR^2}{\hbar} \Delta \hat{p})|\psi\rangle = 0 \quad (\text{H.4})$$

Here R is a constant length scale.

(2) Following our lecture discussions, the ground state $|0\rangle$ of simple harmonic oscillator Hamiltonian

$$\hat{H}_{\text{SHO}} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2}{2} \hat{x}^2 \quad (\text{H.5})$$

must satisfy equation (H.4) where the length R is given by

$$R = \sqrt{\frac{\hbar}{m\omega}} \quad (\text{H.6})$$

Consider the creation and annihilation operators defined as follows for a simple harmonic oscillator (H.5):

$$\hat{a} = \frac{\hat{x}/R + i\hat{p} R/\hbar}{\sqrt{2}}, \quad \hat{a}^\dagger = \frac{\hat{x}/R - i\hat{p} R/\hbar}{\sqrt{2}} \quad (\text{H.7})$$

which simplifies the Hamiltonian (H.5) to the following form

$$\hat{H}_{\text{SHO}} = \hbar\omega(a^\dagger a + \frac{1}{2}) \quad (\text{H.8})$$

Following our lecture discussions, show that in the orthonormal eigenbasis $\{|n\rangle | n = 0, 1, 2, \dots\}$, we have

$$\langle 0 | a^n (a^\dagger)^n | 0 \rangle = n! \quad (\text{H.9})$$

Comment: this means the normalized eigenstate of $\hat{H}_{\text{SHO}}$ can be written as

$$|n\rangle = \frac{(a^\dagger)^n}{\sqrt{n!}}|0\rangle \quad (\text{H.10})$$

(3) Show that

$$\hat{a}^\dagger(t) \equiv e^{iH_{\text{SHO}}t/\hbar} \hat{a}^\dagger e^{-iH_{\text{SHO}}t/\hbar} = e^{i\omega t} \hat{a}^\dagger \quad (\text{H.11})$$

(4) In lectures we showed that the ground state wavefunction satisfying (H.4) is the Gaussian wavepacket:

$$\psi_0(x) \equiv \langle x|0\rangle \propto e^{-x^2/2R^2} \quad (\text{H.12})$$

Obtain the wavefunction $\psi_1(x) = \langle x|1\rangle$ of the first excited state $|1\rangle$.

Hint:

$$\psi_1(x) = \langle x|\hat{a}^\dagger|0\rangle \quad (\text{H.13})$$

(5) Compute the wavefunction $\psi_2(x) = \langle x|2\rangle$ for the 2nd excited state $|2\rangle$.

(6) In this problem we will learn about Bose-Einstein statistics, by considering a simple harmonic oscillator at a finite temperature $T > 0$. In thermal equilibrium, the system is described by a mixed state with a density matrix

$$\hat{\rho} = \frac{e^{-\beta\hat{H}_{\text{SHO}}}}{\text{Tr } e^{-\beta\hat{H}_{\text{SHO}}}}, \quad \beta = \frac{1}{k_B T} \quad (\text{H.14})$$

where k_B is the Boltzmann constant. Show that the expectation value of number operator $\hat{n} = \hat{a}^\dagger \hat{a}$ is given by

$$\langle \hat{n} \rangle = \text{Tr}(\hat{\rho} \hat{a}^\dagger \hat{a}) = \frac{1}{e^{\beta\hbar\omega} - 1} \quad (\text{H.15})$$

H.0.2 Coupled harmonic oscillators

Consider the following quantum-mechanical Hamiltonian describing two particles connected by a spring of spring constant k in one dimension:

$$\hat{H} = \frac{\hat{p}_1^2}{2m} + \frac{\hat{p}_2^2}{2m} + \frac{k}{2}(\hat{x}_1 - \hat{x}_2)^2 \quad (\text{H.16})$$

with canonical commutation relation

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{i,j}, \quad i, j = 1, 2. \quad (\text{H.17})$$

(1) In the Heisenberg picture, derive the equation of motion for positions and momenta of the two particles:

$$\frac{d}{dt} \begin{pmatrix} \hat{p}_1(t) \\ \hat{p}_2(t) \\ \hat{x}_1(t) \\ \hat{x}_2(t) \end{pmatrix} = ? \quad (\text{H.18})$$

(2) Based on results of question (1), show that

$$-\frac{d^2}{dt^2} \begin{pmatrix} \hat{x}_1(t) \\ \hat{x}_2(t) \end{pmatrix} = \mathbf{K} \begin{pmatrix} \hat{x}_1(t) \\ \hat{x}_2(t) \end{pmatrix} \quad (\text{H.19})$$

where $\mathbf{K}$ is a 2×2 matrix. Diagonalize matrix $\mathbf{K}$ to obtain its eigenvalues and eigenvectors. Can you explain their physical meanings?

(3) Show that the center of mass momentum

$$\hat{P}_{\text{COM}} \equiv \hat{p}_1 + \hat{p}_2 \quad (\text{H.20})$$

is a constant of motion, i.e.

$$[\hat{P}_{\text{COM}}, \hat{H}] = 0 \quad (\text{H.21})$$

(4) Show that the Hamiltonian can be written as a summation of two terms: the center of mass part and the relative motion part

$$\hat{H} = \hat{H}_{\text{COM}}(\hat{P}_{\text{COM}}) + \hat{H}_{\text{relative}}(\hat{x}, \hat{p}) \quad (\text{H.22})$$

where $\hat{H}_{\text{COM}}$ only depends on the center of mass momentum (H.20), and $\hat{H}_{\text{relative}}$ only depends on the following variables:

$$\hat{x} = \hat{x}_1 - \hat{x}_2, \quad \hat{p} \equiv \frac{\hat{p}_1 - \hat{p}_2}{2} \quad (\text{H.23})$$

(5) Show that $\hat{x}, \hat{p}$ defined above is a pair of conjugate variables, satisfying

$$[\hat{x}, \hat{p}] = i\hbar \quad (\text{H.24})$$

Using this relation to diagonalize the Hamiltonian $\hat{H}_{\text{relative}}$ of relative motions, show that it is just another simple harmonic oscillator, and find its eigen-frequency ω_{relative} .

(6) **Bonus** Compute the ground state (labeled as $|g\rangle$) wavefunction of the full Hamiltonian (H.16)

$$\psi_g(x_1, x_2) \equiv \langle x_1, x_2 | g \rangle = ? \quad (\text{H.25})$$

You don't have to normalize the wavefunction.

Hint: separate the center of mass and relative motions.

H.0.3 (Bonus) The operator algebra approach to solve a Hamiltonian

The ladder operator approach to algebraically solve simple harmonic oscillator is one special case of the operator algebra approach to solve a Hamiltonian, which we describe below.

Consider a finite-dimensional Hilbert space $\mathbb{V} \simeq \mathbb{C}^N$. In HW1 we learnt tht all linear operators therein also form a Hilbert space $L_{\mathbb{V}}$, where the inner product between two operators $\hat{O}_1$ and $\hat{O}_2$ is given by the Hilbert-Schmidt inner product:

$$\langle O_1 | O_2 \rangle = \text{Tr}(O_1^\dagger O_2) \quad (\text{H.26})$$

(1) **Bonus** Show that one can always find a complete orthonormal basis $\{T_j\}$ for $L_{\mathbb{V}}$, such that each basis vector of linear operator is Hermitian:

$$\langle T_i | T_j \rangle = \delta_{i,j}, \quad T_j^\dagger = T_j, \quad \forall j \quad (\text{H.27})$$

Hint: if $\{|a\rangle\}$ is a complete orthonormal basis for $\mathbb{V}$, then you can show that a complete orthonormal basis for $L_{\mathbb{V}}$ is given by

$$\{O_{ab} \equiv |a\rangle\langle b|; \forall a, b\} \quad (\text{H.28})$$

(2) **Bonus** In the above Hermitian basis of $L_{\mathbb{V}}$, consider the commutator formula

$$[\hat{H}, \hat{T}_i] = \sum_j \hat{T}_j M_{j,i} \quad (\text{H.29})$$

Show that the square matrix $M_{i,j}$ is anti-symmetric and Hermitian:

$$M_{i,j} = -M_{j,i} = M_{j,i}^* \quad (\text{H.30})$$

Hint:

$$M_{j,i} = \langle T_j | [\hat{H}, \hat{T}_i] \rangle \quad (\text{H.31})$$

(3) **Bonus** Show that each right eigenvector $\vec{\xi}$ of matrix M associated with eigenvalue λ , corresponds to an operator $\hat{O} = \sum_j \xi_j \hat{T}_j$, such that

$$[\hat{H}, \hat{O}] = \lambda \hat{O} \quad (\text{H.32})$$

I Homework 9 (due 11/13)

I.0.1 Coherent states

Consider a normalized coherent state $|z\rangle$ of a simple harmonic oscillator of mass m and frequency ω , defined as follows:

$$|z\rangle \equiv e^{-|z|^2/2} e^{za^\dagger} |0\rangle = e^{-|z|^2/2} \sum_{n=0}^{\infty} \frac{z^n}{\sqrt{n!}} |n\rangle, \quad z \in \mathbb{C}. \quad (\text{I.1})$$

- (1) Show that the coherent state defined above is an eigenstate of annihilation operator $\hat{a}$:

$$\hat{a}|z\rangle = z|z\rangle \quad (\text{I.2})$$

- (2) Show that two different coherent states are not orthonormal, with the following inner product:

$$\langle z|z'\rangle = e^{z^*z' - |z|^2/2 - |z'|^2/2}, \quad z, z' \in \mathbb{C}. \quad (\text{I.3})$$

- (3) **Bonus** Prove the completeness relation for coherent states

$$\hat{1} = \int \frac{d^2z}{\pi} |z\rangle \langle z| \quad (\text{I.4})$$

Hint: for a complex variable $z = x + iy \in \mathbb{C}$, we have $d^2z = dx \, dy$.

Comment: since coherent states are not orthogonal to each other, they form an overcomplete set of basis for the Hilbert space.

- (4) Show that under the time evolution by a simple harmonic oscillator Hamiltonian, a coherent state evolves as

$$e^{-i\hat{H}_{\text{SHO}}t/\hbar} |z\rangle = e^{i\phi(t)} |ze^{-i\omega t}\rangle \quad (\text{I.5})$$

Find out the phase $\phi(t)$.

- (5) A displacement operator $D(z)$ is defined as

$$D(z) = e^{za^\dagger - z^*a}, \quad z \in \mathbb{C} \quad (\text{I.6})$$

Show that $D(z)$ is a unitary operator, and further show that

$$D(z)|0\rangle = |z\rangle \quad (\text{I.7})$$

where $|0\rangle$ is the normalized ground state of a SHO.

Hint: according to the Baker-Campbell-Hausdorff formula, if

$$[\hat{A}, \hat{B}] = \alpha \hat{1} \quad (\text{I.8})$$

we have

$$e^{\hat{A} + \hat{B} + \frac{1}{2}[\hat{A}, \hat{B}]} = e^{\hat{A}} e^{\hat{B}} \quad (\text{I.9})$$

I.0.2 Quantization of electromagnetic fields in a one-dimensional cavity

(1) Considering a propagating mode in a one-dimensional optical cavity, with momentum k and frequency $\omega = c|k|$, its electric field is given by

$$\hat{E}_x(y, t) = \sqrt{\frac{\hbar\omega}{\epsilon L}} \left(\hat{a} e^{i(ky - \omega t)} - \hat{a}^\dagger e^{i(\omega t - ky)} \right). \quad (\text{I.10})$$

For phase factor defined below

$$\Phi = \omega t - ky, \quad (\text{I.11})$$

show the electric field can be written as

$$\hat{E}_x(y, t) = \sqrt{\frac{2\hbar\omega}{\epsilon L}} \left(\hat{X} \cos \Phi + \hat{Y} \sin \Phi \right). \quad (\text{I.12})$$

where the quadratures X, Y are defined as

$$\hat{X} = \frac{\hat{a} + \hat{a}^\dagger}{\sqrt{2}}, \quad \hat{Y} = \frac{i(\hat{a}^\dagger - \hat{a})}{\sqrt{2}}. \quad (\text{I.13})$$

(2) In classical electromagnetism, the energy density is given by

$$u = \frac{\epsilon}{2} |\mathbf{E}|^2 + \frac{1}{2\mu} |\mathbf{B}|^2 \quad (\text{I.14})$$

where ϵ is the dielectric constant and μ is the permeability. For a single mode of standing wave in the one-dimensional optical cavity (along y direction), as we discussed in section 4.3.2, the electric and magnetic fields are given by

$$\hat{E}_x(y, t) = i \sqrt{\frac{\hbar\omega}{\epsilon L}} \sin(ky) \left(\hat{a} e^{-i\omega t} - \hat{a}^\dagger e^{i\omega t} \right). \quad (\text{I.15})$$

and

$$\hat{B}_z(y, t) = -k \sqrt{\frac{\hbar}{\epsilon L \omega}} \cos(ky) \left(\hat{a} e^{-i\omega t} + \hat{a}^\dagger e^{i\omega t} \right). \quad (\text{I.16})$$

Show that the total energy of electromagnetic fields is carried by the photon:

$$\int_0^L u(y, t) dy = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) \quad (\text{I.17})$$

(3) In classical electromagnetism, the momentum density of the electromagnetic fields is given by

$$\mathbf{g} = \epsilon \mathbf{E} \times \mathbf{B} \quad (\text{I.18})$$

Again we consider a single mode of standing wave in the one-dimensional cavity with $g_y = -\epsilon E_x B_z$, and compute total momentum $\int_0^L g_y(y, t) dy$ of electromagnetic fields carried by this standing wave.

(4) Show that the expectation values of electric field (I.15) and magnetic field (I.16) in a photon coherent state $|z\rangle$ reproduce their classical values as standing waves. Obtain the amplitude of electric and magnetic fields as functions of z .

Hint: recall that $\hat{a}|z\rangle = z|z\rangle$ in a coherent state $|z\rangle$.

(5) **Bonus** In this problem we discuss the Planck formula of blackbody radiation, by considering electromagnetic fields as standing waves in a 3-dimensional cube of size $L \times L \times L$. The

total energy density of photons (including the two polarization modes) is given by

$$u = \frac{2}{2\pi^2} \int |k|^2 d|k| \langle \hat{n}_k \rangle \omega_k, \quad \omega_k = c|k|. \quad (\text{I.19})$$

where the expectation value of photon number $\hat{n}_k = \hat{a}_k^\dagger \hat{a}_k$ is for thermal equilibrium at temperature $T > 0$. By changing the integration variable from $|k|$ to (angular) frequency $\omega = c|k|$, derive the Planck formula:

$$u = \int d\omega u(\omega), \quad u(\omega) = \frac{1}{\pi^2 c^3} \frac{\omega^3}{e^{\beta \hbar \omega} - 1} \quad (\text{I.20})$$

Comment: in the original version of Planck's formula discussed in section 1.1.1.1, we used the light frequency $\nu = \omega/2\pi$, hence the extra factor of $(2\pi)^4$.

I.0.3 Casimir effect

Consider the quantization of light in a one-dimensional optical cavity of length L , as we discussed in section 4.3.2.

(1) Compute the ground state energy E_0 (i.e. zero point energy) of the system as a function of $L, c, \hbar$.

Hint: by the zeta function regularization we have

$$\sum_{n=1}^{\infty} n = \zeta(-1) = -\frac{1}{12} \quad (\text{I.21})$$

(2) Show that E_0 increases with an increasing cavity length L , and compute the attractive Casimir force between the two cavity walls (separated by a distance L):

$$F_{\text{Cas}} = -\frac{\partial E_0}{\partial L} \quad (\text{I.22})$$

I.0.4 (Bonus) Squeezed states

We consider a generic squeezed state $|\xi, z\rangle$, characterized by two complex numbers $\xi, z \in \mathbb{C}$ as

$$|\xi, z\rangle \equiv \hat{S}(\xi) \hat{D}(z) |0\rangle \propto \hat{S}(\xi) |z\rangle \quad (\text{I.23})$$

where $|0\rangle$ is the normalized ground state of a dimensionless (set $\hbar = m = \omega = 1$) simple harmonic oscillator

$$\hat{H} = \frac{\hat{x}^2 + \hat{p}^2}{2} = a^\dagger a + \frac{1}{2} \quad (\text{I.24})$$

with

$$[\hat{x}, \hat{p}] = i, \quad a = \frac{\hat{x} + i\hat{p}}{\sqrt{2}} e^{i\frac{\theta}{2}}, \quad a^\dagger = \frac{\hat{x} - i\hat{p}}{\sqrt{2}} e^{-i\frac{\theta}{2}}, \quad \forall \theta \in \mathbb{R}. \quad (\text{I.25})$$

The unitary squeezing operator $\hat{S}(\xi)$ is defined as

$$\hat{S}(\xi) \equiv e^{\frac{\xi^* a^2 - \xi (a^\dagger)^2}{2}}, \quad \xi \in \mathbb{C} \quad (\text{I.26})$$

while the displacement operator $\hat{D}(z)$ is defined in (I.6).

(1) **Bonus** Show that for $\xi = re^{i\theta}$ with $r, \theta \in \mathbb{R}$, we have

$$S^\dagger(\xi)aS(\xi) = a \cosh r - e^{i\theta}a^\dagger \sinh r, \quad (\text{I.27})$$

$$S^\dagger(\xi)a^\dagger S(\xi) = a^\dagger \cosh r - e^{-i\theta}a \sinh r \quad (\text{I.28})$$

Hint: use the Baker-Campbell-Hausdorff formula

$$e^A B e^{-A} = B + [A, B] + \frac{1}{2!}[A, [A, B]] + \frac{1}{3!}[A, [A, [A, B]]] + \dots \quad (\text{I.29})$$

(2) **Bonus** Show that the squeezed state (I.23) satisfies:

$$(\hat{x} + i\hat{p}e^{-2r} - C)|\xi, z\rangle = 0. \quad (\text{I.30})$$

where position and momentum operators $\hat{x}, \hat{p}$ are defined in (I.25), and $C \in \mathbb{C}$ is a constant.

Hint: $(a - z)|z\rangle = 0$ for a coherent state $|z\rangle$, which is related to the squeezed state by the squeezing operator $S(\xi)$ as shown in (I.23).

Comment: Eq. (I.30) shows that each squeezed state $|\xi, z\rangle$ is a Gaussian wave packet, with a characteristic radius of $R = e^{-r}$.

(3) **Bonus** Now let's consider the electric field operator in single-mode quantum optics

$$\hat{E}(\chi) = \frac{ae^{-i\chi} + a^\dagger e^{i\chi}}{\sqrt{2}} \quad (\text{I.31})$$

where

$$\chi = \omega t - \vec{k} \cdot \vec{r} + \pi/2 \quad (\text{I.32})$$

is the phase factor. Compute its variance

$$\Delta E^2(\chi) \equiv \langle \xi, z | [\Delta \hat{E}(\chi)]^2 | \xi, z \rangle \quad (\text{I.33})$$

w.r.t. the squeezed state (I.23). For the specific case of $\xi = 2$, plot $\Delta E^2(\chi)$ as a function of χ .

J Homework 10 (due 11/20)

J.0.1 Beta decay of Tritium into Helium ion

(1)-(2) S&N Problem 5.35 (a)-(b), see Fig. J.1.

J.0.2 Quantum quench of a simple harmonic oscillator

Consider the following quantum quench of a simple harmonic oscillator:

$$\hat{H}(t) = \hat{H}_0\theta(-t) + \hat{H}_1\theta(t) \quad (\text{J.1})$$

where $\theta(t)$ is the heaviside step function, and the initial Hamiltonian is a simple harmonic oscillator of frequency ω_0

$$\hat{H}_0 = \frac{\hat{p}^2}{2m} + \frac{m\omega_0^2}{2}\hat{x}^2 \quad (\text{J.2})$$

We denote the ground state of $\hat{H}_0$ to be $|0\rangle$, in the same notation as in lecture notes.

(1) We first consider $\hat{H}_1$ to be a displaced harmonic oscillator of the same frequency:

$$\hat{H}_1 = \frac{\hat{p}^2}{2m} + \frac{m\omega_0^2}{2}(\hat{x} - x_0)^2 \quad (\text{J.3})$$

centered at $x_0 > 0$. Denote the ground state of $\hat{H}_1$ as $|\psi_{x_0}\rangle$.

Compute the transition probability $|\langle\psi_{x_0}|0\rangle|^2$ between the two ground states before and after the quench.

Hint: recall the ground state of a SHO is a Gaussian wavepacket $\langle x|0\rangle \sim e^{-x^2/2R^2}$.

(2) **Bonus** Before the quantum quench (i.e. the sudden change of Hamiltonian) at $t = 0$, assume the initial state of the system is $\hat{H}_0$ ground state $|0\rangle$. After the quantum quench, the system evolves into $e^{-i\hat{H}_1 t/\hbar}|0\rangle$. Show that it is a coherent state.

(3) **Bonus** Now we consider a different scenario: $\hat{H}_1$ is a SHO with a different frequency $\omega \neq \omega_0$. Show that the quantum state $e^{-i\hat{H}_1 t/\hbar}|0\rangle$ after the quench is a squeezed state.

J.0.3 Semiclassical equations of motion

In the Heisenberg picture, if we consider an electron under electromagnetic fields

$$\hat{H} = \frac{|\hat{\Pi}|^2}{2m} + e\phi(\vec{x}), \quad \hat{\Pi} = \hat{\vec{p}} - e\vec{A}(\vec{x}). \quad (\text{J.4})$$

the equations of motion for $\hat{\vec{x}}$ and $\hat{\Pi}$ reproduce the following semiclassical equations of motion:

$$\frac{d\vec{x}}{dt} = \vec{v}, \quad (\text{J.5})$$

$$m\frac{d\vec{v}}{dt} = e\vec{E} + e\vec{v} \times \vec{B} \quad (\text{J.6})$$

5.35 Consider an atom made up of an electron and a singly charged ($Z = 1$) triton (${}^3\text{H}$). Initially the system is in its ground state ($n = 1, l = 0$). Suppose the system undergoes beta decay, in which the nuclear charge *suddenly increases* by one unit (realistically by emitting an electron and an antineutrino). This means that the tritium nucleus (called a triton) turns into a helium ($Z = 2$) nucleus of mass 3 (${}^3\text{He}$).

- (a) Obtain the probability for the system to be found in the ground state of the resulting helium ion. The hydrogenic wave function is given by

$$\psi_{n=1,l=0}(\mathbf{x}) = \frac{1}{\sqrt{\pi}} \left(\frac{Z}{a_0} \right)^{3/2} e^{-Zr/a_0}.$$

- (b) The available energy in tritium beta decay is about 18 keV, and the size of the ${}^3\text{He}$ atom is about $1\text{\AA}$. Check that the time scale T for the transformation satisfies the criterion of validity for the sudden approximation.

Figure J.1: Problem 1.(1)-(2)

where

$$\vec{x} \equiv \langle \hat{x} \rangle, \quad \vec{v} \equiv \frac{1}{m} \langle \hat{\Pi} \rangle \quad (\text{J.7})$$

(1) For a constant electric field $\vec{E}$ and a constant magnetic field $\vec{B}$, if they are perpendicular to each other ($\vec{E} \perp \vec{B}$), solve the semiclassical equation of motion (J.6) to obtain the steady-state velocity $\vec{v}$, which is a constant of time, as a function of $\vec{E}$ and $\vec{B}$.

(2) Consider many electrons constrained in the two-dimensional x - y plane, moving under a constant electric field $\vec{E} = E_y \hat{y}$ and a magnetic field $\vec{B} = |B| \hat{z}$ that are perpendicular to each other. The electric current density of the system is given by

$$\vec{j} = en\vec{v} \quad (\text{J.8})$$

where we assume the electron density n is a constant in the system. The steady-state current density is related to the in-plane electric field by the *Hall conductivity* σ_{xy} :

$$j_x = \sigma_{xy} E_y \quad (\text{J.9})$$

Compute the Hall conductivity σ_{xy} as a function of magnetic field strength $|B|$ and n .

Comment: the perpendicular electric current driven by an electric field under a perpendicular magnetic field is known as the Hall effect. It is an important way to measure the electron density n in a material.

(3) Now consider the case of $\vec{E} = 0$, i.e. a two-dimensional electron moving under a perpendicular magnetic field. Solve the equation of motion (J.6) to show that the electron will move in a circle, known as the cyclotron motion of electrons under a perpendicular magnetic field. Compute the frequency ω_c of the circular motion, known as the cyclotron frequency.

J.0.4 Landau levels

Consider the motion of a two-dimensional electron in the x - y plane, under a perpendicular magnetic field $\vec{B} = B \hat{z}$ with $B < 0$:

$$\hat{H} = \frac{(\hat{\Pi}_x)^2 + (\hat{\Pi}_y)^2}{2m} \quad (\text{J.10})$$

where the kinematic (mechanical) momentum is given by

$$\hat{\Pi}_\alpha = \hat{p}_\alpha - eA_\alpha(\hat{x}, \hat{y}), \quad \alpha = x, y. \quad (\text{J.11})$$

and the vector potential $\vec{A}$ satisfies

$$\partial_x A_y - \partial_y A_x = B \quad (\text{J.12})$$

$e < 0$ is the electron charge.

(1) Prove the commutation relation

$$[\Pi_x, \Pi_y] = i\hbar eB \quad (\text{J.13})$$

where $e = -|e|$ is the electron charge.

(2) Find a dimensionless annihilation operator

$$\hat{a} = a_1 \hat{\Pi}_x + ia_2 \hat{\Pi}_y, \quad a_{1,2} > 0 \quad (\text{J.14})$$

such that

$$[\hat{a}, \hat{a}^\dagger] = 1 \quad (\text{J.15})$$

and the Hamiltonian is diagonalized as a simple harmonic oscillator:

$$\hat{H} = \hbar\omega_c(\hat{a}^\dagger \hat{a} + \frac{1}{2}) \quad (\text{J.16})$$

Here $\omega_c = |eB|/\hbar m$ is the cyclotron frequency.

(3) Note that in addition to mechanical momentum $\hat{\Pi}$, the position operators $(\hat{x}, \hat{y})$ are also gauge-invariant physical observables. Verify the following commutation relation

$$[\hat{x}_a, \hat{\Pi}_b] = i\hbar\delta_{a,b}, \quad a, b = 1, 2. \quad (\text{J.17})$$

Use this relation to find two conserved quantities of the system:

$$\hat{X} = \hat{x} + c_x \hat{\Pi}_y, \quad \hat{Y} = \hat{y} + c_y \hat{\Pi}_x \quad (\text{J.18})$$

so that

$$[\hat{X}, \hat{H}] = [\hat{Y}, \hat{H}] = 0 \quad (\text{J.19})$$

Determine the two constants c_x and c_y .

Comment: $(\hat{X}, \hat{Y})$ are known as the *guiding center coordinates* of electrons under a perpendicular magnetic field.

(4) Show that the guiding center coordinates (J.18) satisfy the following commutation relation:

$$[\hat{X}, \hat{Y}] = -il_B^2, \quad l_B = \sqrt{\frac{\hbar}{|eB|}} \quad (\text{J.20})$$

where l_B is known as the *magnetic length*. Use this commutation relation to find an annihilation operator

$$b = b_1(\hat{X} - i\hat{Y}), \quad b_1 > 0. \quad (\text{J.21})$$

such that

$$[b, \hat{\Pi}_x] = [b, \hat{\Pi}_y] = 0, \quad [b, b^\dagger] = 1. \quad (\text{J.22})$$

In this way, $n_b = b^\dagger b \geq 0$ labels an extra quantum number of the system, in addition to energy $E = \hbar\omega_c(n_a + \frac{1}{2})$ where $n_a = a^\dagger a \geq 0$. Eigenstates $|n_a, n_b\rangle$ with the same n_a but different n_b values are degenerate eigenstates with the same energy.

(5) **Bonus** Denote the projection operators into the n -th Landau level as

$$\hat{P}_n \equiv \sum_{n_b} |n_a = n, n_b\rangle \langle n_a = n, n_b| \quad (\text{J.23})$$

Show that

$$\hat{P}_n \hat{x} \hat{P}_n = \hat{X} \hat{P}_n, \quad \hat{P}_n \hat{y} \hat{P}_n = \hat{Y} \hat{P}_n, \quad \forall n. \quad (\text{J.24})$$

Comment: this problem shows that the guiding center coordinates $\hat{X}, \hat{Y}$ are nothing but the projection of real space position operators $\hat{x}, \hat{y}$ into a single Landau level. The fact that they do not commute with each other, because of Eq.(J.20), demonstrates the extreme quantum nature of Landau levels, even though Ehrenfest's theorem says that the equations of motions do follow the classical result.

(6) **Bonus** Now we choose the symmetric gauge

$$\vec{A} = (-y, x, 0) \frac{B}{2} \quad (\text{J.25})$$

so that the system preserves rotational symmetry:

$$\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} x \cos \theta - y \sin \theta \\ x \sin \theta + y \cos \theta \end{pmatrix}, \quad \begin{pmatrix} p_x \\ p_y \end{pmatrix} \rightarrow \begin{pmatrix} p_x \cos \theta - p_y \sin \theta \\ p_x \sin \theta + p_y \cos \theta \end{pmatrix} \quad (\text{J.26})$$

Show that the angular momentum operator

$$\hat{L}_z = \hat{x} \hat{p}_y - \hat{y} \hat{p}_x \quad (\text{J.27})$$

is related to the two pairs of creation/annihilation operators by

$$\hat{L}_z / \hbar = b^\dagger b - a^\dagger a \quad (\text{J.28})$$

This means the degeneracy within each Landau level (with fixed n_a) comes from different angular momenta of the states.

(7) **Bonus** In the symmetric gauge (J.25), compute the eigenstate wavefunctions

$\langle x, y | n_a = 0, n_b \geq 0 \rangle$ for the lowest Landau level, with $n_a = a^\dagger a = 0$. What do they look like in real space? You don't have to normalize the wavefunction.

Hint: you may want to use complex coordinates:

$$z = x + iy, \quad \bar{z} = x - iy. \quad (\text{J.29})$$

The ground states in the lowest Landau level satisfy

$$\hat{a} |n_a = 0, n_b\rangle = 0 \quad (\text{J.30})$$

K Homework 11 (due 12/4)

K.0.1 (Bonus) Berry's phase for a spin-S particle

(1) **Bonus** Consider an instantaneous eigenstate $|n, \vec{R}\rangle$ of a time-dependent Hamiltonian $\hat{H}(\vec{R})$ with eigenvalue $E_n(\vec{R})$, where parameters $\vec{R}(t)$ change with time. We have learnt in lectures that its *Berry connection* is given by

$$\vec{A}_n(\vec{R}) = -i\langle n, \vec{R} | \vec{\nabla}_R | n, \vec{R} \rangle \quad (\text{K.1})$$

and its *Berry curvature* is given by

$$\vec{B}_n(\vec{R}) = \vec{\nabla}_R \times \vec{A}_n(\vec{R}) = -i(\vec{\nabla}_R \langle n, \vec{R} |) \times (\vec{\nabla}_R | n, \vec{R} \rangle) \quad (\text{K.2})$$

Show that

$$\vec{B}_n(\vec{R}) = -i \sum_{m \neq n} \frac{\langle n, \vec{R} | \vec{\nabla}_R \hat{H}(\vec{R}) | m, \vec{R} \rangle \times \langle m, \vec{R} | \vec{\nabla}_R \hat{H}(\vec{R}) | n, \vec{R} \rangle}{(E_m - E_n)^2} \quad (\text{K.3})$$

Hint:

$$\hat{H}(\vec{R}) | n, \vec{R} \rangle = E_n(\vec{R}) | n, \vec{R} \rangle \quad (\text{K.4})$$

(2) **Bonus** Consider a spin- S particle moving under a time-dependent magnetic field $\vec{R}(t)$:

$$\hat{H}(\vec{R}) = -\vec{S} \cdot \vec{R} \quad (\text{K.5})$$

where $\vec{S} = (S_x, S_y, S_z)$ are angular momentum operators for a particle of spin $j = S$. Without loss of generality, let's consider the magnetic field to be on a unit sphere near the north pole:

$$\vec{R} = (R_x, R_y, \sqrt{1 - R_x^2 - R_y^2}), \quad |R_{x,y}| \ll 1. \quad (\text{K.6})$$

We denote the eigenstate with largest $\hat{S}_z$ eigenvalue as $|0\rangle$, satisfying

$$\hat{S}_z |0\rangle = S |0\rangle, \quad (\hat{\vec{S}} \cdot \hat{\vec{S}}) |0\rangle = S(S+1) |0\rangle. \quad (\text{K.7})$$

Identify the ground state $|\vec{R}\rangle$ of Hamiltonian (K.5) with the following form:

$$|\vec{R}\rangle = \hat{U}(\vec{R}) |0\rangle \quad (\text{K.8})$$

where $\hat{U}(\vec{R})$ is a unitary matrix as a function of $R_{x,y}$ in (K.6). Obtain the unitary operator $\hat{U}(\vec{R})$.

Hint: The commutation relations for angular momenta $\hat{S}_a$, $a = x, y, z$ is given by

$$[S^a, S^b] = i\hbar \epsilon_{abc} S^c \quad (\text{K.9})$$

This leads to the following identity

$$e^{-i\theta\hat{S}_y/\hbar}f(\hat{S}_z)e^{i\theta\hat{S}_y/\hbar} = f(\cos\theta\hat{S}_z + \sin\theta\hat{S}_x). \quad (\text{K.10})$$

where $f(x)$ is any function, and $e^{-i\theta\hat{S}_y/\hbar}$ implements a counter-clockwise rotation along y -axis by angle θ . You also need to make sure that $|\vec{R}\rangle$ is single-valued near the north pole.

(3) **Bonus** Use the instantaneous ground state $|\vec{R}\rangle$ obtained previously to compute the Berry connection

$$\vec{A}_a = -i\langle\vec{R}|\partial_{R_a}|\vec{R}\rangle, \quad a = x, y \quad (\text{K.11})$$

up to lowest order in $R_{x,y}$.

Hint: you can expand $|\vec{R}\rangle$ in Taylor series of variables $R_{x,y}$.

(4) **Bonus** Compute the Berry curvature

$$B(\vec{R}) = \partial_{R_x}A_y - \partial_{R_y}A_x \quad (\text{K.12})$$

at north pole $\vec{R} = (0, 0, 1)$. Use your result to further obtain the Berry flux over the unit sphere of $\vec{R}$, by integrating the Berry curvature over the unit sphere. How does your result depends on S ?

K.0.2 Aharonov-Bohm phase for a quantum particle on a ring

Consider the following Hamiltonian of a quantum particle on a ring, trapped by a delta function potential located at $x = \lambda$:

$$\hat{H} = \frac{(\hat{p} - eA)^2}{2m} - V\delta(\hat{x} - \lambda), \quad V > 0, \quad A = \frac{\Phi}{L} = \frac{2\pi\hbar}{eL}\phi. \quad (\text{K.13})$$

where we have considered a flux of $\phi\Phi_0$ piercing through the center of the ring, $\Phi_0 = 2\pi\hbar/e$ being the flux quantum. We will assume $\lambda = 0$ for problems (1)-(3).

(1) Recall that any valid wavefunction of a particle on a ring must be periodic:

$$\psi(x) = \psi(x + L) \quad (\text{K.14})$$

A complete set of orthonormal basis of periodic functions is naturally given by the plane waves:

$$\{|n\rangle|n \in \mathbb{Z}, \langle x|n\rangle = \frac{e^{2\pi nix/L}}{\sqrt{L}}\}, \quad \langle l|n\rangle = \delta_{n,l} \quad (\text{K.15})$$

Consider an eigenstate $|\psi\rangle$ of Hamiltonian (K.13) satisfying:

$$\hat{H}|\psi\rangle = E|\psi\rangle, \quad |\psi\rangle = \sum_n c_n|n\rangle. \quad (\text{K.16})$$

Show that the above Schrodinger equation leads to the following formula:

$$\left[\left(\frac{2\pi}{L}\right)^2(n - \phi)^2 - \frac{2mE}{\hbar^2}\right]c_n = \frac{2mV}{\hbar^2\sqrt{L}}\psi(0) \quad (\text{K.17})$$

(2) Prove the following identity:

$$\frac{2mL}{\hbar^2V} = \sum_{n \in \mathbb{Z}} \frac{1}{\left(\frac{2\pi}{L}\right)^2(n - \phi)^2 - 2mE/\hbar^2} \quad (\text{K.18})$$

This equation determines the energy eigenvalues of Hamiltonian (K.13).

(3) This problem is a sanity check to see if we reproduce the bound state solution in the infinite space. In the limit of an infinitely long ring i.e. $L \rightarrow \infty$, solve for the energy E in equation (K.18) for a bound state with $E < 0$. Show that

$$E = -\frac{mV^2}{2\hbar^2} \quad (\text{K.19})$$

This reproduces the bound state energy we obtained previously in (3.85).

Hint:

$$\lim_{L \rightarrow \infty} \frac{1}{L} \sum_{n \in \mathbb{Z}} = \int_{-\infty}^{\infty} \frac{dk}{2\pi}, \quad k = \frac{2\pi n}{L} \quad (\text{K.20})$$

(4) **Bonus** Now consider the Hamiltonian (K.13) where the parameter λ changes very slowly in the adiabatic limit. In this case, the ground state (i.e. the aforementioned bound state) of $\hat{H}(\lambda)$ has the following form:

$$\langle x | \psi_\lambda \rangle = \sum_n c_n \frac{e^{\frac{2\pi i n}{L}(x-\lambda)}}{\sqrt{L}} \implies |\psi_\lambda\rangle = \sum_n c_n e^{-\frac{2\pi i n \lambda}{L}} |n\rangle \quad (\text{K.21})$$

Compute the Berry connection

$$A_{\text{Berry}}(\lambda) = -i \langle \psi_\lambda | \partial_\lambda | \psi_\lambda \rangle \quad (\text{K.22})$$

and then show that the Berry phase of adiabatically moving the bound-state particle (localized around $x = \lambda$) around the ring is equal to the AB phase:

$$\gamma = -\oint_0^L A_{\text{Berry}}(\lambda) d\lambda = 2\pi\phi = \frac{e}{\hbar} \oint_0^L A dx \quad (\text{K.23})$$

Hint: you need to use conditions (K.17) and (K.18).

K.0.3 Rabi oscillation

Consider a qubit (i.e. a spin-1/2 particle) coupled to a constant magnetic field along $\hat{z}$ -axis, and a time-dependent magnetic field in the $\hat{x}$ - $\hat{y}$ plane:

$$\hat{H}(t) = \hat{H}_0 + \hat{V}(t) = -\frac{\hbar\omega_{21}}{2}\sigma_z - \hbar\gamma(\sigma_x \cos \omega t - \sigma_y \sin \omega t) \quad (\text{K.24})$$

where $\sigma_{x,y,z}$ are the Pauli matrices.

(1) During the lecture, we have shown that in the interaction picture, the solution of the time-dependent Schrodinger equation is given by

$$|\psi(t)\rangle_I = c_1(t)|\uparrow\rangle + c_2(t)|\downarrow\rangle \quad (\text{K.25})$$

where the coefficients $c_{1,2}$ satisfy the following equations:

$$i\dot{c}_1 = -\gamma e^{i(\omega - \omega_{21})t} c_2, \quad (\text{K.26})$$

$$i\dot{c}_2 = -\gamma e^{i(\omega_{21} - \omega)t} c_1. \quad (\text{K.27})$$

Under the initial condition

$$c_1(t=0) = 1, \quad c_2(t=0) = 0 \quad (\text{K.28})$$

solve the above differential equations to obtain $c_1(t)$ and $c_2(t)$.

- (2) Compute $\langle S_z(t) \rangle$.
- (3) For $\omega = \omega_{21} = \gamma$, plot $\langle S_z(t) \rangle$ as a function of ωt .
- (4) **Bonus** Consider the Schrodinger equation for Rabi's problem

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H}(t) |\psi(t)\rangle \quad (\text{K.29})$$

where the time-dependent Hamiltonian is given in (K.24). Can you find a new “rotating frame”, i.e. a 2×2 unitary matrix $U(t)$ such that the following vector

$$|\tilde{\psi}(t)\rangle = U(t) |\psi(t)\rangle \quad (\text{K.30})$$

evolves under a time-independent Hamiltonian $\tilde{H}$:

$$i\hbar \frac{d}{dt} |\tilde{\psi}(t)\rangle = \tilde{H} |\tilde{\psi}(t)\rangle \quad (\text{K.31})$$

Find such a $U(t)$ and the associated $\tilde{H}$.

- (5) **Bonus** In the rotating frame, under the initial condition

$$|\psi(t=0)\rangle = |\uparrow\rangle \quad (\text{K.32})$$

solve for $|\psi(t)\rangle$ and compute $\langle S_z(t) \rangle$.

(6) Obtain the instantaneous eigenstates $|\pm, t\rangle$ and eigenvalues $E_{\pm}(t)$ for the Hamiltonian (K.24) at a given time $t > 0$. We use the convention that $E_+(t) > 0 > E_-(t)$.

(7) Consider Rabi's Hamiltonian (K.24) as a function of parameter $\lambda = \omega t$. Is it possible to reach the adiabatic limit in Rabi's problem (K.24)? Derive the conditions for the adiabatic limit on $\omega_{21}, \gamma, \omega$.

(8) In the adiabatic limit, compute the Berry's phase for both instantaneous eigenstates $|\pm, \lambda(t)\rangle$ over one period $T = 2\pi/\omega$.

(9) Can you use the Berry's phases obtained above to explain the Rabi oscillation, in the adiabatic limit?

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